Macro-Financial Determinants of the Philippine Stock Exchange Index (PSEi) Returns: A Multiple Linear Regression Analysis, 2018-2025

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Executive Summary

This report examines the connection between international and domestic macro-financial indicators and the monthly log returns of the Philippine Stock Exchange Index (PSEi) from January 2018 through December 2025, using 96 monthly observations. Its nine variables represent four channels: the discount rate (the change in the Bangko Sentral ng Pilipinas (BSP) reverse repurchase rate, the change in the Philippine 10-year government bond yield and consumer price inflation), global risk (the S&P 500 return, the WTI crude oil return and the Volatility Index (VIX)), exchange rate (USD/PHP) and liquidity and real activity (M3 money supply growth and manufacturing production growth). The series come from the Federal Reserve Bank of St. Louis, LSEG Workplace, PSA (Philippine Statistics Authority), and the BSP and were aligned to a common monthly basis. As our dependent variable, the PSEi return is skewed to the left, with 21 months below -5 percent against 16 above +5 percent, and one month, March 2020, with a log return of -0.24, is also the extreme month of the S&P 500, WTI and the VIX.

As our primary modeling approach, multiple linear regression measures the direction, size and significance of each association. Our initial model is significant as a whole ($F(9, 86) = 12.79$) and explains 57.2 percent of the variation in returns (adjusted R-squared 0.5277), yet under conventional standard errors only the USD/PHP exchange rate, the 10-year yield and WTI are individually significant. The money market and exchange rate group is jointly significant, while the domestic macroeconomic group is not, and the global risk group depends on the covariance estimator. Of five candidate models, AIC, AICc and BIC all prefer the three significant regressors, and the other six add no significant explanatory power ($p = 0.507$), although choosing them from the same sample favors them.

Diagnostics found heteroskedasticity (Breusch-Pagan $p = 0.025$), functional-form misspecification (RESET $p = 0.020$) and one dominant influential observation, March 2020, whose Cook's distance of 1.55 is more than five times the next largest. To address these problems, we applied HC3 robust standard errors, a March 2020 dummy variable, Huber and MM robust regression and a reduced regression model. The USD/PHP exchange rate and the 10-year yield stayed significant and negative in every linear fit, whereas WTI weakened under HC3 ($p = 0.147$) and once March 2020 was modeled ($p = 0.171$). The reduced model with the dummy performs best, with an adjusted R-squared of 0.5881 and a cross-validated RMSE of 0.04280 against 0.5277 and 0.04936 for the initial model, a 13 percent lower error.

We extend our work into Logistic regression to turn the question into probabilities: that of an up month (48 of 96 months) and that of a sharp decline, a return below -5 percent (21 months), modeled with a penalized likelihood because such events are rare. A 1 percent peso depreciation multiplies the odds of an up month by 0.39 and the odds of a sharp decline by 2.65, and a 10 basis point rise in the yield multiplies them by 0.78 and 1.22. Both drivers hold without March 2020 and the full models separate months well out of sample (cross-validated AUC 0.80 and 0.83), whereas we observe the oil effect on sharp declines does not hold.

Thus, in summary, we show the following: A weaker peso may signal weak investor confidence and foreign fund outflows and raises the cost of imported goods and foreign-currency commitments, so a 1 percent depreciation goes with a monthly return about 2.0 percentage points lower. A higher yield lowers the present value of future earnings, raises borrowing costs and may make government bonds more attractive than equities, and the policy rate adds no separate signal. The global risk channel appears to act through the peso, and oil is a conditional signal of a global demand shock in an extreme month, and it is not a stable cost channel. Inflation, M3 and manufacturing production are slow year-on-year rates that markets price before release, so they are inconclusive, not irrelevant, since 96 months are too few to measure them precisely. We stress all of this is contemporaneous association, not causation or forecast.

Our recommendation follows the strength of the evidence. In assessing Philippine stocks, investors should monitor the USD/PHP exchange rate and the 10-year government bond yield, and should pay particular attention to companies with substantial foreign-currency liabilities, dependence on imported materials or high debt. If the peso depreciates 1 percent and the yield rises 10 basis points in the same month, the estimated odds put the chance of a sharp decline near 48 percent against 22 percent in the sample (illustrative), so investors should reassess equity exposure, diversify across asset classes, favoring short-dated government securities, and prefer firms with healthy finances and stable cash flows. Crude oil prices should not serve as a standalone indicator, and these signals indicate when to review exposure, not when to buy or sell.

Section 1: Project Overview

Introduction

Equity returns are among the most closely watched outcomes in finance, and among the hardest to explain. Stock share prices may respond to news about interest rates, inflation, currencies, liquidity, global risk appetite and real economic activity, and the weight of each influence shifts with the economic setting and with the level of market uncertainty. Our domain of interest here is that relationship in a small, open emerging market. The Philippine Stock Exchange Index (PSEi) is the local exchange's main index, a free-float-adjusted index of the 30 largest and most actively traded common stocks listed there (The Philippine Stock Exchange, Inc., n.d.), so its monthly return summaries how the market as a whole values Philippine listed companies. N.-F. Chen et al. (1986) showed that macroeconomic innovations, among them industrial production, inflation and the term structure of interest rates, are priced in stock returns, and Wongbangpo & Sharma (2002) examined the interdependence between stock markets and macroeconomic fundamentals in the ASEAN-5 countries, the Philippines among them.

This study examines the macro-financial determinants of the monthly log returns of the PSEi from January 2018 to December 2025, which gives 96 monthly observations. Nine explanatory variables are used: S&P 500 returns, USD/PHP exchange rate movements, changes in the Bangko Sentral ng Pilipinas (BSP) reverse repurchase rate, changes in Philippine 10-year government bond yields, Philippine consumer price inflation, M3 money supply growth, manufacturing production growth, West Texas Intermediate (WTI) crude oil returns and the CBOE Volatility Index (VIX). They follow four channels developed in Section 2: the discount-rate channel (the policy rate, the 10-year yield and inflation), the global risk channel (the S&P 500, the VIX and WTI crude oil), the exchange rate channel (USD/PHP) and the liquidity and real-activity channel (M3 and manufacturing production). The window suits the domain because conditions varied widely across it. The policy rate ranged from 2.00% to 6.50%, the 10-year yield from 3.7% to 8.3% and year-on-year inflation from 0.8% to 8.7%, and the COVID-19 shock produced the worst month of the sample, a log return of -0.24 in March 2020, in line with the unprecedented stock market reaction documented by Baker et al. (2020).

Such variation gives the regressions something to explain, but it also lets a few extreme months dominate the estimates, so our analysis does not stop at estimation. Multiple linear regression measures how each variable is associated with PSEi returns in direction, magnitude and statistical significance while the others are held constant aligned with the principle of ceteris paribus. Statistical inference, model evaluation, regression diagnostics and model improvement then test whether those associations can be trusted, and logistic regression extends the analysis from the size of a return to the probability of a rising or sharply falling month.

Knowing which conditions move the stock market, in which direction and by how much is essential to sound financial decisions. We report in the scope of investment and portfolio analysis with exposure to Philippine equities, for example a fund management benchmarked to the PSEi. Each month must be judged whether conditions justify holding, adding to, hedging or trimming that exposure, and which published indicators deserve attention. Two practical questions follow. The first concerns monitoring: which conditions have moved the market reliably enough to be watched, and how much return is associated with a given change in each of them? The second concerns risk: because large losses are costly and slow to recover, the investor also needs to know whether the same conditions help identify the months in which sharp declines occur, defined here as a monthly log return below -5%.

Likewise, financial institutions that set risk limits and policymakers who track financial conditions face versions of the same problem and can read the results as evidence on how global and domestic developments reach the market. We stress that the report evaluates contemporaneous associations, not forecasts. Monthly equity returns are notoriously hard to predict out of sample (Welch & Goyal, 2008), so the evidence supports decisions about what to monitor and how much risk to carry, and not about market timing. Our decision context therefore reduces to one empirical question.

Research Question

> Which global and domestic macro-financial conditions, individually and as groups, show a statistically and economically meaningful association with the monthly log return of the PSEi between January 2018 and December 2025, and with the probability of a sharp monthly decline, after the regression assumptions are tested and corrected, and what follows from the resulting evidence for how an investor with Philippine equity exposure should monitor and manage the exposure?

We break our core research question down into four sub-questions:

1. **Overall fit.** To what extent are the nine conditions jointly related to monthly PSEi returns (overall F-test), and what share of the month-to-month variation is explained?
2. **Individual effects.** Which conditions have a non-zero association once the others are held constant (t-tests), and how large and how precisely estimated is each (confidence intervals), judged against the smallest effect that matters?
3. **Groups of conditions.** Which of the global factors, domestic macro conditions, and monetary and exchange rate conditions are jointly significant (joint F-tests), and how far is the relationship concentrated in a few individual variables?
4. **Robustness and tails.** How much of the association survives heteroskedasticity-robust inference and a March 2020 adjustment, and how far is it carried over to the probability of a sharp monthly decline (logistic regression)?

Objectives

With 96 observations, nine regressors and an extreme month in the sample, valid inference cannot be assumed, so our objectives are organised around the threats to it:

- Small sample and many regressors: separate individual from joint evidence, and guard against overfitting by comparing specifications on parsimony and cross-validated error rather than in-sample fit.
- Heteroskedasticity, serial dependence and non-normality: assess each directly, and base inference on covariance estimators that remain valid when the classical error assumptions fail.
- Influential observations: establish whether any conclusion depends on a single month by re-estimating with outlier-resistant estimators and with the extreme month modelled explicitly.
- Rare large losses: model sharp declines with a penalised-likelihood binary regression, because the events are too few for conventional maximum likelihood, and validate it out of sample and out of time.

Our report follows the order of our analysis. Sections 2 to 4 describe, prepare and explore the data, Section 5 estimates the baseline multiple regression, Sections 6 and 7 evaluate and diagnose it, Section 8 improves it, Section 9 applies logistic regression, and Sections 10 and 11 interpret the findings and give the financial recommendation.

Section 2: Data

The analysis uses ten series that describe the Philippine equity market and the conditions around it. The dependent variable is the monthly log return of the Philippine Stock Exchange Index (PSEi), and nine regressors represent the factors that prior literature links to equity returns. The unit of observation is the calendar month. The estimation sample runs from January 2018 to December 2025, which gives 96 observations, and December 2017 is collected as well because it is the base month for the first return and the first difference. The series come from three sources: global market indicators from the Federal Reserve Bank of St. Louis (FRED), Philippine macroeconomic indicators and market prices from LSEG Workplace based

from the PSA (Philippine Statistics Authority) and Bangko Sentral ng Pilipinas (BSP), and the policy rate decisions of the BSP. We note that their native frequencies differ, from daily prices to monthly year-on-year rates, and they are transformed to a common monthly basis in the next section.

The regressors were chosen because each represents a channel through which equity prices respond to economic conditions. In the discount-rate channel, the BSP policy rate, the 10-year government bond yield and consumer price inflation change the rate at which future cash flows are discounted and the cost of funds for firms. In the global risk channel, the S&P 500, the VIX and WTI crude oil carry international returns, risk appetite and commodity shocks into a small, open market that relies heavily on imported oil. The exchange rate channel operates through the peso value of foreign-currency earnings and the portfolio decisions of foreign investors. The liquidity and real-activity channel is represented by M3 money supply growth and manufacturing production, which describe the funds available for investment and the strength of the domestic economy. These channels organize the discussion in this section: each variable is introduced under its source, with how its data were obtained, what the loading tables show, and why it is relevant to the model. The table below lists every variable with its channel, source and native frequency, in the order in which the sources and variables are discussed in the subsections that follow.

Table 1: Variables of the study: channel, source and native frequency

Variable	Channel	Source	Native frequency
SP500	Global risk	FRED	Daily
WTI	Global risk	FRED	Daily
VIX	Global risk	FRED	Daily
PH_CPI	Discount rate	LSEG Workplace	Monthly, year-on-year
PH_M3	Liquidity and real activity	LSEG Workplace	Monthly, year-on-year
PH_Manufacturing_Production	Liquidity and real activity	LSEG Workplace	Monthly, year-on-year
PH_10Y_GS_Yield	Discount rate	LSEG Workplace	Daily
USD_PHP	Exchange rate	LSEG Workplace	Daily
BSP_RRP_Rate	Discount rate	BSP	On decision dates
PSEI	Dependent variable	LSEG Workplace	Daily

Source I: FRED — Global Indicators

The three global series come from the Federal Reserve Bank of St. Louis' FRED database and are downloaded through the `quantmod` package: WTI crude oil (DCOILWTICO, in US dollars per barrel), the S&P 500 index (SP500) and the CBOE Volatility Index (VIXCLS). FRED publishes the series on US business days, so a weekday on which a market was closed or no price was published appears as a missing value.

Table 2: FRED (daily): summary

variable	rows	first	last	n_missing	min	mean	max
WTI	2109	2017-12-01	2025-12-31	90	-36.98	67.880	123.64
SP500	2109	2017-12-01	2025-12-31	78	2237.40	4130.800	6932.00
VIX	2109	2017-12-01	2025-12-31	51	9.15	19.639	82.69

Table 3: FRED (daily): first 5 rows

date	WTI	SP500	VIX
2017-12-01	58.35	2642.22	11.43
2017-12-04	57.48	2639.44	11.68
2017-12-05	57.66	2629.57	11.33
2017-12-06	55.79	2629.27	11.02
2017-12-07	56.50	2636.98	10.16

The summary table shows that each of the three series has 2,109 rows, one for every weekday from 1 December 2017 to 31 December 2025. The missing values (90 for WTI, 78 for the S&P 500 and 51 for the VIX) are not errors: they are weekdays on which a value was not published, mainly public holidays, and the calendars differ slightly between the series, since none of the three is available on 50 dates, while 13 dates have an S&P 500 value but no WTI price. These gaps are filled forward in the next section, so that only information available at the time is used. The minimum, mean and maximum describe the period. The VIX ranged from 9.15 to 82.69, the peak being 16 March 2020, and averaged 19.6, and the S&P 500 ranged from 2,237 to 6,932 with a mean of 4,131. WTI ranged from -36.98 to 123.64 US dollars per barrel, and the negative value is 20 April 2020, when the benchmark price turned negative during the collapse in demand. It is a genuine observation, but the logarithm of a negative price does not exist, so oil returns are computed from month-end prices, which are positive in every month of the sample. The first rows show the layout of the file, oldest date first, beginning on 1 December 2017 with a WTI price of 58.35, an S&P 500 level of 2,642.22 and a VIX of 11.43. Together the three series represent the global risk channel, and the subsections below set out why each is expected to matter for Philippine equity returns.

Regressor I: S&P500

The S&P 500 represents the global risk channel in its most direct form. Through its composition of 500 leading companies, it reflects the overall health and performance of the U.S. stock market and acts as a proxy for market-wide movements and investor sentiment, which directly influence stock returns. Its returns are often analyzed to assess the impact of economic factors, investor behavior and market dynamics on stock performance (Dilpriya et al., 2025). Equity markets also tend to move together, and the co-movement strengthens in turbulent periods partly because volatility rises (Forbes & Rigobon, 2002), so a positive relationship between S&P 500 returns and PSEi returns is expected.

The link between the S&P 500 and stock market returns is also grounded in valuation theory. The discounted cash flow (DCF) model and the capital asset pricing model (CAPM) explain how changes in interest rates affect stock prices by changing the present value of expected future earnings. Empirical research on the S&P 500 shows that the effect of interest rates differs between growth and value stocks within the index: nominal returns are positively correlated with interest rate levels and slopes, growth stocks show negative correlations between excess returns and interest rates, and value stocks show positive ones (Sigfrid & Rozkov, 2024).

The distribution of the index's returns and its realized volatility have also been studied to understand financial stability and market behavior during different periods, such as crises. These analyses reveal that return and volatility patterns within the S&P 500 provide insights into market risk and return dynamics, which are essential for predicting near-future market behavior (Akter & Nobi, 2018).

Investor sentiment is a further link between the S&P 500 and stock returns. Studies indicate that firm-specific sentiment influences short-run stock returns within the index, with a noted negative relationship between overall sentiment and market performance, which highlights the behavioral aspects affecting returns (Youssef & Jouini, 2023). The S&P 500 is therefore included as the benchmark for global equity conditions.

Regressor II: WTI

WTI crude oil enters the global risk channel through the commodity market. West Texas Intermediate (WTI) crude oil returns may influence PSEi monthly log returns through production costs, inflation rates, consumer spending and demand, and purchasing power. As the Philippines relies heavily on imported petroleum, rising global oil prices could raise transportation and operating costs and reduce companies' profitability and stock valuations. WTI returns are thus expected to be negatively associated with Philippine stock market returns.

Sadeghi & Roudari (2022) investigated the impact of oil price shocks on the stock market in oil-importing as well as oil-exporting countries and found that the reactions of the markets are contingent on the source of oil shocks and on the economic situation at the time. Likewise, Ziadat et al. (2022) reported that stock markets in oil-importing economies did not exhibit consistent responses to oil shocks, suggesting that the effects may vary across economic conditions.

In addition, according to Escribano et al. (2023), the linkages between crude oil and stock markets tend to intensify in times of economic stress, and shocks are often transmitted from oil to the equity market. Castro & Jiménez-Rodríguez (2024) also confirmed that stock market reactions vary depending on whether the shifts in oil prices are due to supply disturbances, aggregate demand or inventory shocks, so increases in oil prices do not always lead to a decline. The distinction between demand-driven and supply-driven oil price changes is also central to Kilian & Park (2009), who show that the response of U.S. real stock returns to an oil price shock depends on the underlying source of the shock.

Furthermore, Hanif et al. (2024) document that oil demand shocks are significant in explaining the fluctuations in stock returns in key oil exporting and importing countries. These findings support including WTI returns in the regression as a potential independent variable in explaining stock market behavior.

For the Philippines, the 96 monthly observations from 2018 to 2025 suffice for the initial regression estimates, but the COVID-19 pandemic, including the episode in April 2020 in which the WTI price turned negative, and the later rise in energy prices could make the relationship unstable. Although a negative relationship is the theoretical expectation for an oil-importing economy, its sign and significance are therefore left to be estimated empirically.

Regressor III: VIX

The CBOE Volatility Index (VIX) completes the global risk channel by capturing the price of uncertainty rather than the level of returns. It is computed from S&P 500 options and measures expected volatility in the US stock market; it reflects both the expected variability of returns and investors' risk aversion (Bekaert & Hoerova, 2014), which is why it is commonly considered an indicator of investor fear. An increase in the VIX generally reflects greater uncertainty, which may encourage investors to reduce their exposure to riskier assets, including emerging-market equities, and may therefore influence PSEi monthly log returns through investor sentiment, market uncertainty and international capital flows. Consequently, the VIX is expected to have a negative relationship with Philippine stock market returns.

Valadkhani & O'Mahony (2024) examined how changes in the VIX and US dollar index affect emerging stock markets. They found that higher VIX levels significantly reduced emerging-market returns, while decreases in the VIX produced smaller positive effects. Similarly, Kirci Kirci Altinkeski et al. (2024) investigated the relationship between the VIX and global stock markets, finding that higher VIX levels were associated with lower stock returns, particularly during extreme market conditions. These findings suggest that global uncertainty may negatively influence local equity markets.

In India, Narang et al. (2023) found that investor fear, measured through the VIX, played a significant role in stock market declines and subsequent recovery during the COVID-19 pandemic. Sharma & Malik (2022) also observed that rising VIX levels were associated with declining stock returns in several BRIC economies during the pandemic, although these effects weakened over time.

The relationship is, however, not consistently strong across markets. Jurkowska et al. (2024) examined VIX spillovers across the Fragile Five emerging economies and found relatively weak connections between the

VIX and sectoral stock indices, which suggests that global market uncertainty may not always significantly explain local stock market movements. The VIX is therefore included with a negative expected sign, and the strength of the relationship is left to the data.

Source II: PSA via LSEG Workplace — Local Macroeconomics

The three domestic macroeconomic series are economic-indicator series exported from LSEG Workplace as one Excel file each: consumer price inflation (series aPHCPIAR), M3 money supply growth (aPHCMS3YB/A), and manufacturing production growth (aPHCINDYG/CA). LSEG compiles these indicators from national sources, and all three are published monthly as year-on-year percentage changes, which is the form used in the model. The loading function skips the header rows of each file, keeps the date and the value, sets the date to the first day of its month so that the series can be joined by month later, and drops any row without a value.

Table 4: Loaded data, LSEG macroeconomics (monthly, year-on-year %): summary

variable	rows	first	last	n_missing	min	mean	max
PH_CPI	121	2016-09-01	2026-09-01	0	0.80	3.9149	8.70
PH_M3	121	2016-08-01	2026-08-01	0	-8.54	6.6599	22.56
PH_Manufacturing_Production	121	2016-08-01	2026-08-01	0	-80.99	16.7940	497.61

Table 5: LSEG Local macroeconomics: first 5 rows of each series

year_month	PH_CPI	PH_M3	PH_Manufacturing_Production
2026-09-01	7.2	NA	NA
2026-08-01	6.1	1.87	12.23
2026-07-01	6.2	4.94	7.05
2026-06-01	6.4	1.68	11.60
2026-05-01	6.8	2.35	10.48
2026-04-01	7.2	2.14	11.08

Each of the three series has 121 monthly observations, from August or September 2016 to August or September 2026, and none is missing. The files extend beyond the study window on both sides because we downloaded them with a margin: of the 121 months, 97 (December 2017 to December 2025) are used, and the rest is dropped when the series are joined to the monthly calendar in the next section. The extremes in the summary table all fall inside the study window, so they describe the sample as well as the file. Consumer price inflation ranged from 0.80% to 8.70% with a mean of 3.91%, M3 growth from -8.54% to 22.56% with a mean of 6.66%, and manufacturing production growth from -80.99% to 497.61% with a mean of 16.79%. The range of manufacturing production is by far the widest, and it comes from the pandemic years (the lowest value is in 2020 and the highest in 2021), which is consistent with a collapse in output followed by a rebound measured against a low base. These are values reported by the source, not entry errors, and they are examined with the other regressors in the descriptive analysis. The first rows show that the files list the newest month first (September 2026 for consumer prices, August 2026 for M3 and manufacturing production). The three series are already year-on-year rates, so they need no further transformation, and their means differ slightly from the sample means reported later because the file means include the months outside the window. They represent the liquidity and real-activity channel (M3 and manufacturing production) and the inflation side of the discount-rate channel (consumer prices), whose relevance is discussed below.

Regressor IV: Consumer Prices

Consumer price inflation is the first variable of the discount-rate channel. The consumer price index (CPI) measures the rate at which the general price level of the goods and services purchased by consumers changes, and its year-on-year change is used here as the inflation rate. The link between inflation and stock returns can be explained through its impact on corporate earnings, interest rates and investor demand, although the evidence is often contradictory. As inflation rises, costs such as raw materials, labor and transport can increase, and if companies are unable to pass these costs on to consumers, corporate profits could fall and future cash flows become less predictable.

The relationship between inflation and corporate valuations can therefore be explained as follows: the higher the expected inflation, the lower the expected long-run real earnings growth and the higher the real return required by investors, which implies a negative relationship between inflation and stock returns. Fama (1981) showed that this negative relationship in the data can be traced to the link between inflation and real activity, a point taken up again below with the activity variables.

However, inflation can also have a positive relationship with stock returns, depending on its type. Cieslak & Pflueger (2023) show that inflation can be caused by demand-driven factors and by cost-driven factors. Demand-driven inflation can result in economic expansion, so corporate earnings increase and returns are affected positively, whereas cost-driven inflation reduces corporate profits and economic activity. Inflation also depends on monetary policy: Zhang (2021) found that the relationship between stock returns and inflation can be partly attributed to monetary policy, and that the more counter-cyclical monetary policy is, the more negative the relationship between share price returns and inflation becomes.

In emerging markets a link between inflation and share prices is also visible. Folorunso (2023) shows that inflation had a negative and significant impact on Nigerian share prices, although research on emerging Asian markets shows that some defensive industries can act as a hedge against inflation (Isnandari & Chalid, 2017). CPI inflation should therefore be considered as a theoretical determinant of PSEi monthly log returns, as it can have an impact on corporate profits, discount factors, investors' purchasing power and monetary policy.

Regressor V: M3 Money Supply

M3 money supply growth belongs to the liquidity and real-activity channel. The money supply indicates how much money is circulating within the economy, so its growth rate shows whether domestic liquidity is expanding or tightening, and when liquidity increases, investors may have more money to invest in financial assets. Portfolio balance theory describes how investors change the allocation of their assets depending on the liquidity in the economy, so that monetary policy affects asset prices through liquidity and risk premia (Drechsler et al., 2018).

Bhattacharjee & Das (2021) observed a long-run connection between the money supply and the stock market index and stated that the connection is positive. They also showed that an increase in the money supply can lead to an immediate positive impact on stock prices, although there was no evidence of a short-term causal effect, which means that the effect of the money supply can be more apparent in the long run. In addition, Nguyen et al. (2022) found that increasing liquidity leads to positive returns in the stock market across 34 emerging economies, and that this relationship is stronger in countries with higher quality institutions. This implies that more money in the economy can support the stock market, but that the effect depends on the financial situation of the country.

Not all studies find a positive relationship, and the evidence differs by market. Mallik et al. (2024) found an inverse relationship between money supply and stock prices in Bangladesh in the short term, whereas Schrank (2024) found a positive relationship between money supply and stock prices in Thailand. Different market conditions may therefore lead to different effects.

The impact of M3 growth on monthly log returns of the PSEi can thus be explained by the availability of funds to invest and the conditions of the market, but the studies show that it does not have a consistent relationship with stock returns. Increasing liquidity can lead to more investment and higher stock prices,

but if the extra money does not translate into demand for stocks, the relationship can be weaker or even negative. M3 growth is therefore included in the regression as an important variable.

Regressor VI: Manufacturing Production

Manufacturing production completes the liquidity and real-activity channel as an indicator of economic activity. An increase in manufacturing can raise sales and profits for businesses, which can raise investor confidence and lift the stock market, and Fama (1981) found that stock returns are positively related to measures of real activity such as industrial production. However, manufacturing growth is not always reflected in high stock returns, because expectations, interest rates, inflation and other economic factors also affect stock prices.

Endres (2020) used a long-run ARDL model to study the impact of industrial production and a number of other macroeconomic variables on the PSEi, with data between January 2009 and December 2019, and the results suggest a negative relationship between manufacturing growth and the PSEi stock return in the long run, so that increasing manufacturing does not by itself improve stock returns. This shows how the relationship between macroeconomic factors and stock returns may change over time.

Another study, by Camino et al. (2023), investigated the relationship between stock indices and economic growth within the ASEAN-4 nations and revealed that GDP growth was not statistically significant in explaining the variations in the stock indices of those countries. Although they focused on economic growth, they suggest that an improvement in the real economy may not have a significant effect on the stock market.

While there may be many complexities to the relationship between manufacturing growth and stock market returns, in general, higher growth levels can point to increased levels of economic activity. This will largely depend on investors' perception of how sustainable the growth is and if it will translate into increased company profits. As there are many studies surrounding manufacturing, and overall economic growth in the Philippines, further analysis on the sector may provide the most insightful information on whether this growth factor can be related to monthly log returns on the PSEi.

Source III: BSP via LSEG Workplace or Direct — Local Financial Environment

The PSEi (series .PSI), the USD/PHP spot rate (PHP=) and the 10-year government bond yield (PHBD2509=RRPS) are daily price-history exports from LSEG Workplace, one Excel file each. In each file a block of statistics precedes the daily history, so the loading code starts reading at the row where the history begins (row 33 for the PSEi, row 29 for USD/PHP and row 19 for the bond), converts the Excel serial dates to dates, and keeps the closing value of the PSEi, the mid price of USD/PHP and the yield to maturity of the bond.

The BSP policy rate is different, because it is a decision rather than a market price. It was built from the text of the announcements of the BSP Monetary Board, which state each decision and the resulting rate (64 announcements from 14 December 2017 to 11 December 2025, kept in the data folder as a text table). The date and the rate were extracted from the announcement texts into a two-column table with the help of an AI assistant (Claude Sonnet 5.5).

Table 6: Loaded data, LSEG financial markets + BSP: summary

variable	rows	first	last	n_missing	min	mean	max
PH_10Y_GS_Yield	2109	2017-12-01	2025-12-31	0	3.5605	5.8730	8.631
PSEI	1975	2017-12-01	2025-12-29	0	89.8620	129.5000	177.500
USD_PHP	2109	2017-12-01	2025-12-31	0	47.6470	53.5030	59.262
BSP_RRP_Rate	64	2017-12-14	2025-12-11	0	2.0000	4.1719	6.500

Table 7: Local Financial Environment: first 5 rows of each series

date	PH_10Y_GS_Yield	PSEI	USD_PHP	BSP_RRP_Rate
2025-12-31	6.195000	NA	58.8555	NA
2025-12-30	6.195000	NA	58.8610	NA
2025-12-29	6.195000	102.9916	58.8280	NA
2025-12-26	6.156000	103.3399	58.7545	NA
2025-12-25	6.143000	NA	58.7950	NA
2025-12-23	6.143000	102.8358	58.8000	NA
2025-12-22	6.133000	102.9175	58.7435	NA
2025-12-19	6.114000	101.0232	58.6455	NA
2018-06-20	6.805099	136.2993	53.3005	3.50
2018-05-10	6.693177	145.8795	51.9295	3.25
2018-03-22	6.591733	154.9906	52.4395	3.00
2018-02-08	6.235111	167.1516	51.7450	3.00
2017-12-14	5.463038	167.6786	50.4650	3.00

The summary table shows how the four series are built. The bond yield and the USD/PHP rate have a row for every weekday between 1 December 2017 and 31 December 2025 (2,109 rows) and no missing values, while the PSEi has 1,975 rows, one for each day the exchange traded, which is 134 fewer, and its last row is 29 December 2025. The yield ranged from 3.56% to 8.63% with a mean of 5.87%, and the peso traded between 47.65 and 59.26 to the US dollar with a mean of 53.50. The policy rate table is not a daily series: it has one row for each of the 64 Monetary Board decisions, with rates between 2.00% and 6.50%, and the rate changed at 31 of the decisions and was kept unchanged at the others. The extracted rates were compared with the percentage stated in each announcement text, and 64 of the 64 agree. In the table of first rows, NA marks a date on which a series has no value. This comes from joining series with different calendars (the PSEi is not observed on Philippine holidays, and the policy rate only on decision dates) and is not missing data. The files list the newest date first, while the policy rate table is ordered from the oldest decision. The four series are the dependent variable (the PSEi), the exchange rate (USD/PHP) and the discount-rate channel (the bond yield and the policy rate), and the subsections below explain why each is relevant.

Regressor VII: 10-year Government Bond Yield

The 10-year government bond yield is the second discount-rate variable. It reflects the returns demanded by investors on long-term government securities, acts as a benchmark in the country, and can also be viewed as a forecast of interest rates, inflation and economic growth. Changes in the Philippine 10-year yield might influence the PSEi monthly log returns through borrowing costs, investments and stock valuation: a rise in the yield increases the discount rate used for stock valuation and is expected to influence stock market returns negatively.

Zaremba et al. (2023) examined how interest rates move, change, and whether they have any relation with market performance across 60 countries. The authors found that countries experiencing larger increases in government bond yields tended to have lower subsequent equity returns than those experiencing smaller increases. Prabheesh et al. (2024) investigated financial markets in eight emerging economies during the COVID-19 pandemic. The authors observed that asset purchase announcements generally coincided with rising stock returns and declining bond yields. Together, the results suggest that movements in government bond yields may contribute significantly to explaining equity market performance.

However, empirical evidence shows the relationship isn't consistently negative. McMillan (2022) identified that the relationship between bond yields and stock returns is different and changes over time, with predominantly negative correlations becoming positive under certain economic conditions. Molenaar et al. (2024) identified inflation, real interest rates, and government creditworthiness as the key determinants of the stock-bond correlations. Additionally, Schrank (2024) found that increases in monetary policy interest rates in

Thailand were associated with higher stock returns and government bond yields, particularly during periods of political or economical crisis in the country.

For the Philippines, these findings justify testing whether changes in the 10-year government bond yield are related to PSEi monthly log returns; because the yield is a level, it enters the model as its monthly change. The 96 monthly observations from 2018 to 2025 are sufficient for the regression analysis, although the stability of the association may be affected by economic uncertainty and shocks, such as the COVID-19 pandemic and inflation. Although a negative relationship is theoretically expected, its direction and significance are therefore established through empirical testing.

Regressor VIII: Dollar Value against Local Currency

The USD/PHP exchange rate is the exchange rate channel of the model. It is quoted in pesos per US dollar, so an increase means a depreciation of the peso. Exchange rate fluctuations influence stock market returns by affecting firm valuations, particularly those of firms with foreign currency exposure, and by shaping investor expectations; currency movements reach firm valuations through earnings expectations (Bartov & Bodnar, 1994). The theoretical relationship can be explained through the International Fisher Effect, interest rate parity, and portfolio theory, which describe how exchange rates, interest rate differentials, and international investment decisions interact (Umoru et al., 2024). Studies on emerging markets suggest that exchange rate movements can significantly affect stock returns through macroeconomic factors such as inflation and interest rates, and that these effects may be nonlinear, varying according to prevailing economic conditions (Bonga-Bonga & Kabwe wa Kabwe, 2026).

Empirical evidence generally suggests that currency appreciation can positively influence stock market performance, while depreciation may negatively affect returns, particularly when accompanied by high inflation or economic instability (Bonga-Bonga & Kabwe wa Kabwe, 2026; El-Diftar, 2023). However, Umoru et al. (2024) found that exchange rate changes had positive effects on stock returns at several quantiles, while their overall Bayesian Vector Autoregression (BVAR) results indicated relatively weak direct effects. Interest rate changes, meanwhile, had a more significant influence over the long term. The relationship can also operate in the opposite direction, with stock market returns predicting exchange rate movements (Abd-Allah, 2026).

Exchange rate volatility may further influence stock return volatility, although its effects vary across countries and currency pairs (Karunanayake, 2014; Sreenu, 2023). Additionally, cointegration findings in emerging markets suggest possible long-run equilibrium relationships between exchange rates and stock returns (El-Diftar, 2023). These findings provide a basis for examining whether USD/PHP exchange rate changes influence Philippine stock market returns, considering the country's emerging market status. A negative sign is expected, because a depreciation of the peso tends to weigh on returns, but the strength and direction of this relationship must be established through empirical analysis.

Regressor IX: Target Reverse Repurchase Rate

The Bangko Sentral ng Pilipinas (BSP) Reverse Repurchase Rate (RRP) is the policy variable of the discount-rate channel. Central bank policy rates such as the BSP RRP influence liquidity conditions and the cost of borrowing in the economy, which in turn affect investor behavior and asset prices, including stock market returns. A reduction in the policy rate typically signals an easing monetary stance, which can lower financing costs and potentially stimulate investment and consumption, thereby supporting stock market performance. Conversely, a rate increase signals tightening, raises borrowing costs, and can weigh on valuations and investor sentiment. In the United States, unexpected cuts in the policy rate have been found to raise stock prices, mainly by lowering expected excess returns (Bernanke & Kuttner, 2005).

Within the sample the BSP policy rate moved in both directions: it was raised from 3.00 percent in December 2017 to 4.75 percent by the end of 2018, lowered to 2.00 percent by November 2020, raised to a peak of 6.50 percent in October 2023 and lowered to 4.50 percent by December 2025. The rate stood at 5.75 percent in early 2025 and was lowered to 5 percent by August 2025 as part of an easing cycle that continued beyond the sample, to 4.25 percent in February 2026. Beginning in April 2026, the BSP reversed course and raised the

rate to 4.50 percent, then 4.75 percent in June and 5 percent by August 2026, citing inflationary pressures linked to the Middle East conflict (Statista Ltd, 2026). Because the series contains easing phases and a tightening phases, the effect of RRP changes on stock returns may differ by direction, and we examine it empirically.

From a theoretical perspective, studies on stock market returns and their correlations with macroeconomic variables and financial market indicators suggest that monetary policy changes impact stock market volatility and returns. For example, research on the US markets shows that structural changes in policy and economic shocks, such as those from quantitative easing, affect return and volatility spillovers between markets, including stocks (Su, 2022). This implies that changes in central bank rates like the BSP RRP can alter the risk-return dynamics in the stock market.

Moreover, the relationship between stock market returns and interest rates is often mediated by investor expectations and market sentiment. The study on US stock market returns highlights that economic cycles and investor sentiment influence the predictive power of financial indicators, including correlations among industry portfolios, which can be affected by monetary policy shifts (Li et al., 2020). Thus, changes in the BSP RRP may influence stock market returns through their impact on investor expectations and market co-movements. The expected sign is negative, but it may differ between rate increases and rate cuts.

Dependent Variable: PSEI

The dependent variable of the study is the monthly log return of the Philippine Stock Exchange Index (PSEi). The PSEi is the benchmark index of the Philippine Stock Exchange (PSE), representing the performance of 30 selected publicly listed companies based on criteria such as market capitalization, liquidity, and free-float requirements (The Philippine Stock Exchange, Inc., n.d.). As a primary indicator of the Philippine equity market, it provides an overview of stock market performance and reflects changes in investor confidence and market conditions of the country as-a-whole. Its movements are particularly important to investors and financial analysts because they offer insights into the performance of publicly traded companies and the broader investment environment in the Philippines.

Historically, the PSEi has experienced significant fluctuations in response to various domestic and international developments. In January 2018, the index reached a record high of over 9,000 points before experiencing a notable decline amid concerns involving inflation, currency depreciation, global trade tensions, and monetary tightening (Fitch Solutions Group Limited, 2019). Similarly, the COVID-19 pandemic in 2020 and subsequent global economic uncertainties contributed to further fluctuations in the Philippine stock market. These developments illustrate how the PSEi responds to changing economic and financial conditions, making it relevant to examine the factors associated with its movements (Fitch Solutions Group Limited, 2022).

The performance of the PSEi may be associated with several macroeconomic and financial indicators, including gross domestic product (GDP), inflation rates, interest rates, exchange rates, and international stock market performance (Gallardo & Viray, 2025). These variables may influence the Philippine equity market through different channels, such as changes in corporate profitability, borrowing costs, purchasing power, foreign investment flows, and investor sentiment. Since these factors can operate simultaneously, examining their individual relationships with the PSEi can provide a more comprehensive understanding of the forces associated with Philippine stock market performance.

Given these relationships, multiple linear regression is an appropriate statistical method for examining how several explanatory variables are simultaneously associated with the PSEi. It allows researchers to estimate the direction and magnitude of each explanatory variable's relationship with the index while holding other variables constant (Yan & Su, 2009). Furthermore, the method provides a basis for testing the statistical significance of individual variables and assessing their combined explanatory power. Through this approach, the researchers can identify which economic and financial indicators have stronger statistical relationships with the Philippine stock market.

Moreover, since the PSEi is observed over time, applying regression analysis to financial time-series data requires evaluating whether the estimated relationships satisfy the relevant statistical assumptions. Diagnostic procedures can help identify potential issues such as multicollinearity, heteroskedasticity, autocorrelation,

and influential observations (Krispin, 2019). Addressing these concerns through appropriate model improvements can strengthen the reliability of the regression results and the conclusions drawn from them. Before the regression can be estimated, the daily and monthly series described above are combined into a single monthly dataset, which is the subject of the next section.

Section 3: Data Preprocessing

The series collected in the previous section cannot be used in a regression as they are. They differ in frequency: the PSEi, the USD/PHP exchange rate, the 10-year government bond yield, WTI crude oil, the S&P 500 and the VIX are daily, the BSP Target Reverse Repurchase (RRP) rate is a list of dated rates rather than a regular series, and consumer prices (CPI), M3 money supply and manufacturing production are monthly year-on-year rates. The daily series also follow different trading calendars, since the Philippine and U.S. markets are closed on different holidays. Because the macroeconomic series are published monthly, the calendar month is the unit of analysis. This section combines the sources into one monthly dataset from January 2018 to December 2025, transforms the price and rate series into returns or changes so that the model does not rest on persistent price levels, and then checks the quality of the result. The descriptive statistics and graphs of the finished dataset follow in Section 4.

Table 8: Analysis dataset: 96 monthly observations (Jan 2018 to Dec 2025), 10 variables

variable	role	transformation
PSEI	Dependent	Monthly log return
SP500	Regressor	Monthly log return
USD_PHP	Regressor	Monthly log change
BSP_RRP_Rate	Regressor	Month-end first difference
PH_10Y_GS_Yield	Regressor	Month-end first difference
PH_CPI	Regressor	YoY % change
PH_M3	Regressor	YoY % growth
PH_Manufacturing_Production	Regressor	YoY % growth
WTI	Regressor	Monthly log return
VIX	Regressor	Monthly average

Table 9: Analysis dataset: first 6 months, one row per variable

variable	2018-01	2018-02	2018-03	2018-04	2018-05	2018-06
PSEI	-0.0032	-0.0478	-0.0643	-0.0080	-0.0592	-0.0559
SP500	0.0547	-0.0397	-0.0273	0.0027	0.0214	0.0048
USD_PHP	0.0266	0.0140	0.0016	-0.0099	0.0175	0.0148
BSP_RRP_Rate	0.000	0.000	0.000	0.000	0.250	0.250
PH_10Y_GS_Yield	0.585	0.395	0.377	-0.012	-0.038	0.265
PH_CPI	4.00	3.90	4.30	4.50	4.60	5.20
PH_M3	9.61	10.10	10.35	10.54	7.66	5.68
PH_Manufacturing_Production	-2.62	-3.82	-3.37	20.81	19.17	26.56
WTI	0.0696	-0.0537	0.0545	0.0553	-0.0233	0.1014
VIX	11.06	22.46	19.02	18.27	14.12	13.68

We first join the four daily files on the date column. A full join keeps every date on which at least one series was observed, which gives 2109 dates from 1 December 2017 to 31 December 2025. Where a series has no value on one of these dates, the last previous value is carried forward (`tidyr::fill()`), so that only

information available on that date is used. The VIX is not filled, since its monthly average uses the days on which it was observed. The daily table is then collapsed to one row per month. For the PSEi, USD/PHP, the 10-year yield, WTI and the S&P 500, the value of the last available day of the month is kept (`last_valid()`, which returns a missing value instead of an empty result when a month has no observation), and the VIX is averaged over the month.

The RRP rate is handled differently because a policy rate stays in force until a new one is set. For each month, the code takes the latest rate dated on or before the last day of that month (`findInterval()` on the dates), which turns the list of dated rates into a month-end series. CPI, M3 and manufacturing production are already monthly and are joined without change. All series are attached to a calendar with one row per month from December 2017 to December 2025. December 2017 is kept only as the base month, so that the first return and the first difference, for January 2018, can be computed.

The variables are then transformed. The PSEi, S&P 500 and WTI become monthly log returns, the log of the month-end level minus the log of the previous month-end level, and the USD/PHP rate becomes the monthly log change in the same way. The RRP rate and the 10-year yield are already in percentage points, so they enter as month-end first differences. The three year-on-year series and the VIX are used as they are. After 2017 is dropped, the analysis dataset has 96 monthly observations and 10 variables, the PSEi return as the dependent variable and nine regressors, as the variable guide above sets out. The first six months show the result: the PSEi log return was -0.0032 in January 2018 and -0.0478 in February 2018, and the RRP rate was unchanged until it rose by 0.25 percentage points in May 2018 and again in June 2018.

Each transformation follows from how the variable behaves. Price levels such as the PSEi, the S&P 500, WTI and the exchange rate follow persistent paths, and a regression of one such series on another can show strong relationships that arise from the persistence alone (Granger & Newbold, 1974), so each is replaced by its return. The logarithmic return is used because a gain and a loss of the same log size cancel exactly, because it adds up over successive months, and because it is close to the simple percentage change for moves of the size seen in most months (Hudson & Gregoriou, 2015). It exists only for positive prices, which is why oil is measured at month-end, when its price is positive in every month. Month-end values are used so that each return covers exactly one calendar month, from one month-end to the next. The RRP rate and the 10-year yield are already rates and move slowly and persistently, so their change over the month, in percentage points, is the new information that the month brings and is the natural measure of a policy move or a shift in yields. The VIX is a volatility index that fluctuates around a long-run level, so it is kept in levels and averaged over the month, which describes the risk climate of the whole month and not the sentiment of a single day.

The three year-on-year series are already rates of change and are used as reported, because differencing them again would discard information and make their interpretation harder. They are, however, persistent, since each value shares eleven months of change with the previous one: the first-order autocorrelation is 0.95 for consumer prices, 0.94 for M3 and 0.73 for manufacturing production, against values close to zero for the PSEi, S&P 500, WTI and USD/PHP returns (all between -0.14 and 0.13). Whether this persistence leaves autocorrelation in the model residuals is tested in Section 7, which finds no evidence of it. The units of the final dataset are mixed, and the coefficients later in the report are read accordingly: log returns and changes are in decimal form (0.01 is one percent), the RRP rate and the yield in percentage points, the year-on-year rates in percent and the VIX in index points.

Table 10: Structure and month coverage

check	value
rows	96
expected_rows	96
first_month	2018-01-01
last_month	2025-12-01
duplicate_months	0
missing_months	0
is_sorted	TRUE

Table 11: Data types, missing observations and non-finite values

variable	type	n_na	pct_na	n_inf	n_unique
year_month	Date	0	0	0	96
PSEI	numeric	0	0	0	96
USD_PHP	numeric	0	0	0	96
PH_10Y_GS_Yield	numeric	0	0	0	96
WTI	numeric	0	0	0	96
SP500	numeric	0	0	0	96
VIX	numeric	0	0	0	96
BSP_RRP_Rate	numeric	0	0	0	6
PH_CPI	numeric	0	0	0	54
PH_M3	numeric	0	0	0	95
PH_Manufacturing_Production	numeric	0	0	0	96

Table 12: Months with missing values

year_month	missing_vars

Three groups of checks were run on the analysis dataset. The first checks its structure: the number of rows against the number of months expected between January 2018 and December 2025, the first and last month, and whether any month is duplicated, missing or out of order. The second reports the type of each column and counts its missing values, its infinite values, which a logarithm of zero would produce, and its distinct values, where a single distinct value would mean a constant column. The third lists the months with a missing value and names the variables that are missing in each.

The dataset passes every structural check. It has 96 rows against 96 expected, runs from January 2018 to December 2025, and has no duplicated or missing months. No column has a missing or infinite value, so the table of months with missing values has no rows and no month needed to be imputed or removed. The month column is stored as a date and all ten variables as numeric values, so the loading steps left no variable in the wrong type. No column is constant. The BSP RRP change takes only 6 distinct values and is zero in 65 of the 96 months, because the rate is held unchanged in most months and, when it does move, moves in multiples of 0.25 percentage points. The other variables take between 54 distinct values (PH_CPI, which is reported to one decimal place) and 96.

A dataset that passes these checks is complete, but this does not show that every value is plausible. The year-on-year series, manufacturing production in particular, take very large values in 2020 and 2021 (from -81% to 498%), which coincide with the pandemic period. No observation inside the study window is corrected or excluded, for three reasons. The values are reported by the source and have an economic explanation, the

collapse of output in the pandemic and the rebound measured against a low base (Section 2). Removing the most unusual months would discard the episodes that test the model most. And whether a month distorts the estimates is better judged from the fitted model, through the influence diagnostics of Section 7 and the remedies of Section 8, than by a cut-off applied beforehand. The next section screens the values with plausibility checks alongside the descriptive statistics, distributions and correlations of the dataset.

Section 4: Exploratory Data Analysis

This section describes the analysis dataset before any model is estimated. The unit of observation is the calendar month, and the 96 observations run from January 2018 to December 2025. The dependent variable is the PSEi monthly log return, and the nine regressors are in the form set out in the previous section. They fall into three kinds: market prices and yields (the S&P 500, WTI, VIX, USD/PHP and the 10-year yield), the change in the BSP policy rate, and the year-on-year macroeconomic rates (CPI, M3 and manufacturing production). We first identify the descriptive statistics and plausibility checks, followed by visualizing box-plots, histograms, time-series plots, scatter plots and correlation heat-maps. We show three observations run through the discussion: how the PSEi behaves from month to month, how far one month, March 2020, shapes almost every display, and how the three kinds of regressor differ in their link to the PSEi. Together they show which patterns the regression has to explain and which features of the data may call for caution.

Descriptive Statistics

Table 13: Descriptive statistics (excess kurtosis; normal = 0; outliers = |z| > 3)

variable	n	mean	median	min	max	range	sd	skewness	ex_ kurtosis	n_outliers
PSEI	96	-0.0053	0.0005	-0.2417	0.1222	0.3639	0.0597	-0.9380	1.9935	1
USD_PHP	96	0.0017	0.0001	-0.0367	0.0478	0.0846	0.0159	0.4831	0.3935	0
PH_10Y_GS_Yield	96	0.0089	0.0063	-1.0832	0.8807	1.9640	0.3683	-0.3845	0.6885	0
WTI	96	-0.0006	0.0056	-0.7820	0.6150	1.3970	0.1356	-1.1059	13.8619	2
SP500	96	0.0098	0.0194	-0.1337	0.1194	0.2531	0.0477	-0.5967	0.2852	1
VIX	96	19.7319	18.3355	11.0624	57.7368	46.6744	6.7634	2.3679	9.4983	2
BSP_RRP_Rate	96	0.0156	0.0000	-0.5000	0.7500	1.2500	0.2201	0.9022	2.4853	2
PH_CPI	96	3.9333	3.8500	0.8000	8.7000	7.9000	1.8714	0.5449	-0.2633	0
PH_M3	96	6.6395	6.1200	-8.5400	22.5600	31.1000	6.4730	0.1307	-0.1210	0
PH_Manufacturing_Production	96	19.7386	0.4650	-80.9900	497.6100	578.6000	97.6942	3.7061	13.9863	4

Table 14: Plausibility checks

check	n_flagged
VIX <= 0	0
PH_CPI outside [-5, 25]%	0
PH_M3 outside [-20, 40]%	0
PH_Manufacturing_Production outside [-50, 100]%	14
|monthly log return| > 0.25 (PSEI, SP500, USD_PHP, WTI)	3

The descriptive statistics show that the PSEi return is centred near zero but skewed to the left. Its mean monthly log return is -0.53%, its median 0.05% and its standard deviation 5.97%, with 48 up months and 48 down months. Skewness is -0.94 and excess kurtosis 1.99, so large losses are larger and more frequent than large gains: 21 months fall below -5% against 16 above +5%. Much of this shape comes from one month: without March 2020 the skewness falls to -0.50 and the excess kurtosis to 0.47, so the PSEi is closer to normal than the full sample suggests. The only month beyond three standard deviations is March 2020,

at -24.2% (z = -3.96). The same month is also the outlier of the S&P 500 (-13.4%), WTI (-78.2%) and the VIX (57.7, z = 5.62), which is our first sign of how much one month shapes the data.

The regressors differ in scale and in shape. The S&P 500 is also left-skewed (-0.60) but has a positive mean (0.98%, standard deviation 4.77%), and the USD/PHP return (mean 0.17%, standard deviation 1.59%) is mildly right-skewed (0.48), its largest values being months of peso depreciation. The monthly change in the 10-year yield averages 0.01 percentage points with a standard deviation of 0.37, and ranges from -1.08 (January 2023) to +0.88 (September 2022). WTI has the heaviest tails (excess kurtosis 13.86), but two months, March and May 2020, account for almost all of it: without them the excess kurtosis falls to 0.22. The VIX is right-skewed (2.37), as expected for a gauge that jumps in stress months. The RRP change has a median of exactly zero because the rate is unchanged in 68% of months. Among the year-on-year rates, CPI runs from 0.8% to 8.7% (mean 3.93%, standard deviation 1.87) and M3 from -8.5% to 22.6% (mean 6.64%, standard deviation 6.47), whereas manufacturing production has a median of 0.47% but a mean of 19.7% and a standard deviation of 97.7, because of a few extreme year-on-year values.

The plausibility checks agree: there is no VIX value who is zero or below, and CPI and M3 stay inside the ranges set for them, but 14 months of manufacturing production lie outside -50% to 100% (April 2020 to March 2022) and three monthly log returns, all of WTI (November 2018, March 2020 and May 2020), exceed 25% in absolute value. These values are kept as reported. They are real movements caused by the COVID-19 shock, not recording errors, and dropping them would remove the most informative months from a sample of only 96. The regressors need not be normal in any case: the distribution that matters for the regression is that of the residuals, and both the normality of the residuals and the influence of these months on the fitted model are assessed in the diagnostics.

Box-plots

The box-plots show where each variable is concentrated and which months lie outside it. A point is marked as an outlier when it lies more than 1.5 interquartile ranges beyond the box, a looser rule than the $|z| > 3$ rule of the statistics table, so more months are flagged. The PSEi box runs from -4.0% to 2.7% around a median of about zero, and all three of its outliers are losses: March 2020, June 2022 and September 2022. The extremes of the market series have the same character. The S&P 500 has one outlier, March 2020, the VIX has two, March and April 2020, and the USD/PHP has three depreciation months, among them June and September 2022, the same two months in which the PSEi fell sharply. March 2020 is an outlier in the PSEi, S&P 500, WTI and VIX at once, so market stress appears as losses, depreciations and volatility spikes that arrive together. The 10-year yield and WTI have extremes on both sides. The RRP change has a collapsed box at zero, so each of the 31 months in which it moved is drawn as an outlier. Manufacturing production is the most extreme case: its box spans only -5.1% to 10.7%, yet 17 months lie outside it, from -81% to 498%. CPI and M3 are tighter, with two months above the box for CPI (January and February 2023) and one for M3 (August 2020).

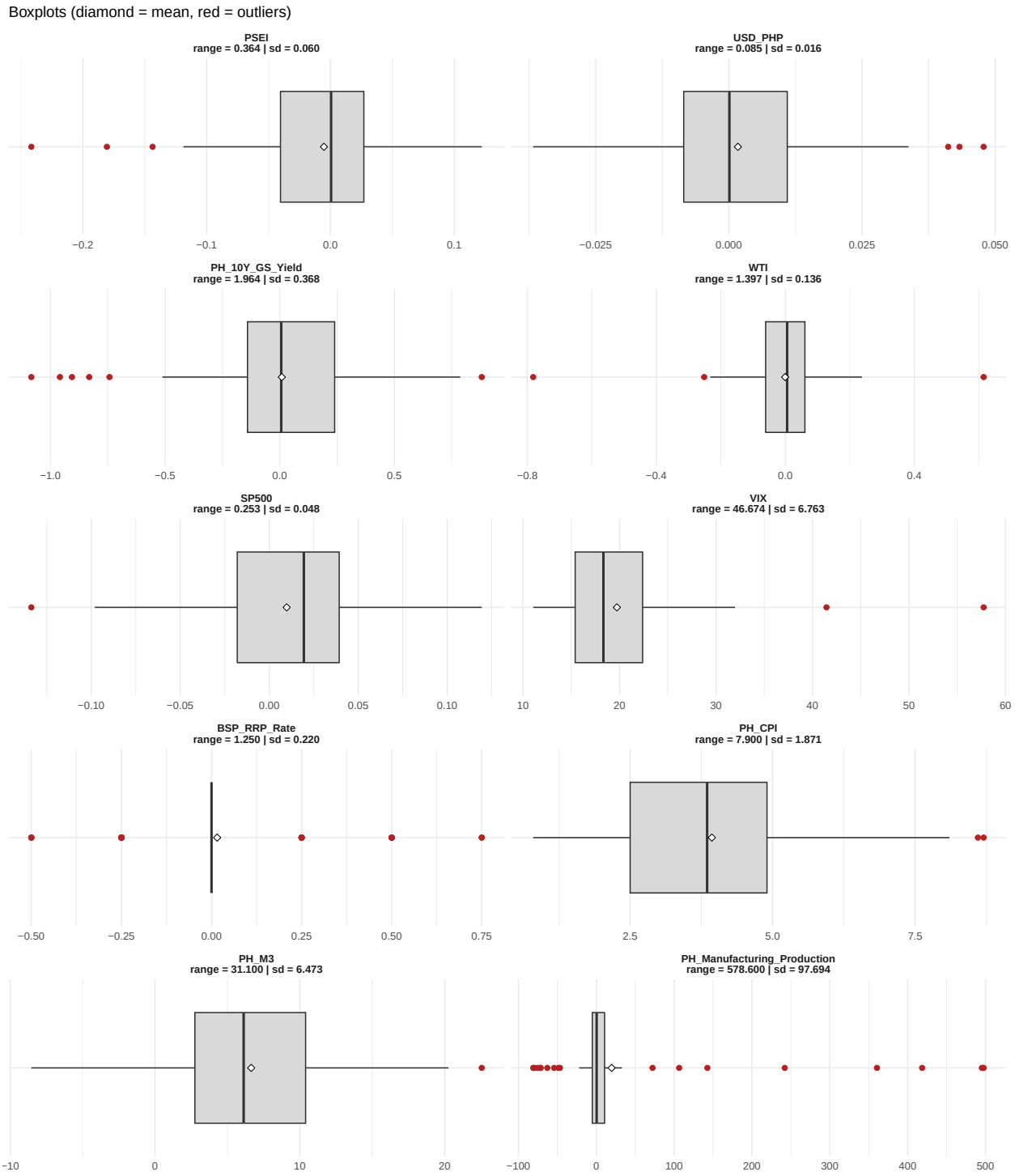


Figure 1: Boxplot of each variable (diamond = mean, red points = outliers)

Histograms



Figure 2: Histogram of each variable, with its mean and median marked

The histograms add the mean (solid line) and the median (dashed line) of each variable, which makes asymmetry and the pull of extreme months easy to see. For the USD/PHP, WTI and the 10-year yield the

two lines almost coincide, so a typical month is well described by its mean. For the PSEi and the S&P 500 the mean lies below the median (-0.53% against 0.05%, and 0.98% against 1.94%), the sign of a left tail. Most PSEi months lie between about -10% and +10%, with the tallest bars close to zero, and the few bars below -10% are the isolated months identified earlier. Among the macroeconomic series, the VIX mean (19.7) sits above its median (18.3), as expected for a right-skewed measure, whereas CPI and M3 have a mean close to the median (3.93 against 3.85, and 6.64 against 6.12) and spread-out, roughly bell-shaped bars. Two histograms are dominated by one bar. The RRP change is concentrated at zero, with a mean of 0.016 and a median of 0, and the small bars beside it fall at multiples of 0.25 percentage points, the size of a standard rate move. Manufacturing production has most of its months within a few points of zero (an interquartile range of -5.1% to 10.7%) and a few isolated bars out to 500%, so its mean (19.7%) is far above its median (0.47%). For this variable the median describes a typical month and the mean does not.

Time-Series

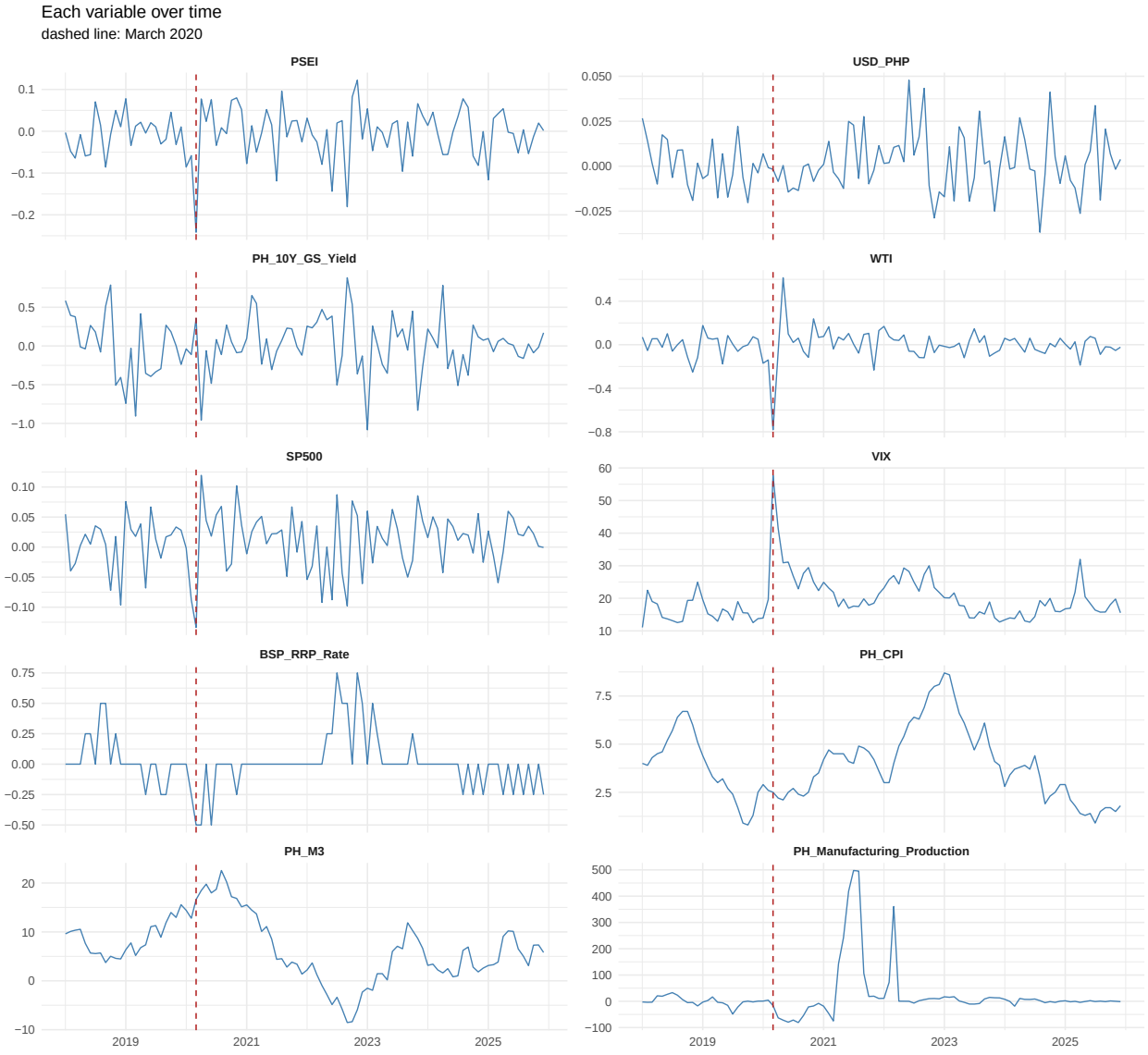


Figure 3: Each variable over time, January 2018 to December 2025 (dashed line = March 2020, the COVID-19 shock)

The time-series panels show when these patterns occurred. The dashed line marks March 2020, which is the sharpest move in four panels at once: the PSEi, S&P 500, WTI and VIX all reach their extremes in that month (-24.2%, -13.4%, -78.2% and 57.7), and the RRP rate was cut by 0.5 percentage points. Volatility is not constant. The standard deviation of the PSEi return is 9.4% in 2020 and 8.6% in 2022, against about 3% in 2019 and about 5% in 2018 and 2023 to 2025, and the VIX averages 29.1 in 2020 and 25.6 in 2022 against 15 to 17 in 2018, 2019, 2023 and 2024. These are two separate stress episodes. In 2022 the PSEi had its second and third worst months (September and June), the 10-year yield had its largest rise (0.88 percentage points in September) and the RRP rate rose by 0.75 percentage points in July and again in November. This clustering of calm and turbulent periods means that the variance of the errors in a regression may not be constant, which the diagnostics test. The RRP panel shows the policy cycles: hikes in 2018, cuts from 2019 to 2020, a long run of hikes from May 2022 to October 2023, and cuts again from August 2024. The year-on-year series are smooth and persistent. CPI peaks at 6.7% in late 2018, falls to 0.8% in October 2019 and reaches 8.7% in January 2023, M3 growth rises to 22.6% in August 2020 and falls to -8.5% in September 2022, and manufacturing production is flat apart from the collapse of 2020 and the surge of 2021 to 2022, which is consistent with comparison against a depressed base period.

Scatter Plots

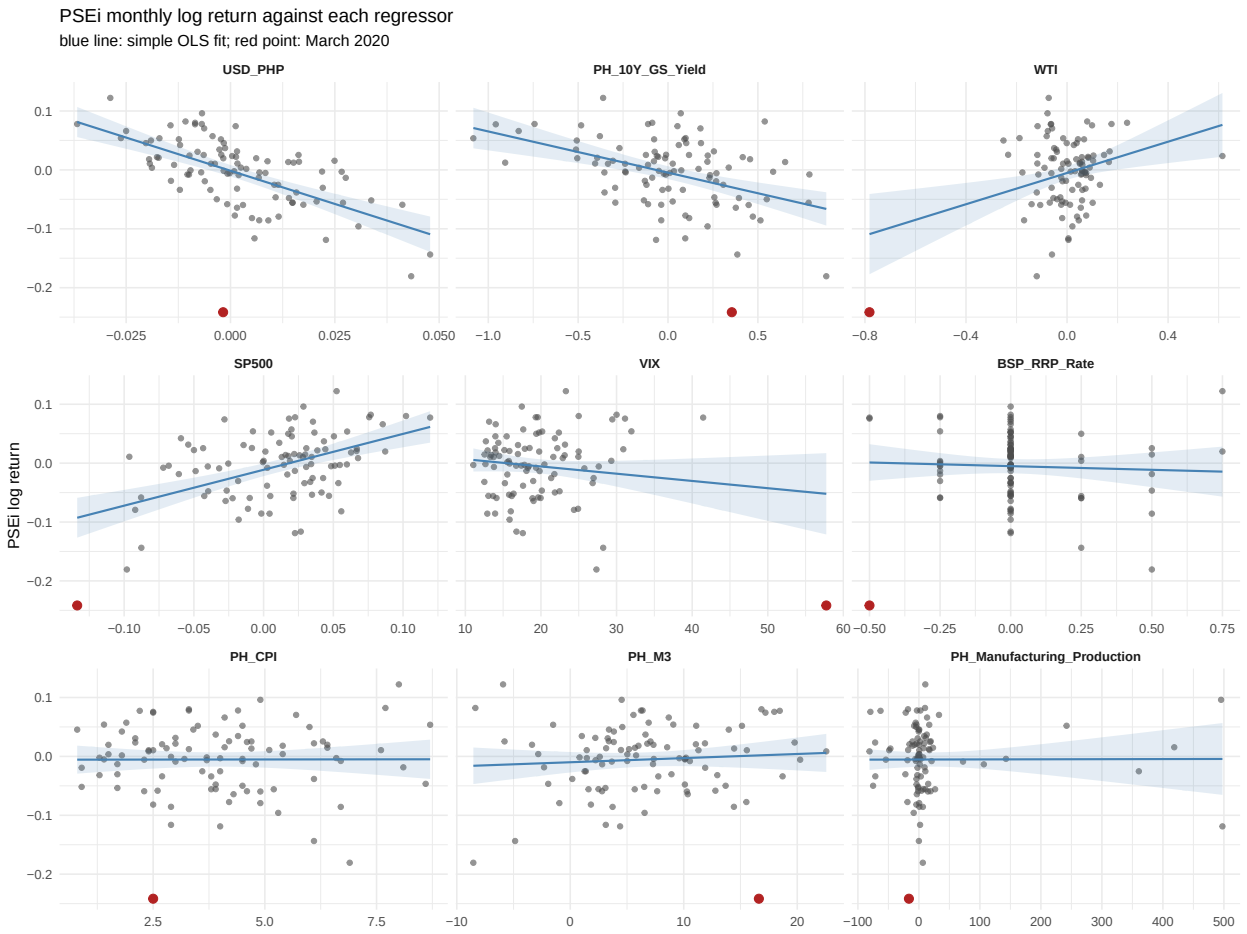


Figure 4: PSEi monthly log return against each regressor

The scatter plots relate the PSEi return to each regressor, with the simple OLS line, its 95% band and March 2020 marked in red. Four regressors show a visible linear relationship. The USD/PHP has the strongest,

with a negative slope (r = -0.60): months in which the dollar bought more pesos coincided with lower PSEi returns. The S&P 500 (r = 0.49) and WTI (r = 0.30) slope upwards and the 10-year yield slopes downwards (r = -0.43), so PSEi returns were higher in months of rising U.S. equities and oil and lower in months of rising local yields. All four relationships are significant at the 5% level. The other five panels are flat: the VIX (r = -0.14), the RRP change (-0.05), CPI (0.00), M3 (0.08) and manufacturing production (0.00) have lines close to horizontal and bands that include a zero slope. March 2020 matters most for WTI and the VIX. Without it the WTI correlation falls from 0.30 to 0.08 (p = 0.44), so the apparent oil effect rests on one month, and the VIX correlation changes sign from -0.14 to 0.13, neither being significant. The USD/PHP (-0.67), 10-year yield (-0.43) and S&P 500 (0.42) correlations do not depend on it. These are bivariate associations with the other regressors ignored. They show which relationships the multiple regression should recover, not what remains when all regressors are held constant together.

Pearson Correlation Heat-maps

Table 15: Pearson correlation matrix

	PSEI	USD_PHP	PH_10Y_GS_Yield	WTI	SP500	VIX	BSP_RRP_Rate	PH_CPI	PH_M3	PH_Manufacturing_Production
PSEI	1.000	-0.600	-0.432	0.302	0.487	-0.139	-0.046	0.003	0.076	0.003
USD_PHP	-0.600	1.000	0.281	0.050	-0.322	-0.070	0.139	0.043	-0.249	0.147
PH_10Y_GS_Yield	-0.432	0.281	1.000	0.019	-0.377	0.062	0.111	0.068	-0.092	0.009
WTI	0.302	0.050	0.019	1.000	0.413	-0.303	0.069	-0.043	0.086	0.012
SP500	0.487	-0.322	-0.377	0.413	1.000	-0.255	-0.021	-0.092	0.158	0.026
VIX	-0.139	-0.070	0.062	-0.303	-0.255	1.000	-0.187	0.000	0.171	-0.106
BSP_RRP_Rate	-0.046	0.139	0.111	0.069	-0.021	-0.187	1.000	0.658	-0.511	0.055
PH_CPI	0.003	0.043	0.068	-0.043	-0.092	0.000	0.658	1.000	-0.559	0.124
PH_M3	0.076	-0.249	-0.092	0.086	0.158	0.171	-0.511	-0.559	1.000	-0.180
PH_Manufacturing_Production	0.003	0.147	0.009	0.012	0.026	-0.106	0.055	0.124	-0.180	1.000

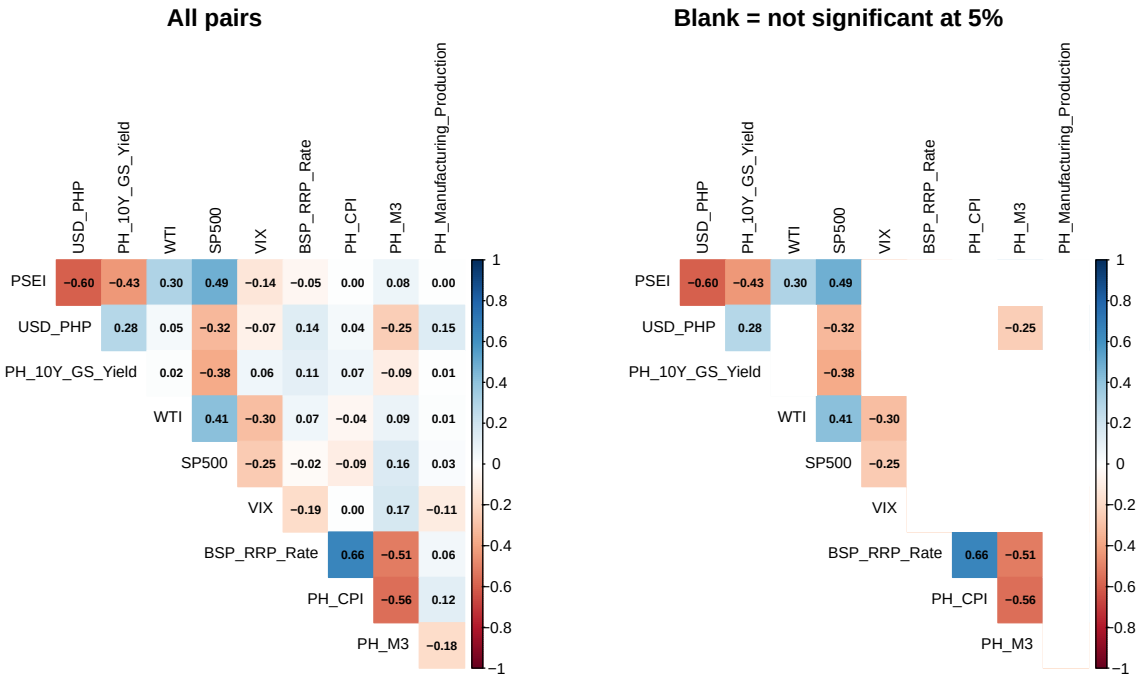


Figure 5: Correlation heatmaps of the PSEi return and the regressors: all pairs (left) and with the cells left blank where the correlation is not significant at the 5% level (right)

Table 16: Ten strongest pairwise correlations

var1	var2	r
BSP_RRP_Rate	PH_CPI	0.658
PSEI	USD_PHP	-0.600
PH_CPI	PH_M3	-0.559
BSP_RRP_Rate	PH_M3	-0.511
PSEI	SP500	0.487
PSEI	PH_10Y_GS_Yield	-0.432
WTI	SP500	0.413
PH_10Y_GS_Yield	SP500	-0.377
USD_PHP	SP500	-0.322
WTI	VIX	-0.303

The correlation matrix and its two heatmaps summarise all pairwise relationships. The second heatmap leaves a cell blank when the correlation is not significant at the 5% level, which removes 31 of the 45 pairs. The PSEi row repeats the scatter-plot result: it is significantly correlated with the USD/PHP (-0.60), the S&P 500 (0.49), the 10-year yield (-0.43) and WTI (0.30), and with none of the VIX, RRP change, CPI, M3 and manufacturing production (p-values from 0.18 to 0.98). Most pairs among the regressors are weak: of the 36, only seven have |r| of at least 0.30 and three have |r| of at least 0.50. The largest among the market variables are WTI with the S&P 500 (0.41) and the 10-year yield with the S&P 500 (-0.38). The exception is the domestic macroeconomic group. The RRP change is correlated with CPI (0.66) and with M3 growth (-0.51), and CPI with M3 (-0.56), three of the four strongest pairs in the table of ten. This is consistent with a policy rate that rises when inflation is high and money growth is slowing. None of the three is related to the PSEi, so they overlap with one another without explaining the index, which can inflate their standard errors in a regression and is checked with variance inflation factors in the diagnostics. Taken with the bivariate patterns, the evidence points to the exchange rate, global equity returns, local yields and, less reliably, oil as the main candidates, with March 2020 the observation most likely to influence the fit.

Section 5: Baseline Multiple Linear Regression

Our baseline multiple linear regression (MLR) relates monthly PSEi log returns to the nine macro-financial regressors introduced in the data section, estimated by ordinary least squares (OLS) on the 96 months from January 2018 to December 2025. The model specifies the conditional mean of the return: each slope measures the association between one regressor and the return while the other eight are held constant, a partial association in the ceteris paribus sense and not a causal effect. The section follows the baseline as a ladder of inferential steps, each answering a narrower question than the one before it. We begin with the overall fit and the joint significance of the model, move to the individual coefficients through hypothesis tests and confidence intervals, ask whether the estimated effects matter in economic terms and not only in statistical terms, and end with joint F-tests on groups of regressors. Together, our modeling explores an initial account of the factors associated with PSEi returns and set the agenda for the diagnostic testing and robustness checks that follow.

OLS is the natural starting point for three reasons. Its coefficients are transparent partial associations that can be read directly in return units, which keeps the financial interpretation close to the estimates. Under the Gauss-Markov conditions, it is the best linear unbiased estimator, and when the errors are normally distributed the t and F statistics have exact distributions, so hypotheses about single coefficients and about groups of coefficients are tested within one framework. And because nested linear models are simple to compare, the baseline serves as the benchmark against which every later specification is judged.

The same method has well-known weaknesses, and they shape how we read our results. The conditional mean is assumed to be linear and additive, so asymmetries and interactions are ignored. The standard errors are

valid only if the errors have constant variance and are serially uncorrelated; when they are not, conventional tests can mislead, which is why heteroskedasticity-consistent covariance estimators were developed (Long & Ervin, 2000; White, 1980). Least squares gives disproportionate weight to extreme observations (Huber, 1973), and in a sample of 96 months a single episode can move the estimates. Correlated regressors inflate standard errors, an omitted variable biases the included coefficients, and a contemporaneous regression on observational data supports statements about association, not about causation nor forecasting (Welch & Goyal, 2008). In-sample fit also rewards every added regressor whether or not it carries information. The baseline results are therefore provisional: the classical assumptions are tested in the diagnostics section and relaxed through corrections in the model-improvement section, where the final specification is chosen.

```
PSEI ~ USD_PHP + PH_10Y_GS_Yield + WTI + SP500 + VIX + BSP_RRP_Rate + PH_CPI + PH_M3 +
PH_Manufacturing_Production
```

Table 17: Initial MLR: PSEI on 9 regressors (OLS)

term	estimate	std_error	t_stat	p_value	ci_lower	ci_upper	signif
(Intercept)	0.0065800	0.0192800	0.3412	0.734	-0.0317500	0.0449100	
USD_PHP	-2.0450000	0.3038000	-6.7330	<0.0001	-2.6490000	-1.4410000	***
PH_10Y_GS_Yield	-0.0405200	0.0128100	-3.1650	0.00215	-0.0659800	-0.0150700	**
WTI	0.1297000	0.0364800	3.5550	0.000617	0.0571700	0.2022000	***
SP500	0.1342000	0.1144000	1.1730	0.244	-0.0931600	0.3615000	
VIX	-0.0002430	0.0006980	-0.3482	0.729	-0.0016310	0.0011440	
BSP_RRP_Rate	-0.0120300	0.0271400	-0.4433	0.659	-0.0659800	0.0419200	
PH_CPI	0.0006358	0.0033080	0.1922	0.848	-0.0059400	0.0072120	
PH_M3	-0.0011240	0.0008602	-1.3060	0.195	-0.0028340	0.0005865	
PH_Manufacturing_Production	0.0000330	0.0000446	0.7411	0.461	-0.0000556	0.0001217	

Table 18: Model fit and overall F-test (H0: all slope coefficients = 0)

statistic	value
n	96
r_squared	0.5724
adj_r_squared	0.5277
residual_std_err	0.04102
f_statistic	12.79
f_df_numerator	9
f_df_denominator	86
f_p_value	<0.0001

We first ask whether the nine regressors, taken together, carry information about the return at all. The coefficient table reports each estimate with its standard error, t statistic, p-value and 95% confidence interval, with stars marking p-values below 0.001, 0.01 and 0.05 and a dot marking those below 0.10. The fit table reports R-squared, adjusted R-squared, the residual standard error and the overall F-test, whose null hypothesis is that all nine slopes are zero against the alternative that at least one is not. R-squared is the share of the sample variation in returns accounted for by the fitted equation; adjusted R-squared penalizes the loss of degrees of freedom and is the more honest guide when some regressors may be redundant. Both describe fit within the estimation sample and say nothing about performance on new data.

The initial model accounts for about 57.2 percent of the variation in monthly PSEi returns (R-squared = 0.572; adjusted R-squared = 0.528), with a residual standard error of about 4.1 percentage points across 96 observations. The overall F-test, $F(9, 86) = 12.79$ with a p-value below 0.0001, rejects the null hypothesis

that all slope coefficients are simultaneously zero, so the regressors carry statistically significant information about returns when taken together. The gap of about 0.045 between R-squared and adjusted R-squared is a first hint that some of the nine regressors add little, a possibility that the individual and joint tests below examine directly.

Three of the nine regressors are significant at the 1 percent level. The exchange rate has the strongest association with returns (coefficient = -2.045, $p < 0.0001$). Because both variables are measured as log changes, the estimate reads as an elasticity: holding the other regressors constant, a one percent rise in USD/PHP, a one percent depreciation of the peso, is associated with a monthly return about 2.0 percentage points lower. The 10-year government bond yield also has a significant negative coefficient (-0.0405, $p = 0.00215$), so that each 10 basis point rise in the yield is associated with a return about 0.4 percentage points lower, a pattern that fits competition between fixed income and equities as yields rise. WTI crude oil has a significant positive coefficient (0.1297, $p = 0.000617$): a 10 percent rise in oil is associated with a return about 1.3 percentage points higher. A positive sign is notable for an economy that relies heavily on imported oil, and Kilian & Park (2009) suggest one reading, that oil price increases driven by strong global demand coincide with higher equity returns, so that WTI may proxy for global activity and not for an oil cost shock. The relationship warrants the robustness checks that follow, particularly with respect to the influence of March 2020.

The remaining six regressors, namely the S&P 500, the VIX, the BSP reverse repurchase rate, consumer price inflation, M3 growth and manufacturing production, are not significant even at the 10 percent level. The S&P 500 is the most informative case. Its simple correlation with PSEi returns is 0.487, yet in the multiple regression its coefficient is small relative to its standard error ($p = 0.244$) and its standardized size falls to 0.107. The contrast illustrates the difference between a marginal and a partial association: the S&P 500 moves with oil (correlation 0.413), with the yield (-0.377) and with the exchange rate (-0.322), so much of what it appears to explain alone is already carried by those regressors, and little remains once they are held constant. The variance inflation factors examined in the diagnostics section are all small (the largest is 2.16), so this is shared variation and not a collinearity problem. Manufacturing production is also insignificant ($p = 0.461$), although the extreme values identified in the exploratory analysis may be distorting its estimate. Those observations remain untreated in the baseline model.

Coefficient Tests

Table 19: (A) Two-sided t-tests, alpha = 0.05

H0	H1	estimate	std_error	t_stat	df	t_crit	p_value	decision
b(USD_PHP) = 0	b(USD_PHP) != 0	-2.0450000	0.3038000	-6.7330	86	1.988	<0.0001	Reject H0
b(PH_10Y_GS_Yield) = 0	b(PH_10Y_GS_Yield) != 0	-0.0405200	0.0128100	-3.1650	86	1.988	0.00215	Reject H0
b(WTI) = 0	b(WTI) != 0	0.1297000	0.0364800	3.5550	86	1.988	0.000617	Reject H0
b(SP500) = 0	b(SP500) != 0	0.1342000	0.1144000	1.1730	86	1.988	0.244	Fail to reject H0
b(VIX) = 0	b(VIX) != 0	-0.0002430	0.0006980	-0.3482	86	1.988	0.729	Fail to reject H0
b(BSP_RRP_Rate) = 0	b(BSP_RRP_Rate) != 0	-0.0120300	0.0271400	-0.4433	86	1.988	0.659	Fail to reject H0
b(PH_CPI) = 0	b(PH_CPI) != 0	0.0006358	0.0033080	0.1922	86	1.988	0.848	Fail to reject H0
b(PH_M3) = 0	b(PH_M3) != 0	-0.0011240	0.0008602	-1.3060	86	1.988	0.195	Fail to reject H0
b(PH_Manufacturing_Production) = 0	b(PH_Manufacturing_Production) != 0	0.0000330	0.0000446	0.7411	86	1.988	0.461	Fail to reject H0

Table 20: (B) One-sided tests against the theoretically expected sign

H0	H1	rationale	estimate	t_stat	t_crit	p_one_sided	decision
b(USD_PHP) >= 0	b(USD_PHP) < 0	peso depreciation -> foreign outflows	-2.0450000	-6.7330	-1.663	<0.0001	Reject H0
b(PH_10Y_GS_Yield) >= 0	b(PH_10Y_GS_Yield) < 0	higher yields lower equity valuations	-0.0405200	-3.1650	-1.663	0.00107	Reject H0
b(SP500) <= 0	b(SP500) > 0	global equity co-movement	0.1342000	1.1730	1.663	0.122	Fail to reject H0
b(VIX) >= 0	b(VIX) < 0	risk aversion	-0.0002430	-0.3482	-1.663	0.364	Fail to reject H0
b(BSP_RRP_Rate) >= 0	b(BSP_RRP_Rate) < 0	policy tightening raises discount rates	-0.0120300	-0.4433	-1.663	0.329	Fail to reject H0
b(PH_CPI) >= 0	b(PH_CPI) < 0	inflation erodes real earnings	0.0006358	0.1922	-1.663	0.576	Fail to reject H0
b(PH_M3) <= 0	b(PH_M3) > 0	more liquidity supports prices	-0.0011240	-1.3060	1.663	0.903	Fail to reject H0
b(PH_Manufacturing_Production) <= 0	b(PH_Manufacturing_Production) > 0	stronger real activity	0.0000330	0.7411	1.663	0.23	Fail to reject H0

Table 21: (C) Confidence intervals for the slope coefficients

term	estimate	90% low	90% high	95% low	95% high	99% low	99% high	95% width	0 in 95% CI?
USD_PHP	-2.05e+00	-2.55e+00	-1.540000	-2.65e+00	-1.440000	-2.85e+00	-1.250000	1.210000	No (significant)
PH_10Y_GS_Yield	-4.05e-02	-6.18e-02	-0.019200	-6.60e-02	-0.015100	-7.43e-02	-0.006790	0.050900	No (significant)
WTI	1.30e-01	6.90e-02	0.190000	5.72e-02	0.202000	3.36e-02	0.226000	0.145000	No (significant)
SP500	1.34e-01	-5.60e-02	0.324000	-9.32e-02	0.362000	-1.67e-01	0.435000	0.455000	Yes (not significant)
VIX	-2.43e-04	-1.40e-03	0.000917	-1.63e-03	0.001140	-2.08e-03	0.001600	0.002770	Yes (not significant)
BSP_RRP_Rate	-1.20e-02	-5.72e-02	0.033100	-6.60e-02	0.041900	-8.35e-02	0.059500	0.108000	Yes (not significant)
PH_CPI	6.36e-04	-4.86e-03	0.006140	-5.94e-03	0.007210	-8.08e-03	0.009350	0.013200	Yes (not significant)
PH_M3	-1.12e-03	-2.55e-03	0.000307	-2.83e-03	0.000586	-3.39e-03	0.001140	0.003420	Yes (not significant)
PH_Manufacturing_Production	3.31e-05	-4.11e-05	0.000107	-5.56e-05	0.000122	-8.44e-05	0.000151	0.000177	Yes (not significant)

Second, we test each coefficient separately. For every slope the two-sided test takes the null hypothesis that the coefficient is zero against the alternative that it is not; the t statistic is the estimate divided by its standard error and is compared with the critical value of the t distribution with 86 degrees of freedom, 1.988 at the 5 percent level. Where theory gives an expected sign, a one-sided test is also reported, with the null hypothesis that the coefficient is zero or has the opposite sign and a critical value of 1.663. It is more powerful against the stated direction but cannot detect an effect of the opposite sign, so it is used only for regressors whose sign has a clear rationale. WTI is left to the two-sided test, because an oil price increase can raise or lower the equity market of an oil-importing economy depending on whether it signals global demand or an input-cost shock. A rejection or a failure to reject is a statement about the sampling uncertainty of one estimate under its null; it says nothing about how large the effect is, which is why the next two approaches examine intervals and magnitudes.

The two-sided tests in panel (A) reject the null hypothesis for the exchange rate, the 10-year yield and WTI and fail to reject it for the other six regressors, in agreement with the p-values in the coefficient table. The one-sided tests in panel (B) support the expected negative signs for the exchange rate ($p < 0.0001$) and the 10-year yield ($p = 0.001$), consistent with the possibility that peso depreciation contributes to capital outflows and that higher government bond yields reduce the relative attractiveness of equities. The S&P 500, the VIX, the policy rate and manufacturing production carry their expected signs but are not significant, with one-sided p-values between 0.122 and 0.364. Inflation and M3 growth carry signs opposite to the theoretical expectation and are also insignificant, so their estimated effects cannot be distinguished from zero. WTI is excluded from the one-sided tests for the reason given above.

The confidence intervals in panel (C) add the dimension of precision. The 95 percent intervals for the exchange rate (-2.649 to -1.441), the 10-year yield (-0.0660 to -0.0151) and WTI (0.0572 to 0.2022) exclude zero and stay on the same side of zero at the 99 percent level, which reinforces the three rejections. The intervals for the other six regressors contain zero at all three levels. An interval carries more information than a significance decision, because it reports the range of coefficient values that the data cannot reject; an interval that contains zero therefore marks the limit of the evidence and does not show that no effect exists (Greenland et al., 2016). The S&P 500 illustrates this: its interval (-0.0932 to 0.3615) is wide, which reflects imprecise estimation and not conclusive evidence of no relationship.

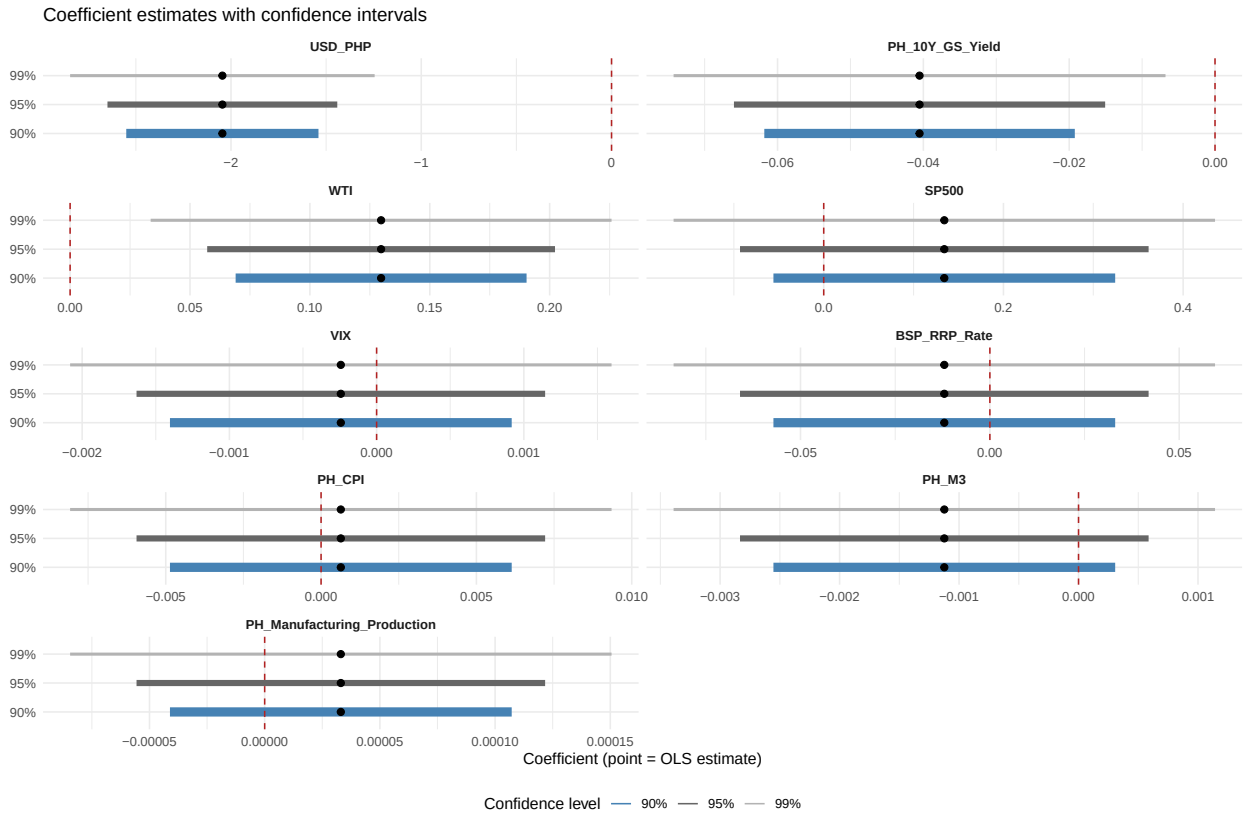


Figure 6: Initial model: coefficient estimates with 90%, 95% and 99% confidence intervals (dashed line = 0)

The confidence-interval figure shows the same information graphically, with each coefficient plotted at the 90, 95 and 99 percent levels and a dashed line marking the null value of zero. The nested intervals show how the verdict on a coefficient depends on the significance level chosen: a coefficient whose 99 percent interval excludes zero is significant under any conventional threshold, whereas one whose interval crosses zero between the 90 and 95 percent levels is a borderline case. Because each panel is drawn on its own horizontal scale, the figure is suited to judging how far each interval sits from zero and not to comparing interval widths across regressors, a comparison that belongs to the next rung, where the effects are put on a common scale.

In the figure, the exchange rate (about -2.05) and the 10-year yield (about -0.041) show negative associations with PSEi returns and WTI (about +0.13) a positive one. They are the only regressors whose intervals exclude zero at all three levels, as the two-sided tests and the interval table already indicated. Of the three, the yield lies closest to zero: even its 99 percent interval remains below the reference line, but only just.

The intervals for the S&P 500, the VIX, the policy rate, inflation, M3 growth and manufacturing production all contain zero, so none of these coefficients can be distinguished from zero in the baseline model. M3 growth comes closest to significance, with zero falling only just inside its 90 percent interval. The S&P 500 interval stays wide (-0.093 to 0.362 at the 95 percent level), which points to uncertainty about the coefficient and not to the absence of a relationship. Manufacturing production has numerically small bounds only because of the scale of the variable, and its interval also contains zero. All of these intervals rest on the classical regression assumptions and may change if the residual diagnostics call for robust standard errors, a question taken up in the sections that follow.

Economic Significance

Table 22: Statistical vs economic significance: effect of a one-standard-deviation move in each regressor on the PSEi monthly return (the PSEi return has a standard deviation of 5.97 percentage points)

term	sd_of_ regressor	slope	effect_pp	ci_low_pp	ci_high_ pp	std_beta	p_value	significant
USD_PHP	0.0159	-2.05e+00	-3.250	-4.210	-2.290	-0.5440	<0.0001	yes
PH_10Y_GS_Yield	0.3680	-4.05e-02	-1.490	-2.430	-0.555	-0.2500	0.00215	yes
WTI	0.1360	1.30e-01	1.760	0.775	2.740	0.2950	0.000617	yes
SP500	0.0477	1.34e-01	0.640	-0.444	1.720	0.1070	0.244	no
VIX	6.7600	-2.43e-04	-0.164	-1.100	0.774	-0.0275	0.729	no
BSP_RRP_Rate	0.2200	-1.20e-02	-0.265	-1.450	0.923	-0.0444	0.659	no
PH_CPI	1.8700	6.36e-04	0.119	-1.110	1.350	0.0199	0.848	no
PH_M3	6.4700	-1.12e-03	-0.727	-1.830	0.380	-0.1220	0.195	no
PH_Manufacturing_Production	97.7000	3.31e-05	0.323	-0.543	1.190	0.0541	0.461	no

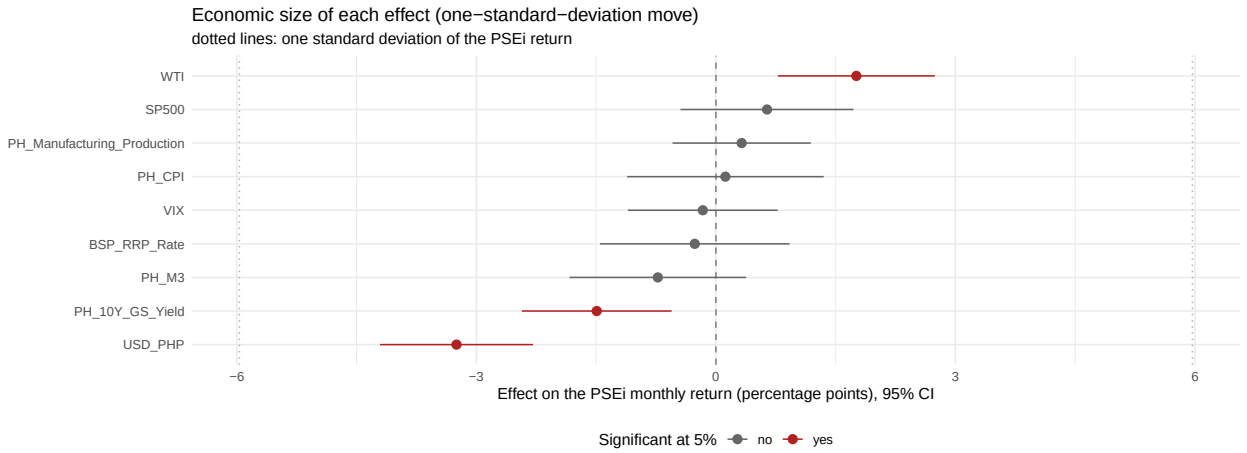


Figure 7: Economic size of each effect: change in the PSEi monthly return, in percentage points, for a one-standard-deviation move in the regressor, with 95% confidence intervals (dotted lines = one standard deviation of the PSEi return)

A coefficient can be statistically significant without being large enough to matter, and a coefficient that is not significant can still be economically important if it is estimated imprecisely (Wasserstein & Lazar, 2016; Ziliak & McCloskey, 2004). We therefore restate every slope in units that can be compared. Because a slope depends on the unit in which its regressor is measured, each effect is expressed as the change in the monthly PSEi return, in percentage points, associated with a one-standard-deviation move in the regressor, together with its 95 percent confidence interval and its standardized coefficient, which measures the same move as a share of the standard deviation of the return itself (5.97 percentage points). Reading the effect in percentage points ties the regression to an investor's own question, how much return is at stake, and the dotted lines in the figure mark a move of one standard deviation of the return as a benchmark for what counts as large.

The exchange rate has the largest economic effect. A one-standard-deviation rise in USD/PHP (0.0159) is associated with a decrease of 3.25 percentage points in the monthly return, with a 95 percent confidence interval from -4.21 to -2.29 percentage points. Its standardized coefficient of -0.544 means that a one-standard-deviation move in the exchange rate corresponds to a move of just over half a standard deviation of the return in the opposite direction. WTI and the 10-year yield are also economically meaningful. A one-standard-deviation rise in WTI (0.136) is associated with a 1.76 percentage-point higher return (standardized coefficient 0.295), and a one-standard-deviation rise in the yield (0.368) with a 1.49 percentage-point lower

return (standardized coefficient -0.250). The conservative ends of their intervals, 0.78 and 0.56 percentage points, are still roughly a tenth of the monthly volatility of the index or more.

Among the regressors that fail the significance test, the S&P 500 has an estimated one-standard-deviation effect of +0.64 percentage points (standardized coefficient 0.107), with an interval from -0.44 to 1.72 that includes zero. M3 growth has an estimated effect of -0.73 percentage points (standardized coefficient -0.122, $p = 0.195$), and manufacturing production has the smallest estimated effect, a rise of 0.32 percentage points per standard deviation (standardized coefficient 0.054), despite its far wider unstandardized range.

The intervals of these six regressors carry a further insight. Each of them reaches, at one end or the other, between 1.1 and 1.8 percentage points for a one-standard-deviation move, a magnitude comparable to the point estimates for WTI and the yield. The data therefore cannot exclude economically meaningful effects for any of the six; what they show is that a sample of 96 months is too noisy, with nine regressors to estimate, to measure them precisely. The six are better described as inconclusive than as irrelevant, and the model-improvement section returns to this when it considers removing them.

Joint Tests

We then move from single coefficients to groups. A joint F-test takes the null hypothesis that every coefficient in a group is zero against the alternative that at least one is not. It is the appropriate tool when regressors work together or overlap, since a group can be informative even when no member passes an individual test, and a single joint test also controls the overall error rate that nine separate t-tests would not. Each test is reported in three forms: with the classical covariance matrix, which assumes constant variance and uncorrelated errors, and with the HC3 and Newey-West covariance matrices, which relax those assumptions (Long & Ervin, 2000; Newey & West, 1987; White, 1980). Agreement across the three is evidence that a conclusion does not rest on the error assumptions, and divergence marks a result that does. Four groups are examined: the global factors, the domestic macroeconomic indicators, the money market and foreign exchange indicators, and the three regressors that were individually significant. The first three correspond to channels through which financial conditions may reach the equity market, and the subsections below set out what the literature leads one to expect from each before the results are read.

Global Factor Indicators

The global factors group contains the S&P 500, WTI crude oil and the VIX. The joint null hypothesis is that all three coefficients are zero, so that global equity returns, oil returns and market-implied volatility carry no information about PSEi returns once the domestic regressors are controlled. In the baseline model only WTI is individually significant, so the joint test shows whether the group has explanatory power beyond that single variable or whether WTI carries it alone. Our literature review below suggests why the three variables may matter together: global equity, oil and volatility shocks transmit to one another and to local markets, and the strength of that transmission changes with market conditions.

The joint impact of international market variables such as the S&P 500, WTI returns, and the VIX on local equity market performance can be understood through their interconnected spillover effects and volatility transmission mechanisms. One study focusing on six Latin American stock markets (Argentina, Brazil, Chile, Colombia, Mexico, and Peru) found that the S&P 500 acts as a consistent sender of return spillovers across different quantiles, indicating its influence on these local markets. In contrast, the oil market (WTI) was identified as the highest receiver of spillovers, suggesting that shocks in WTI returns are more likely to be influenced by other markets rather than driving local equity returns directly. The VIX exhibited a dynamic role, alternating between being a sender and receiver of spillovers depending on market conditions, with stronger spillover effects during extreme market quantiles and bullish phases where the VIX became a notable sender of shocks to local markets (Kruel & Ceretta, 2025).

Further, the volatility of WTI crude oil prices is influenced by shocks in the VIX and other financial indicators, with the VIX having a significant effect on oil price volatility. This relationship is time-varying and more pronounced during economic crises, which can indirectly affect local equity markets through changes in oil

price volatility and associated uncertainty (Le et al., 2022). Additionally, the interaction between the VIX and WTI spot prices is complex, with evidence of unidirectional influence from the VIX to WTI but no straightforward Granger causality, implying that their joint impact on local markets may involve intricate feedback loops (K. Chen, 2026).

Geopolitical risk is a further link between oil, volatility and equity markets. Using a daily geopolitical risk index, it has been shown that geopolitical risk plays an important role in determining oil price volatility and, to a lesser extent, stock market volatility, that higher geopolitical risk goes with higher oil returns and lower stock returns, and that volatility spillovers run from oil to stock returns, so a geopolitical shock may reach a local equity market through oil prices and market volatility (Smales, 2021).

Domestic Macroeconomic Indicators

The domestic macroeconomic group contains consumer price inflation, M3 growth and manufacturing production, and the joint null hypothesis is that none of them is related to returns. None is individually significant, so the joint test shows whether three weak signals, taken together, add up to a detectable one. Two features of these series argue against a strong monthly relationship. They are year-on-year rates and therefore change slowly, with first-order auto-correlations of 0.95 for inflation, 0.94 for M3 growth and 0.73 for manufacturing production, so each month adds little news; and equity prices respond to unanticipated information (N.-F. Chen et al., 1986), which a slowly moving year-on-year rate conveys only sluggishly. The literature below nevertheless documents links between consumer activity, money and the Philippine stock market, so the joint test is a fair way to give the group its chance.

The joint impact of domestic macroeconomic variables such as the Philippine Consumer Price Index (PH_CPI), money supply (PH_M3), and manufacturing production on the local equity market performance can be understood through their influence on various sectors and overall stock market indices. Consumer activity, which is closely related to consumer spending and inflation measures like CPI, has been identified as a significant driver of stock market performance in the Philippines. Specifically, consumer activity and monetary policy variables, including money supply (M3), have a direct and indirect effect on the performance of all sectors in the Philippine Stock Market, with consumer activity being a primary determinant of stock market increases (Fernandez & Li, 2020).

Monetary policy, which encompasses money supply (PH_M3), shows a significant direct effect on sectors such as Financial, Holding Firms, Services, and Mining-Oil, although it does not directly affect Industrial and Property sectors. Business activity, which can be proxied by manufacturing production, is also a significant factor but mainly for the Financial, Industrial, and Property sectors (Fernandez & Li, 2020). This suggests that manufacturing production influences stock market performance, particularly in sectors related to industrial output.

Inflation, as measured by CPI, is among the macroeconomic indicators that have a significant relationship with the Philippine Stock Exchange Index (PSEi). Studies using Granger causality analysis indicate that inflation, along with GDP, exchange rate, and interest rates, impacts the stock market performance, reflecting the sensitivity of the PSEi to changes in these macroeconomic variables (Gallardo & Viray, 2025).

Money Market and FX Indicators

The money market and foreign exchange group contains the BSP reverse repurchase rate, the 10-year government bond yield and USD/PHP, and the joint null hypothesis is that all three coefficients are zero. Two of the three are individually significant and the policy rate is not, so a joint rejection would mostly reflect the exchange rate and the yield, and the test is informative chiefly about whether the policy rate adds anything once the market-determined prices are present.

The joint impact of changes in monetary and foreign exchange variables, such as BSP's RRP, the yield on Philippine 10-year government securities, and the USD/PHP exchange rate, on equity markets can be explained by the combined influence of these variables on the cost of capital, investment decisions, and stock

prices. Changes in monetary policy may influence bond yields and forex rates that, in turn, will affect investor decision-making and stock market performance. According to Checo et al. (2024), in their study of the effects of monetary policy transmission in emerging economies, it was discovered that tighter monetary policies result in persistent increases in government bond yields and, at the same time, appreciation of the domestic currency against foreign exchange rates and a decline in stock prices.

Moreover, similar research was conducted by Sy (2020), where he looked at the association of monetary and fiscal policies with the performance of the Philippine Stock Exchange Index from 2001 to 2017. Sy determined that a long-run equilibrium exists among the monetary policy rates, other macroeconomic factors, and the stock market returns. It was interesting to note that there is a positive long-term association between the domestic policy rates and stock market prices. In this sense, the effect of interest rates on stock market performance differs under different economic circumstances and periods of interest.

Furthermore, Tala (2025) studied the impact of macroeconomic fluctuations on the Financial Index of the Philippine Stock Exchange from 2009-2024, considering the exchange rate and the short-term and long-term interest rates. In this case, statistically significant long-run relationships were found between the macroeconomic indicators and the financial index, while short-run effects varied depending on economic conditions. The exchange rate and long-term interest rates remained significant during the COVID-19 pandemic, which indicates that the monetary and foreign exchange indicators can impact the stock exchange at the same time, even in the case of a period of economic uncertainty. These findings provide a basis for examining the joint relationship of BSP policy rate changes, Philippine 10-year government securities yield changes, and USD/PHP exchange rate movements with PSEi monthly log returns.

Set of Independently Significant Regressors

The last group is not a technical channel but a check on parsimony. We collect the three regressors that were individually significant, the exchange rate, the 10-year yield and WTI, and test whether they remain jointly significant when the null restricts only these three coefficients and leaves the other six free. Because the three were chosen after seeing the individual tests, the joint test is subject to selection, and its p-value should be read as descriptive and not as independent confirmation.

The joint impact of independently influential variables on the local equity market could be explained in terms of the collective effects of macroeconomic and financial variables on stock returns. Economic variables which are found to have significant influences on stock performance independently could also be able to add more insights about stock movements when considered collectively. In their study about the connection between macroeconomic variables and stock market performance among five Asian countries including the Philippines, Babar et al. (2024) discovered that economic variables such as gross domestic product, interest rates, exchange rate, and money supply were found to have positive relationships with stock market returns.

Furthermore, Alloul & Ferrouhi (2024) studied the effect of a number of macroeconomic variables, such as inflation rate, employment rate, foreign direct investments, and economic growth on stock market returns in developed, emerging and frontier markets. It was found out that there existed statistically significant long-term links between some macroeconomic variables and stock returns, but at the same time, the significance of some variables depended on the market types under consideration. Short-run relationships also varied across different market categories. This suggests that the relationships between macroeconomic variables and stock returns may vary across different economies.

In addition, Cakici & Zaremba (2024) analyzed the determinants of stock returns in various economies employing machine learning approaches. They discovered that specific financial variables such as momentum, earnings yield, market size, and long-term reversal were among key factors predicting stock returns; however, the importance of those varied in different forecast horizons. Meanwhile, particular variables measuring country risk did not contribute much to the forecast. These findings suggest that economic and financial variables may differ in their importance when considered together. Overall, these findings provide a basis for examining independently significant regressors jointly to determine their combined relationship with the Philippine Stock Exchange Index (PSEi) monthly log returns.

Table 23: Joint F-tests on groups of coefficients (alpha = 0.05)

test_group	H0	q	df_resid	F_ classical	p_ classical	p_hc3	p_newey_ west	decision_ classical
All slopes (overall significance)	b = 0 for all 9 coefficients	9	86	12.790	<0.0001	<0.0001	<0.0001	Reject H0
Global factors	b = 0 for all 3 coefficients	3	86	8.368	<0.0001	0.0901	0.00303	Reject H0
Domestic macro	b = 0 for all 3 coefficients	3	86	1.070	0.366	0.644	0.355	Fail to reject H0
Monetary and FX	b = 0 for all 3 coefficients	3	86	21.440	<0.0001	<0.0001	<0.0001	Reject H0
Individually significant three	b = 0 for all 3 coefficients	3	86	21.650	<0.0001	<0.0001	<0.0001	Reject H0

The joint-test table reports, for each group, the F statistic under the classical covariance matrix together with the p-values under the classical, HC3 and Newey-West matrices.

The test of all nine slopes ($q = 9$, residual degrees of freedom = 86) gives $F = 12.79$ ($p < 0.0001$) and remains significant under both HC3 and Newey-West standard errors, so the baseline model is jointly significant however the covariance matrix is estimated. The money market and foreign exchange group (the policy rate, the 10-year yield and USD/PHP) is also strongly significant under all three methods ($F = 21.44$, $p < 0.0001$). This mainly reflects the individual significance of the exchange rate and the yield, since the policy rate is not significant on its own. In financial terms, the price of foreign currency and the price of domestic government borrowing together carry much of the explanatory power in the baseline model.

The global factors group (the S&P 500, WTI and the VIX) is jointly significant under the classical covariance matrix ($F = 8.368$, $p < 0.0001$) and under Newey-West standard errors ($p = 0.00303$), but its p-value rises to 0.0901 under HC3, which supports significance only at the 10 percent level. The sensitivity may reflect heteroskedasticity or the influence of extreme observations, particularly in March 2020, and because WTI is the group's only individually significant member, much of the group's strength may rest on that one variable. In financial terms, the evidence that global conditions matter beyond oil is weak in the baseline model.

The domestic macroeconomic group (inflation, M3 growth and manufacturing production) is not jointly significant under the classical ($F = 1.070$, $p = 0.366$), HC3 ($p = 0.644$) or Newey-West ($p = 0.355$) covariance matrices. These variables add little explanatory power in the baseline specification, which is consistent with the slow-moving character of year-on-year series noted above. A failure to reject the null hypothesis does not establish that their effects are absent, however, since the intervals reported earlier cannot rule out effects of one percentage point or more.

Finally, the exchange rate, the 10-year yield and WTI are jointly significant under all three methods ($F = 21.65$, $p < 0.0001$), so their combined significance is robust to the adjustments for heteroskedasticity and autocorrelation. Because these three were singled out by the individual tests, however, their joint p-value describes the selected set and does not independently confirm it.

Taken together, the five joint models give a consistent but provisional picture. The initial model is significant as a whole and accounts for a little over half of the variation in monthly returns. The exchange rate, the 10-year yield and WTI are the only regressors with individually significant coefficients, and of these the exchange rate has the largest economic effect; the other six are inconclusive and not demonstrably irrelevant. The group tests narrow the evidence to the prices of foreign currency and domestic borrowing, show that the case for the global factors depends on the covariance estimator, and find little in the domestic macroeconomic indicators. None of this has yet been tested against the assumptions on which the standard errors rest. March 2020 is a recurring concern, the extreme manufacturing production observations remain untreated, and the divergence between the HC3 and classical results already shows that inference is sensitive to the error assumptions. The following sections therefore evaluate the model against alternatives, diagnose it

formally and correct what the diagnostics reveal before any of these associations is carried into the financial interpretation.

Section 6: Model Evaluation

We note that a regression model is evaluated by what it adds, not by how much it fits. R-squared never falls when a regressor is added, so the specification with the most regressors always has the largest R-squared, whether or not the added regressors carry any information about returns. This section therefore asks of every candidate model what R-squared cannot answer: whether the extra complexity is necessary, in the sense that matters to an investor, which is the ability to describe returns that were not used to fit the model. The tools are the ones the problem calls for. Adjusted R-squared and a leave-one-out predicted R-squared correct fit for complexity; the Akaike and Bayesian information criteria (AIC and BIC) price each additional parameter; nested F and Wald tests examine whether a block of coefficients is jointly zero; and a step-wise search checks whether an algorithm that knows nothing about finance can recover the same specification.

The candidate models are not arbitrary subsets of the data. Each one is a statement about what moves the Philippine equity market. The intercept-only model says that nothing observable does. The core model says that three market prices, the exchange rate, the 10-year yield and oil, carry the information. The domestic macro model says that fundamentals, namely the policy rate, inflation, money growth and manufacturing output, do. The market-related model adds global equity returns and volatility to the core, and the full model includes everything. Comparing them is therefore a comparison of economic hypotheses, and the result has a direct use: it decides which variables an investor should monitor and how much of the variation in monthly returns can be attributed to them.

Candidate Models

The five candidates form a nested family, which is what makes formal comparison possible. Every model is estimated on the same 96 monthly observations, so differences in fit reflect differences in specification and not differences in sample. The intercept-only model is nested in all the others; the core model is nested in the market-related and full models; the market-related model is nested in the full model; and the domestic macro model is nested in the full model but not in the core or market-related models, which makes it a rival hypothesis and not a subset. The subsections below state each hypothesis, its members and its economic reasoning before any statistics are shown.

We show of two features of the construction. First, the core model was assembled from the three regressors that were individually significant in the full model, so it is a model chosen after looking at the data. A preference for it over the full model is partly mechanical, because a set selected for strong t statistics will tend to look better than the set that remains, and the comparisons that follow cannot fully remove that advantage. Second, the number of parameters matters at this sample size: the full model spends 11 parameters, the nine slopes, the intercept and the error variance, on 96 observations, roughly one for every nine, which is why the criteria that penalize complexity carry significant weight in this section.

Intercept Only

The intercept-only model is the hypothesis that none of the nine regressors helps to describe monthly returns, so that the best description of any month's return is the average return. It is the benchmark that every other model must beat, and it is the counterpart of the historical-mean forecast that Welch & Goyal (2008) show to be hard to outperform in equity return prediction. By construction its R-squared is zero and its in-sample error equals the standard deviation of the return, about 5.9 percentage points. Its leave-one-out predicted R-squared is slightly negative (-0.021), which records that predicting each month with the mean of the other 95 is marginally worse than using the full-sample mean, a reminder that even the simplest model carries estimation error. In financial terms it is the hypothesis that a month's return is unrelated to that

month's macro-financial conditions. Because the regressors are measured in the same month as the return, rejecting it shows that returns move with contemporaneous conditions.

Core Regressors

The core model is the financial-prices hypothesis: monthly returns on the Philippine equity market are explained by three market prices, the USD/PHP exchange rate, the 10-year government bond yield and WTI crude oil. Each stands for a distinct transmission channel. The exchange rate captures the peso value of foreign-currency earnings and the portfolio decisions of foreign investment. The yield is the discount rate applied to future cash flows and the opportunity cost of holding equities. Oil carries commodity and global-demand information into a market that relies heavily on imported fuel, with the sign ambiguity noted in the coefficient analysis (Kilian & Park, 2009). These are the three regressors whose coefficients were individually significant in the baseline model, and the same three that the joint test found significant under every covariance estimator. With three slopes, an intercept and the error variance, the model spends five parameters, fewer than half of the full model's eleven.

Domestic Macro 4

The domestic macro model is the fundamentals hypothesis: returns follow the domestic policy and real-economy environment, measured by changes in the BSP reverse repurchase rate, consumer price inflation, M3 money growth and manufacturing production growth. It deliberately differs from the three-variable domestic group tested in the baseline section, because it includes the policy rate, which a fundamentals story cannot omit. The hypothesis has long standing in finance, since Fama (1981) tied stock returns to inflation, real activity and money. It is also the hypothesis that a monthly frequency treats least kindly: inflation, money growth and production enter as year-on-year rates that change slowly and carry little new information from one month to the next, while equity prices respond to news (N.-F. Chen et al., 1986). The model is not nested in the core or market-related specifications, so it is a genuine rival and not a restricted version of them, and its nested comparison with the full model shows what the financial variables add to it.

Market-related Regressor

The market-related model is the global risk-appetite hypothesis: on top of the three core prices, Philippine returns also respond to the S&P 500, which carries global equity returns, and to the VIX, which measures expected volatility and risk aversion in global markets. There is good reason to expect both to matter. Equity markets are interdependent across borders (Forbes & Rigobon, 2002), and the VIX is closely tied to the price of risk (Bekaert & Hoerova, 2014). The model nests the core model as the restriction that the two additional coefficients are zero, so the evidence on this hypothesis is the incremental contribution of the S&P 500 and the VIX once the exchange rate, the yield and oil are already in the model. What is at stake is whether global sentiment has an independent effect on the Philippine market or reaches it through the prices already in the core, since a global risk-off episode typically weakens the peso and lifts local yields at the same time.

Full-set Regressor

The full model is the kitchen-sink hypothesis: all nine regressors enter and nothing is excluded in advance. It is the most flattering specification, because adding regressors can only raise R-squared, and for that reason it serves as the reference against which the more parsimonious models are judged and not as a candidate that wins by default. Its role in the comparison is to show whether the regressors left out of the smaller models, the S&P 500, the VIX, the policy rate, inflation, M3 growth and manufacturing production, contribute explanatory power that the smaller models lack. The cost of including them is statistical as well as practical. With 11 parameters on 96 observations the coefficients are estimated with more noise, and the baseline

analysis has already shown that each of the six extra coefficients is individually indistinguishable from zero. The comparison below weighs the full model's slightly higher fit against that noise.

Table 24: In-sample fit of the candidate models

model	n	n_ regressors	r_squared	adj_r_ squared	predicted_ r_squared	rmse_in_ sample	residual_ std_err	f_ statistic	f_p_value
M0: intercept only	96	0	0.00000	0.00000	-0.02116	0.05937	0.05968	NA	NA
M1: core 3	96	3	0.54590	0.53110	0.45320	0.04001	0.04087	36.8700	<0.0001
M2: domestic macro	96	4	0.01072	-0.03276	-0.17800	0.05905	0.06065	0.2465	0.911
M3: market-related	96	5	0.55570	0.53100	0.39310	0.03957	0.04087	22.5100	<0.0001
M4: full 9	96	9	0.57240	0.52770	0.33750	0.03882	0.04102	12.7900	<0.0001

The in-sample fit table reports four measures that answer different questions. R-squared is the share of the sample variation in returns that the fitted equation accounts for, and it cannot fall when a regressor is added. Adjusted R-squared charges for each added regressor and rises only if the added regressor's t statistic exceeds one in absolute value, or, for a block of regressors, if the block's F statistic exceeds one. Predicted R-squared uses the leave-one-out prediction error, the PRESS statistic (Allen, 1974), so that each month is predicted from a model fitted to the other 95; it is the closest of the four to a test on data the model has not seen. The in-sample root mean squared error restates fit in return units.

The ranking by R-squared and the ranking by the other measures point in opposite directions. R-squared rises from 0.546 for the core model to 0.556 for the market-related model and 0.572 for the full model, and the in-sample error falls from 4.00 to 3.88 percentage points, a gain of about 3 percent. Adjusted R-squared does not move (0.531, 0.531 and 0.528), and predicted R-squared falls steadily, from 0.453 to 0.393 and 0.338. The six extra regressors therefore buy a 0.027 gain in R-squared and a 3 percent lower in-sample error, and they cost 0.116 of predicted R-squared. Selecting the full model because it has the largest R-squared would reward exactly the noise that the leave-one-out measure penalizes. The domestic macro model confirms the point from the other side: with an R-squared of 0.011, a negative adjusted R-squared (-0.033) and a negative predicted R-squared (-0.178), it cannot be distinguished from the intercept-only model (F-test p = 0.911).

The figures also have a financial reading. R-squared is the share of the variance of monthly returns that is associated with the model's regressors, so the core model attributes about 55 percent of return variance to the exchange rate, the yield and oil in sample, and about 45 percent on the leave-one-out measure. If the estimated associations were stable, a hedge that neutralized those three exposures would remove roughly that share of the variance, and what remains, a residual standard deviation of about 4.1 percentage points a month, is risk that these three prices do not explain. The leave-one-out figure is the more cautious of the two, although it is still optimistic for the core model because its regressors were chosen with the same data.

AIC-BIC

Table 25: AIC, AICc and BIC (lower is better)

model	k	logLik	AIC	AICc	BIC	delta_AIC	delta_AICc	delta_BIC	weight_ AICc	weight_BIC	rank_AIC	rank_BIC
M0: intercept only	2	134.888	-265.776	-265.647	-260.647	69.787	69.249	62.094	0.000	0.000	4	4
M1: core 3	5	172.781	-335.562	-334.896	-322.741	0.000	0.000	0.000	0.771	0.971	1	1
M2: domestic macro	6	135.405	-258.811	-257.867	-243.424	76.752	77.029	79.316	0.000	0.000	5	5
M3: market-related	7	173.827	-333.654	-332.381	-315.703	1.909	2.515	7.037	0.219	0.029	2	2
M4: full 9	11	175.668	-329.335	-326.192	-301.127	6.227	8.704	21.613	0.010	0.000	3	3

Table 26: Model preferred by each criterion

criterion	preferred
AIC	M1: core 3
AICc	M1: core 3
BIC	M1: core 3

The information criteria give the comparison a probabilistic form. AIC is minus twice the maximized log-likelihood plus two for each estimated parameter, and BIC replaces the two with the natural logarithm of the sample size, about 4.6 for 96 observations, so BIC penalizes complexity more heavily (Akaike, 1974; Schwarz, 1978). The parameter count includes the error variance. AICc adds a small-sample correction that becomes material when the number of parameters is large relative to the sample, as it is for the full model (Hurvich & Tsai, 1989). Only differences between models matter, and the tables report each model's distance from the best together with the weight that follows from it, a relative probability that the model is the best of those compared (Burnham & Anderson, 2004).

All three criteria choose the core model. The market-related model is 1.9 AIC points and 2.5 AICc points behind, a gap that leaves it with some support but clearly second: its AICc weight is 0.22, against 0.77 for the core model. The full model is 6.2 AIC points and 8.7 AICc points behind, with an AICc weight of 0.01, and its BIC distance of 21.6 points leaves it with essentially no weight at all. The BIC weight of the core model is 0.97. The intercept-only and domestic macro models are more than 60 points behind on every criterion and have no support. Differences of this size reflect a consistent pattern: the likelihood gained by adding the S&P 500, the VIX and the four domestic variables to the core model is smaller than the price the criteria charge for the extra parameters.

Our economic reading is the three market prices already summarize what the other variables could add. Global equity returns and volatility, which might be expected to matter for a small open market, do not improve the description of Philippine returns once the exchange rate, the yield and oil are in the model, which is consistent with the view that global risk reaches the local market through the peso and domestic yields.

Anova F-Test

Table 27: Nested-model F-tests (alpha = 0.05)

restricted	full	extra_ terms	q	df_resid	RSS_ restricted	RSS_full	F	p_value	decision
M0: intercept only	M4: full 9	USD_PHP, PH_ 10Y_GS_Yield, WTI, SP500, VIX, BSP_RRP_Rate, PH_CPI, PH_M3, PH_ Manufacturing_ Production	9	86	0.3383	0.1447	12.7900	<0.0001	Reject H0 (keep extra terms)
M1: core 3	M4: full 9	SP500, VIX, BSP_ RRP_Rate, PH_ CPI, PH_M3, PH_ Manufacturing_ Production	6	86	0.1536	0.1447	0.8883	0.507	Fail to reject H0 (restriction holds)
M2: domestic macro	M4: full 9	USD_PHP, PH_ 10Y_GS_Yield, WTI, SP500, VIX	5	86	0.3347	0.1447	22.5900	<0.0001	Reject H0 (keep extra terms)
M3: market-related	M4: full 9	BSP_RRP_Rate, PH_CPI, PH_M3, PH_ Manufacturing_ Production	4	86	0.1503	0.1447	0.8404	0.503	Fail to reject H0 (restriction holds)
M1: core 3	M3: market-related	SP500, VIX	2	90	0.1536	0.1503	0.9911	0.375	Fail to reject H0 (restriction holds)

The nested F-tests give the sharpest answer to whether a block of coefficients is jointly zero, by comparing the residual sums of squares of a restricted and a full model, and the comparisons were chosen to match the economic hypotheses. The test of the intercept-only model against the full model reproduces the overall F-test of the baseline model ($F = 12.79$, $p < 0.0001$): the regressors matter together. The remaining tests show which of them are responsible. Moving from the core model to the full model adds six regressors and reduces the residual sum of squares from 0.1536 to 0.1447, a fall of 5.8 percent, which gives $F = 0.89$ ($p = 0.507$); the six cannot be distinguished from zero jointly. Adding the four domestic variables to the market-related model leaves the verdict unchanged ($F = 0.84$, $p = 0.503$), and so does adding the S&P 500 and the VIX to the core model ($F = 0.99$, $p = 0.375$).

The test in the opposite direction is as informative. Adding the five financial variables, the exchange rate, the yield, oil, the S&P 500 and the VIX, to the domestic macro model is overwhelmingly significant ($F = 22.59$, $p < 0.0001$). Together the tests say that market prices contain information about the equity market that domestic fundamentals lack, and that domestic fundamentals and global sentiment add nothing once the market prices are present. This is the economic content of the nested comparison: the full model always has the highest fit, so the useful result is which hypotheses the data can reject.

We maintain two cautions. A failure to reject does not prove that the omitted coefficients are zero, as the intervals in the baseline analysis showed: with 96 observations the test has limited power against effects of one or two percentage points. And because the core model was assembled from regressors that were significant in the full model, failing to reject the other six is the expected result of that selection, and selection of this kind distorts p-values (Heinze et al., 2018). The cross-validation in the model-improvement section therefore carries extra weight.

Wald Test

Table 28: Wald tests of nested restrictions (classical, HC3 and Newey-West)

restricted	full	q	chisq_ classical	p_chisq_ classical	F_ classical	p_F_ classical	F_hc3	p_F_hc3	F_nw	p_F_nw
M0: intercept only	M4: full 9	9	115.100	<0.0001	12.7900	<0.0001	11.0700	<0.0001	22.6400	<0.0001
M1: core 3	M4: full 9	6	5.330	0.502	0.8883	0.507	0.4143	0.868	0.7238	0.632
M2: domestic macro	M4: full 9	5	113.000	<0.0001	22.5900	<0.0001	16.3200	<0.0001	22.1400	<0.0001
M3: market-related	M4: full 9	4	3.362	0.499	0.8404	0.503	0.4203	0.794	0.8371	0.505
M1: core 3	M3: market-related	2	1.982	0.371	0.9911	0.375	0.6844	0.507	0.8627	0.425

The Wald tests repeat the same five comparisons with covariance matrices that relax the classical error assumptions. With the classical covariance matrix the Wald F statistic equals the nested F statistic, and the chi-square version, which rests on large-sample theory, gives almost the same p-value (0.502 against 0.507 for the core model against the full model). The robust versions use the HC3 and Newey-West estimators (Long & Ervin, 2000; Newey & West, 1987; White, 1980), which remain valid under heteroskedasticity and serial correlation. The verdicts do not change. The six extra regressors remain jointly insignificant (HC3: $F = 0.41$, $p = 0.868$; Newey-West: $F = 0.72$, $p = 0.632$), as do the four domestic variables added to the market-related model ($p = 0.794$ and 0.505) and the S&P 500 and VIX added to the core model ($p = 0.507$ and 0.425). The financial variables remain overwhelmingly significant when added to the domestic macro model (HC3: $F = 16.32$; Newey-West: $F = 22.14$; both $p < 0.0001$).

The robust p-values for the extra regressors are, if anything, larger than the classical ones, so the case for the smaller model does not depend on an assumption of constant error variance. This is useful but not conclusive. We show that the diagnostic tests that follow examine whether the error assumptions hold at all, and the sensitivity of the global-factor group to the covariance estimator in the baseline analysis is a reminder that robust and classical answers do not always agree.

Stepwise Selection

Table 29: Stepwise selection results (bidirectional, starting from the full model)

direction	criterion	selected	n_ regressors	adj_r_ squared	AIC	BIC
both	AIC	USD_PHP, PH_10Y_GS_ Yield, WTI, PH_M3	4	0.5402	-336.5	-321.1
both	BIC	USD_PHP, PH_10Y_GS_ Yield, WTI	3	0.5311	-335.6	-322.7

Table 30: Selection path: both directions, AIC

Step	Df	Deviance	Resid. Df	Resid. Dev	criterion_ value
	NA	NA	86	0.1447	-603.8
- PH_CPI	1	0.0000621	87	0.1447	-605.7
- VIX	1	0.0001688	88	0.1449	-607.6
- BSP_RRP_Rate	1	0.0002419	89	0.1451	-609.5
- PH_Manufacturing_Production	1	0.0010880	90	0.1462	-610.7
- SP500	1	0.0027970	91	0.1490	-610.9

Table 31: Selection path: both directions, BIC

Step	Df	Deviance	Resid. Df	Resid. Dev	criterion_ value
	NA	NA	86	0.1447	-578.1
- PH_CPI	1	0.0000621	87	0.1447	-582.7
- VIX	1	0.0001688	88	0.1449	-587.1
- BSP_RRP_Rate	1	0.0002419	89	0.1451	-591.5
- PH_Manufacturing_Production	1	0.0010880	90	0.1462	-595.4
- SP500	1	0.0027970	91	0.1490	-598.1
- PH_M3	1	0.0046080	92	0.1536	-599.7

Stepwise selection shows whether an algorithm, starting from the full model and removing or re-adding one regressor at a time, arrives at the same place as the economic hypotheses. It does, with one qualification. Under BIC the search removes inflation, the VIX, the policy rate, manufacturing production, the S&P 500 and then M3 growth, and stops at the core model of the exchange rate, the yield and oil (adjusted R-squared 0.531). Under AIC the order of removal is the same until M3 growth, which AIC keeps, so that the selected model is the core three plus M3 growth (adjusted R-squared 0.540). The order of removal is exactly the order of descending p-values in the baseline model (0.848, 0.729, 0.659, 0.461, 0.244, 0.195), so the search adds little to what the individual tests already showed. The criterion values in the selection-path tables are on the scale the search routine uses internally, around -600, and are not comparable with the values in the earlier AIC table; only the differences between steps carry meaning.

The disagreement about M3 growth is a disagreement about the price of a parameter and not about the data. Adding a regressor lowers AIC when its t statistic exceeds about 1.4 in absolute value, lowers BIC when it exceeds about 2.1 at this sample size, and raises adjusted R-squared when it exceeds 1. M3 growth keeps its place under AIC and loses it under BIC, so its t statistic in the four-regressor model lies between the two thresholds, and the AIC-selected model is less than one AIC point better than the core model, a gap too small to separate them. The choice between the criteria is a choice of loss function. An analyst who fears omitting a real exposure, as a risk manager might, tolerates a spurious variable and leans toward AIC; an analyst who wants a short, stable list of indicators to monitor leans toward BIC.

Stepwise search has well-known weaknesses. The p-values and confidence intervals of the selected model are too optimistic because they ignore the search (Heinze et al., 2018), and choosing among many specifications on the same data invites overstating the evidence, a problem that Harvey et al. (2016) document at length for factors in the cross-section of returns. No single criterion is therefore decisive, and the weight of the conclusion rests on their agreement.

The evidence converges. Every measure other than unadjusted R-squared prefers the core model of the exchange rate, the yield and oil, or at most adds one weak variable to it. Market prices carry the explanatory power, domestic fundamentals do not, and global sentiment adds nothing beyond the prices through which it arrives. The baseline specification is therefore the working model for the sections that follow, with three

caveats: it was selected on the same data, its fit is an in-sample association and not a forecast, and the error assumptions behind its inference remain untested. The diagnostics test those assumptions next, and the model-improvement section returns to the comparison with cross-validated error, a fairer test of predictive value than any in-sample measure.

Section 7: Regression Diagnostics

The inference reported so far rests on assumptions about the errors of the regression. The standard errors, t statistics and F tests are valid only if the error variance is constant and the errors are serially uncorrelated; exact small-sample inference also needs normal errors; the coefficients are unbiased only if the functional form is right; and each coefficient is informative only if the regressors are not too collinear and no single month dominates the fit. This section examines each assumption on the residuals of the nine-regressor baseline model, using the monthly PSEi returns of January 2018 to December 2025.

In testing, we have two working questions. The first is which assumptions hold, and most do: the regressors are not collinear, the residuals are serially uncorrelated and close to normal, and the coefficients are stable across the sample. The second is whether the exceptions are separate defects or one. Three checks flag a problem, namely unequal error variance, a functional form that the data reject and a few influential observations, and a sample that contains the COVID-19 market crash of March 2020 invites the question whether they share a source. The answer decides the remedy, since three independent defects would call for three corrections and one extreme month for a single targeted one. The tests are reported in the order of the assumptions, alongside the residual plots that show where in the sample a problem lies when a test says only that it exists, and the question is taken up when the influential observations are examined.

Heteroskedascticity

Table 32: Heteroskedasticity tests (H0: homoskedastic errors)

test	statistic	df	p_value	decision
Breusch-Pagan (studentized / Koenker)	18.970	9	0.0254	Reject H0
Breusch-Pagan (non-studentized)	21.670	9	0.01	Reject H0
White-type (fitted, fitted^2)	4.163	2	0.125	Fail to reject H0
White (regressors, squares, cross-products)	53.670	54	0.487	Fail to reject H0

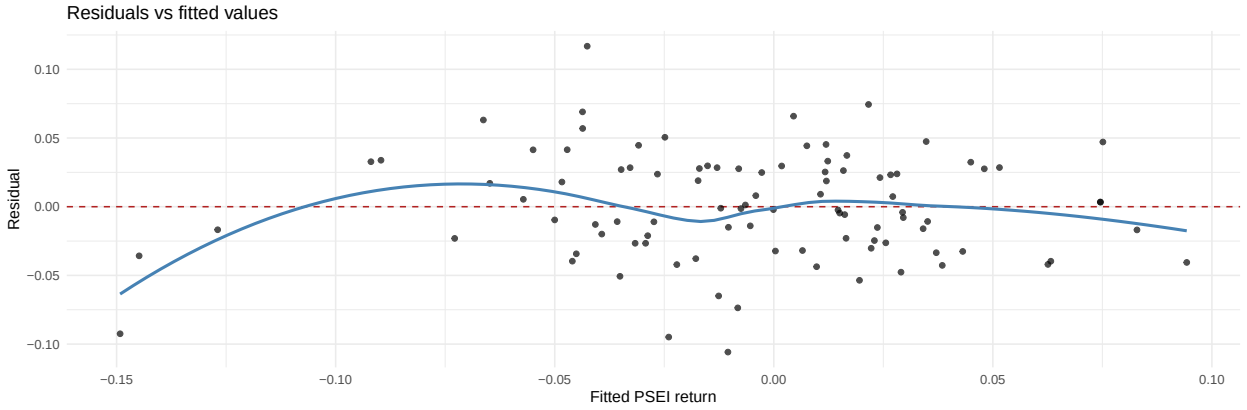


Figure 8: Initial model: residuals against fitted values

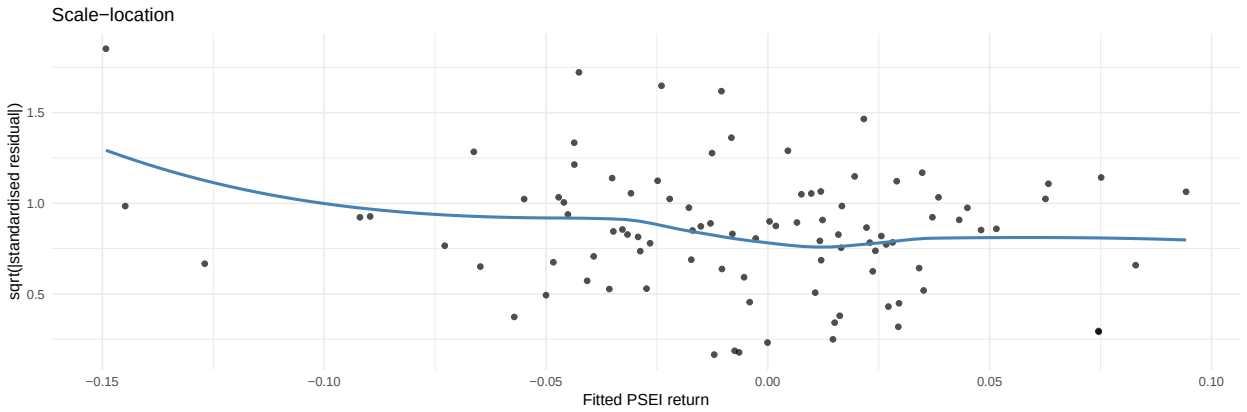


Figure 9: Initial model: scale-location plot, square root of the absolute standardised residuals against fitted values

Constant error variance underpins the classical standard errors. The Breusch-Pagan test (Breusch & Pagan, 1979) in its studentized form (Koenker, 1981) rejects at the 5 percent level, with a statistic of 18.97 on 9 degrees of freedom (p = 0.0254), and so does the non-studentized version (21.67, p = 0.010). The White test (White, 1980) on squares and cross-products of the regressors does not reject (p = 0.487), and neither does a White-type test on the fitted values and their square (p = 0.125).

The disagreement is informative. The White test regresses the squared residuals on 54 terms with only 96 observations, so it has little power against a moderate departure, whereas the Breusch-Pagan test uses 9 degrees of freedom and is the more sensitive test here. Together, they point to unequal variance of moderate strength and not to a gross failure. The plots show where the extra variance lies. The residuals scatter around zero with no strong pattern, but the spread is widest, and the points sparsest, at the lowest fitted returns, and the scale-location curve rises only at its left end. The variance is therefore not growing steadily with the level of the return; it is larger in the few months with the most negative fitted returns, and in the three months from March to May 2020 the residual standard deviation is 6.3 percentage points against 3.8 in all other months.

Unequal variance of this kind is what is expected of equity returns, whose volatility rises in turbulent markets and clusters in time (Engle, 1982). Its consequence is for inference, not estimation: the coefficients remain unbiased, but the classical standard errors can understate their uncertainty, so that a variable looks more reliable than it is. That is a risk above all for exposures estimated from a crisis month, and it is why robust standard errors are applied to this model in the model-improvement section. Whether the extra variance belongs to the crisis months alone is taken up with the influential observations.

Autocorrelation

Table 33: Autocorrelation tests (H0: no serial correlation). DW = 2.105; implied lag-1 correlation ≈ -0.052

test	statistic	df	p_value	decision
Durbin-Watson (H1: positive autocorrelation)	2.1050		0.552	Fail to reject H0
Durbin-Watson (H1: two-sided)	2.1050		0.896	Fail to reject H0
Breusch-Godfrey, order 1	0.4906	1	0.484	Fail to reject H0
Breusch-Godfrey, order 3	1.0310	3	0.794	Fail to reject H0
Breusch-Godfrey, order 6	5.8060	6	0.445	Fail to reject H0
Breusch-Godfrey, order 12	12.2600	12	0.425	Fail to reject H0
Ljung-Box, 6 lags (residuals)	4.9070	6	0.556	Fail to reject H0
Ljung-Box, 12 lags (residuals)	9.1320	12	0.692	Fail to reject H0

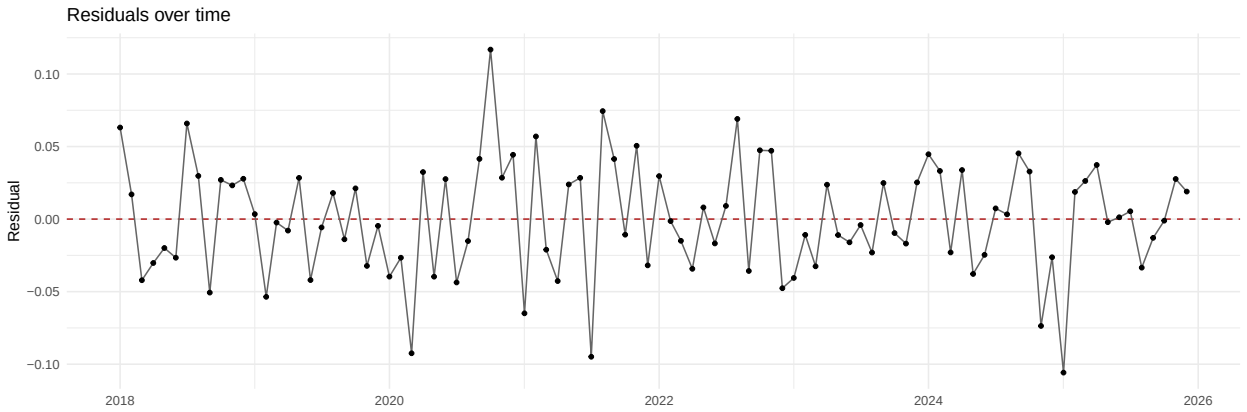


Figure 10: Initial model: residuals in month order

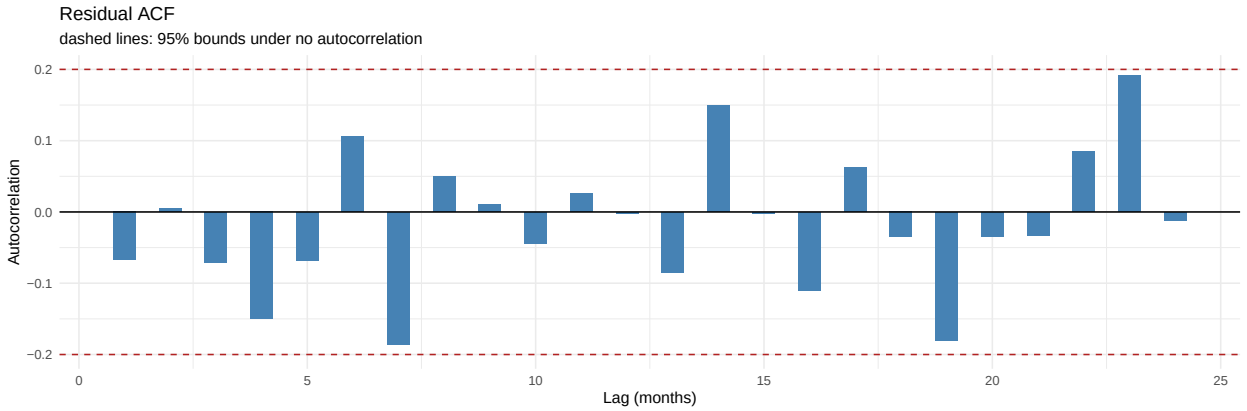


Figure 11: Initial model: autocorrelation function of the residuals

Autocorrelation is relevant here because the data are a monthly time series. If returns, or the drivers of returns that the model omits, persisted from one month to the next, the residuals would be serially correlated and the standard errors would be biased in a direction set by the sign of the correlation. The Durbin-Watson statistic is 2.105 (Durbin & Watson, 1950), against 2 for no autocorrelation, and implies a first-order correlation of about -0.05; its p-values are 0.552 against positive autocorrelation and 0.896 against either sign. The Breusch-Godfrey tests (Breusch, 1978; Godfrey, 1978) at lags 1, 3, 6 and 12 give p-values of 0.484, 0.794, 0.445 and 0.425, and the Ljung-Box tests (Ljung & Box, 1978) at 6 and 12 lags give 0.556 and 0.692. The Durbin-Watson test is designed for first-order dependence, which is why the higher-order Breusch-Godfrey tests and the Ljung-Box tests are added to cover the longer lags that monthly data can show.

The plots agree. The residuals in month order fluctuate around zero without runs of one sign, and none of the first 24 sample auto-correlations lies outside the 95 percent band; the largest, 0.19 at lag 23, is just inside the band of 0.20 and has no economic interpretation.

The result is what is expected of a market that incorporates information quickly. If last month's unexplained return helped to predict this month's, an investor could trade on the pattern. Serial dependence in index returns has been documented at short horizons (Lo & MacKinlay, 1988), but at the monthly frequency used here the residuals show none, so the nine regressors leave nothing systematic of that kind behind. No correction for autocorrelation is needed as a remedy.

Non-Normality

Table 34: Normality tests on residuals (skewness = -0.091, excess kurtosis = 0.284)

test	statistic	df	p_value	decision
Shapiro-Wilk	0.9863		0.421	Fail to reject H0
Jarque-Bera	0.4545	2	0.797	Fail to reject H0

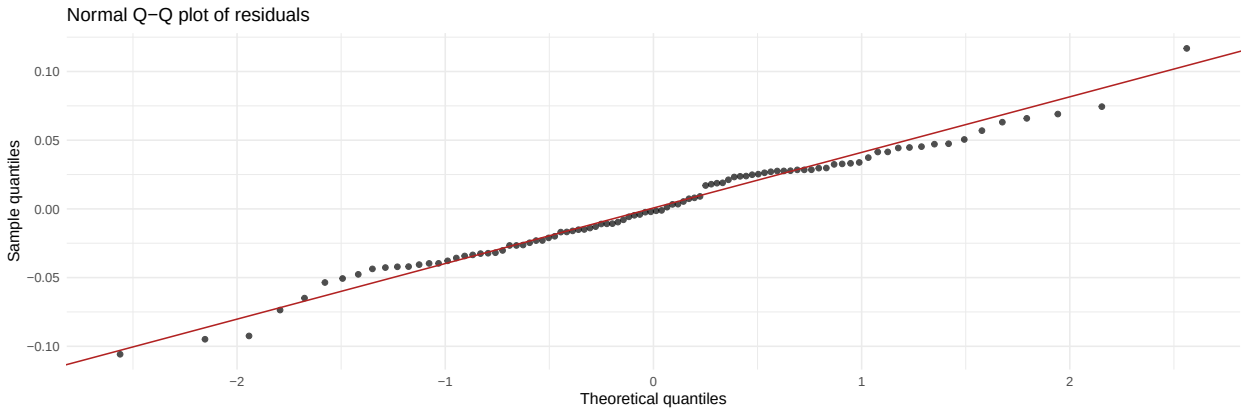


Figure 12: Initial model: normal Q-Q plot of the residuals

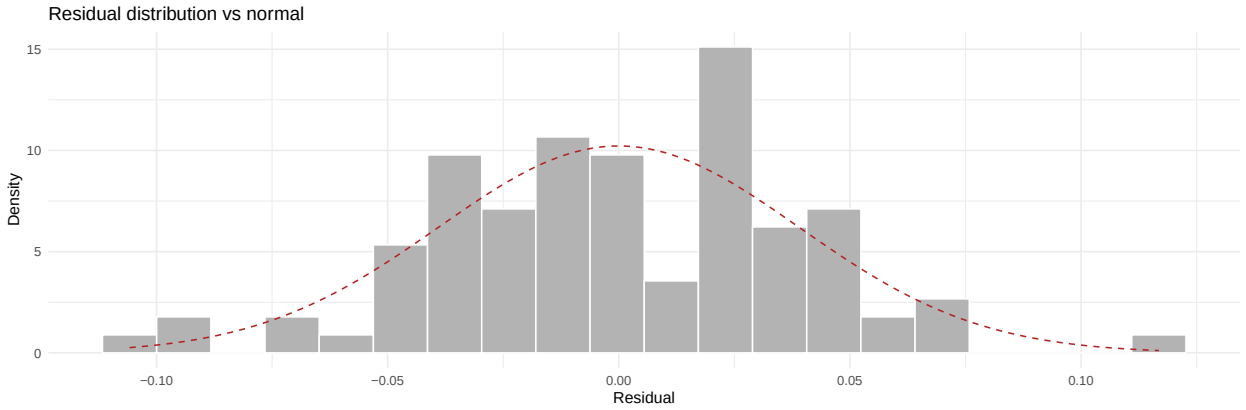


Figure 13: Initial model: residual density against a normal curve

Normality matters for exact t and F inference in a sample of 96 months. The Shapiro-Wilk test (Shapiro & Wilk, 1965) gives 0.9863 (p = 0.421) and the Jarque-Bera test (Jarque & Bera, 1980) gives 0.4545 (p = 0.797); neither rejects. Non-normal errors would not bias the coefficients, but they would make the t and F p-values approximate and could signal the fat tails that undermine risk statements built on normality. The residual skewness is -0.09 and the excess kurtosis 0.28, and the normal Q-Q plot follows the reference line closely, with small departures only at the two extremes.

These figures are striking beside those of the raw returns, whose skewness is -0.94 and excess kurtosis 2.0. The long left tail of the PSEi return, the mark of crash months, has almost disappeared from the residuals, which means that much of that tail is associated with the regressors, chiefly the exchange rate, the yield and oil, which move with the index in a sell-off. A large part of the left-tail risk is therefore systematic with

respect to observable prices and not idiosyncratic, and for an investor, that is the part that can be monitored and hedged.

The distribution of the residuals raises no concern for inference. The largest studentized residual, -3.7, belongs to March 2020, but one month does not disturb the tests, and the t and F distributions are reliable for 96 observations with errors this close to normal. Nothing in this check calls for a remedy.

Functional Form (Ramsey RESET)

Table 35: Ramsey RESET specification tests (H0: correct functional form)

test	statistic	df	p_value	decision
RESET: fitted^2	4.346	1, 85	0.0401	Reject H0
RESET: fitted^2, fitted^3	4.093	2, 84	0.0201	Reject H0
RESET: fitted^2 to fitted^4	5.407	3, 83	0.0019	Reject H0
RESET: squares of regressors	1.289	9, 77	0.257	Fail to reject H0

The RESET test (Ramsey, 1969) asks whether powers of the fitted values add explanatory power. If they do, the linear specification has missed non-linearity or an omitted variable, and the coefficients may be biased. Adding the square of the fitted values gives $F = 4.35$ ($p = 0.040$), adding the square and cube gives 4.09 ($p = 0.020$), and adding powers two to four gives 5.41 ($p = 0.002$), so all three reject linearity. Adding the squares of the regressors themselves gives 1.29 ($p = 0.257$) and does not. RESET detects a misspecification without saying what the right form is, so a rejection has to be read together with the plots and the influence measures.

The split in the results is informative. The tests that reject work through the fitted values, which are dominated by the exchange rate, the yield and oil, while the test on the squares of the individual regressors finds no curvature in any one relationship, such as a convex response to oil. A rejection of the first kind can reflect a genuinely different functional form or a few observations that a linear fit cannot accommodate. The residual plot favors the second reading: its smoother bends mainly at the sparse left end, where a handful of months with the most negative fitted returns determine its shape.

The distinction matters for what an investor should do. If the relation between returns and the regressors were truly non-linear, for example if the index responded more than proportionally to large exchange-rate moves, exposures estimated from ordinary months would understate the risk in stress. If instead the rejection comes from one episode, the exposures are sound for ordinary months and the episode needs separate treatment. The two readings call for different remedies, a transformation or interaction terms in the first case and an adjustment for the episode in the second, and the result is carried to the model-improvement section.

Multicollinearity (VIF)

Table 36: Variance inflation factors (mean VIF = 1.544)

variable	VIF	tolerance	R2_on_ others	flag
PH_CPI	2.164	0.462	0.538	low (< 5)
BSP_RRP_Rate	2.014	0.496	0.504	low (< 5)
PH_M3	1.751	0.571	0.429	low (< 5)
SP500	1.680	0.595	0.405	low (< 5)
WTI	1.382	0.723	0.277	low (< 5)
USD_PHP	1.315	0.761	0.239	low (< 5)
VIX	1.258	0.795	0.205	low (< 5)
PH_10Y_GS_Yield	1.256	0.796	0.204	low (< 5)
PH_Manufacturing_Production	1.072	0.933	0.067	low (< 5)

Table 37: Condition number of the standardised regressors

statistic	value
Largest / smallest singular value (condition number)	2.89

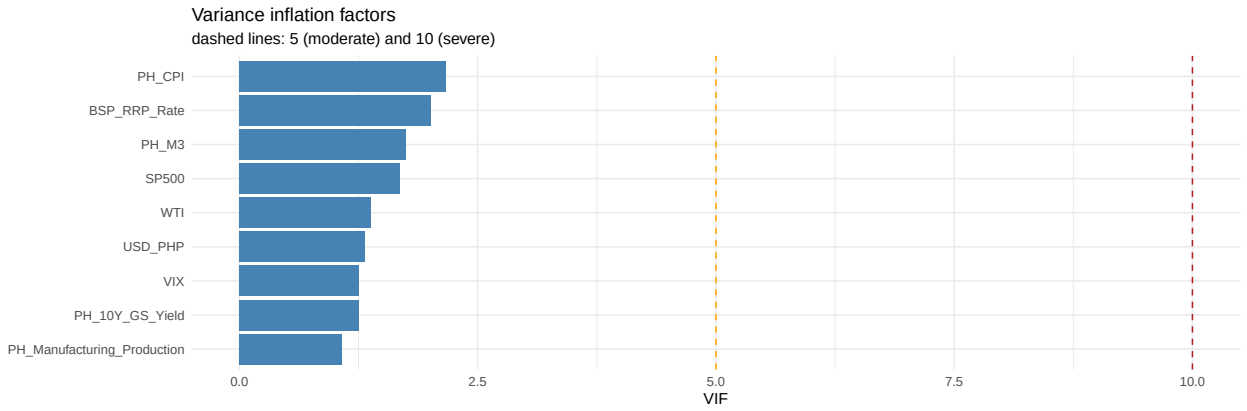


Figure 14: Variance inflation factor of each regressor (dashed lines = 5, moderate, and 10, severe)

Multicollinearity does not bias the coefficients but inflates their standard errors, so that a regressor that matters can look insignificant. Two kinds of evidence are used. The correlations among the regressors are modest: the largest, 0.66, is between inflation and the change in the policy rate, while M3 growth correlates at -0.56 with inflation and -0.51 with the policy rate. The variance inflation factors confirm this. Their mean is 1.54, the largest are inflation (2.16), the policy rate (2.01) and M3 growth (1.75), and all nine lie well below the conventional warning level of 5, as does the condition number of the standardized regressors (2.9, against a warning level of 30). A factor of 2.16 means that the variance of that coefficient is 2.16 times what it would be if the regressor were uncorrelated with the others, a standard error about 47 percent larger. These thresholds are rules of thumb and not tests (O'Brien, 2007), but the margin here is wide.

The three largest factors belong to the three domestic monetary variables, the group that the exploratory correlation analysis singled out, and the overlap has a clear economic source: a central bank raises its policy rate when inflation is high and money growth is slowing, so the three series carry overlapping information

about the monetary regime. The overlap is real but mild. The imprecise coefficients of these variables in the baseline model therefore reflect how little they say about monthly returns, not an inability to separate them from one another, and no regressor needs to be dropped or combined to cure multicollinearity.

Influential Observations

Table 38: Influence measures: cut-offs and number of flagged months

measure	cutoff	n_flagged	max_value
Cook's distance	0.04167	7	1.5530
Leverage	0.20830	10	0.5686
|Studentised residual|	2.00000	5	3.6740
|DFFITS|	0.64550	7	4.2180
Any |DFBETAS|	0.20410	22	2.3260

Table 39: Ten most influential months (by Cook's distance)

year_month	cooks_d	leverage	rstudent	dffits	n_dfbetas_ flagged	flag_cook	flag_hat	flag_rstu	flag_ dffits	n_flags
2020-03-01	1.5500	0.5690	-3.67	-4.220	7	TRUE	TRUE	TRUE	TRUE	4
2021-07-01	0.2810	0.2750	-2.83	-1.740	2	TRUE	TRUE	TRUE	TRUE	4
2021-08-01	0.1840	0.2860	2.19	1.390	2	TRUE	TRUE	TRUE	TRUE	4
2020-05-01	0.0916	0.3780	-1.23	-0.960	3	TRUE	TRUE	FALSE	TRUE	3
2020-10-01	0.0750	0.0785	3.11	0.909	3	TRUE	FALSE	TRUE	TRUE	3
2022-10-01	0.0741	0.2840	1.37	0.865	7	TRUE	TRUE	FALSE	TRUE	3
2022-11-01	0.0497	0.2260	1.31	0.708	2	TRUE	TRUE	FALSE	TRUE	3
2020-04-01	0.0404	0.3090	0.95	0.635	3	FALSE	TRUE	FALSE	FALSE	1
2018-01-01	0.0403	0.1290	1.67	0.641	4	FALSE	FALSE	FALSE	FALSE	0
2023-01-01	0.0393	0.2350	-1.13	-0.628	3	FALSE	TRUE	FALSE	FALSE	1

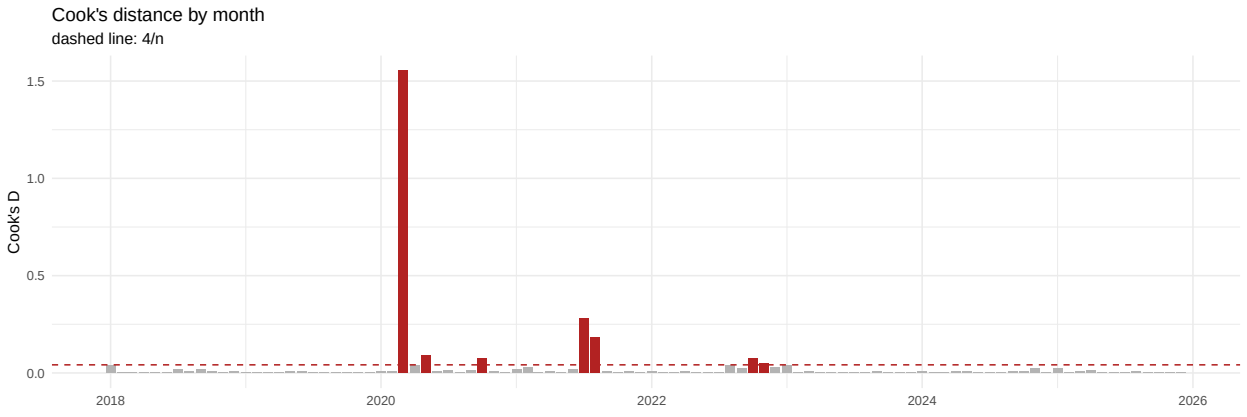


Figure 15: Initial model: Cook's distance by month (dashed line = 4/n)

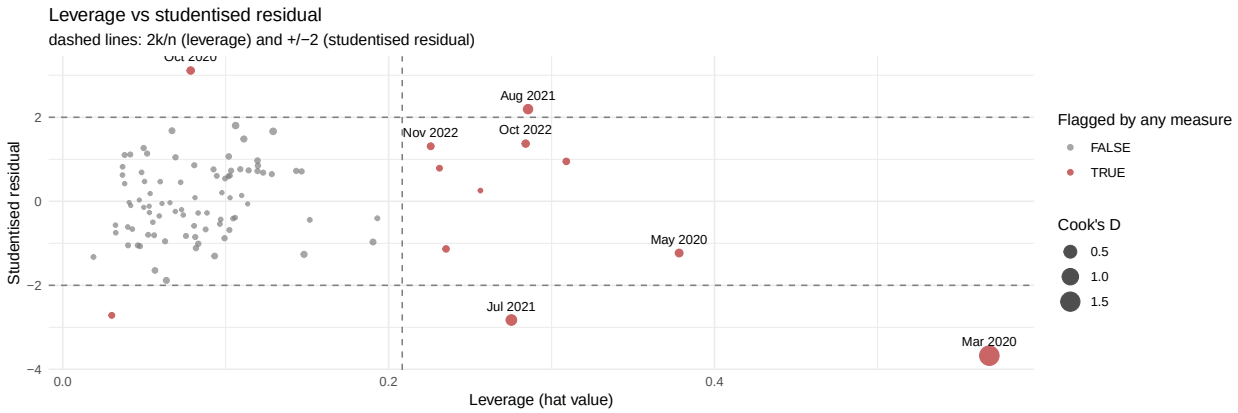


Figure 16: Initial model: leverage against studentised residual, point size = Cook's distance (dashed lines = 2k/n and +/-2)

A regression can be sound on average and still be dominated by a few observations. Four measures are used: Cook's distance (Cook, 1977), which gauges how far the fitted values move when a month is dropped; leverage, which measures how unusual a month's regressors are; the studentized residual, which measures how unusual its return is given the regressors; and DFFITS, which scales the change in a month's own fitted value. At the usual cut-offs, seven months exceed the Cook's distance cut-off of 0.042, ten exceed the leverage cut-off of 0.208, five have studentized residuals beyond 2 in absolute value, and seven exceed the DFFITS cut-off of 0.65.

One month stands apart. March 2020 has a Cook's distance of 1.55, more than five times the next largest (0.28, July 2021); its leverage of 0.57 is nearly three times the cut-off and five times the average of 0.10; and its studentized residual is -3.67 and its DFFITS -4.22. It is not the month with the largest raw residual. The model already assigns it a fitted return of about -15 percent against an actual -24 percent, so its residual of about -9 points is smaller than those of three other months. It is influential because of its leverage and not because it is badly fitted: the fit has bent toward it.

The reason is financial. In March 2020 the S&P 500 had its largest fall, the VIX its highest level and WTI its largest fall of the whole sample, all in the same month, so the month lies far from the other 95 in the space of the regressors. Global shocks arrive together in a crisis, and the regression must estimate its response to them from a single observation. The plot of leverage against studentized residual shows the result, with the other months in a cluster at low leverage and March 2020 alone at the right. The remaining flagged months, May and October 2020 and July and August 2021, are far less influential, with Cook's distances below 0.3.

Table 40: DFBETAS by coefficient (cut-off = 2/sqrt(n) = 0.204)

term	n_flagged	max_abs	worst_month
(Intercept)	7	1.510	2020-03
USD_PHP	7	0.355	2018-01
PH_10Y_GS_Yield	6	0.493	2020-03
WTI	3	2.330	2020-03
SP500	6	0.408	2022-10
VIX	7	1.970	2020-03
BSP_RRP_Rate	6	0.441	2020-03
PH_CPI	6	0.397	2023-01
PH_M3	8	0.521	2020-10
PH_Manufacturing_Production	3	1.610	2021-07



Figure 17: Initial model: DFBETAS by month for each coefficient (dashed lines = the cut-off of +/-2/sqrt(n))

DFBETAS show which coefficients a month moves. Twenty-two months move at least one coefficient by more than the cut-off of 0.20 standard errors, but March 2020 is in a class of its own: it is the most influential month for five of the ten coefficients, and dropping it would shift the WTI coefficient by 2.3 standard errors and the VIX coefficient by 2.0. The exceptions show that the picture is not about one month alone. The manufacturing production coefficient is moved most, by 1.6 standard errors, by July 2021, and the exchange-rate coefficient, the largest in economic terms, is moved by at most 0.35 standard errors by any month.

Two readings follow forward. The exposures to the exchange rate, and to a lesser extent the yield, are robust to any single month. The apparent sensitivity of returns to oil and to the VIX, by contrast, is largely a statement about March 2020, since dropping that one month would move those coefficients by about two standard errors, enough to change a conclusion. These are exactly the exposures an investor might hedge on the strength of the regression, and they rest on one single observation.

The influential months should not be deleted. March 2020 is a real event, the COVID-19 crash, and not a

data error, and deleting it would remove the episode in which risk matters most. The question is how to let the model account for the month without letting it dictate the other coefficients.

The three problems found so far now line up. Unequal variance is concentrated in the crisis months, the fitted-value RESET rejections coincide with a fit bent toward a single point, and the only observation of real influence is that same month. Whether they are three symptoms of one episode is a hypothesis and not yet a result. It is tested directly in the model-improvement section, where a single treatment of March 2020 is tried against all three.

Parametric Stability

Table 41: Stability tests (H0: stable coefficients)

test	statistic	df	p_value	decision
OLS-CUSUM	0.5089		0.958	Fail to reject H0
Recursive CUSUM	0.7022		0.246	Fail to reject H0
Chow: after Mar 2020 (COVID-19 shock)	1.6550		0.107	Fail to reject H0
Chow: after Dec 2021 (sample midpoint)	0.5490		0.85	Fail to reject H0
Chow: after Apr 2022 (start of BSP tightening)	0.4456		0.919	Fail to reject H0
sup-F (unknown break date)	20.1400		0.306	Fail to reject H0

Table 42: Estimated break (first segment ends in this month)

statistic	value
Most likely break date from the sup-F search	2020-08

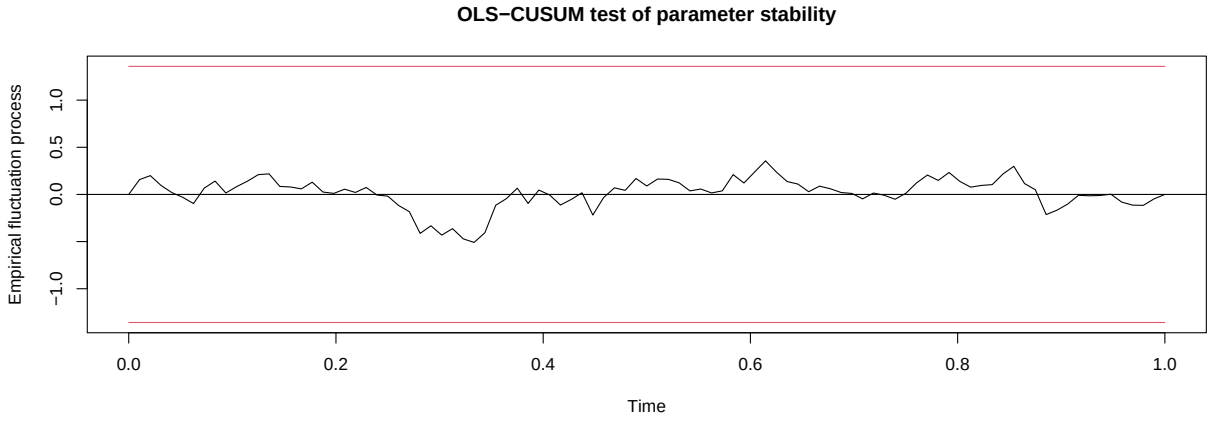


Figure 18: Initial model: OLS-CUSUM test of parameter stability

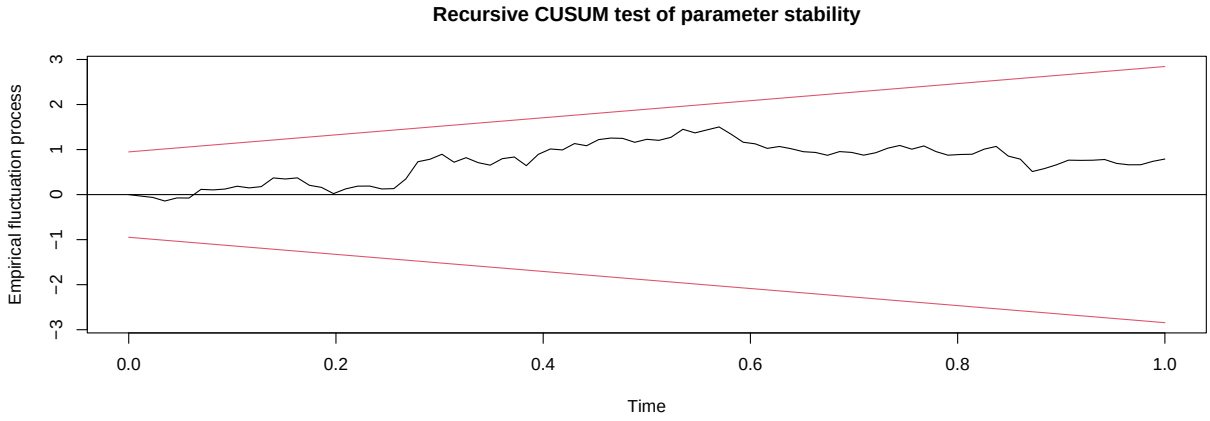


Figure 19: Initial model: recursive CUSUM test of parameter stability

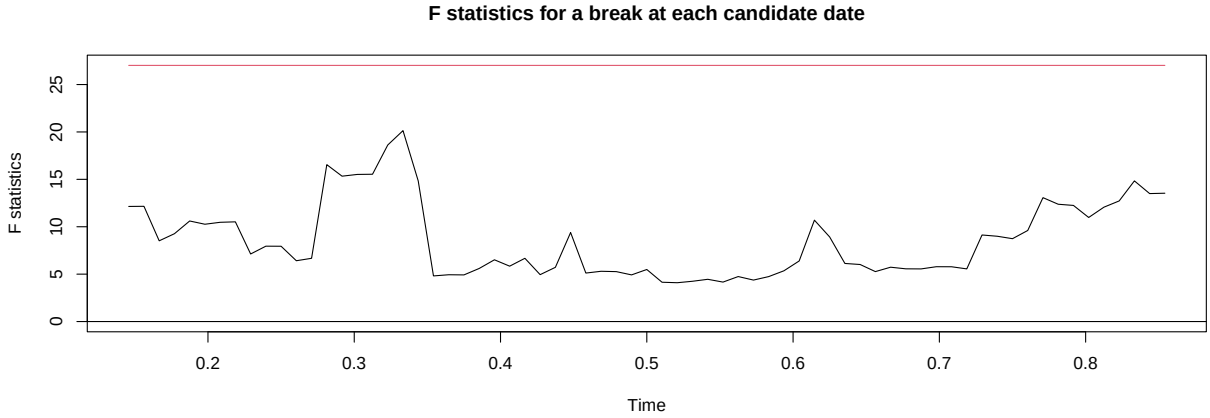


Figure 20: Initial model: F statistics for a structural break at each candidate date

The last assumption is that the relationship between the regressors and returns is the same throughout the eight years. If it changed, for instance because the pandemic or the tightening of monetary policy from 2022 altered how the market responds to the exchange rate or to yields, one set of coefficients would describe none of the sub-periods. The tests cover different ways in which a relationship can change: the CUSUM tests are sensitive to gradual drift, the Chow test to a break at a known date and the sup-F test to a break at an unknown date. Neither CUSUM test comes close to rejecting (OLS-CUSUM p = 0.958; recursive CUSUM p = 0.246). The Chow tests at the three dates where a break is economically plausible, after March 2020, after December 2021 and after April 2022, give p-values of 0.107, 0.850 and 0.919, and the sup-F test (Andrews, 1993) gives a statistic of 20.14 and p = 0.306. The search statistic peaks in August 2020, in the aftermath of the crash, but stays well below the 5 percent critical value at every candidate date.

The stability and influence results fit together. Had the pandemic changed the way the market responds to the regressors, the tests would reject and the sample would need a break dummy or a split. They do not, so we read the crash as an exceptional observation within a stable relationship and not as a new regime, and a one-month adjustment is a better remedy than a split sample. For the investor the implication is reassuring: the exposures estimated over 2018 to 2025 show no sign of having changed across the pandemic and the tightening cycle.

Diagnostics Summary

Table 43: Diagnostics summary for the initial model (alpha = 0.05)

area	test	result	verdict	action
Multicollinearity	Largest VIF (flag at 5)	2.16	No problem	None needed
	Condition number (flag at 30)	2.9	No problem	None needed
Heteroskedasticity	Breusch-Pagan (studentized)	p = 0.0254	Problem	HC3 robust standard errors (Section 8)
	White (squares and cross-products)	p = 0.487	No problem	None needed
Autocorrelation	Durbin-Watson (two-sided)	p = 0.896	No problem	None needed
	Breusch-Godfrey, order 12	p = 0.425	No problem	None needed
	Ljung-Box, 12 lags	p = 0.692	No problem	None needed
Normality	Shapiro-Wilk	p = 0.421	No problem	None needed
Specification	RESET (fitted^2, fitted^3)	p = 0.0201	Problem	March 2020 dummy (Section 8)
Influence	Cook's distance above 4/n	7 months, largest 1.55	Problem	March 2020 dummy and robust regression (Section 8)
Stability	OLS-CUSUM	p = 0.958	No problem	None needed
	sup-F (unknown break date)	p = 0.306	No problem	None needed

Section 8: Model Improvement Approaches

The diagnostics of Section 7 found the initial model, the nine-regressor OLS fit of Section 5, sound in most respects. There is no multicollinearity (the largest variance inflation factor is 2.16), no autocorrelation (the Durbin-Watson, Breusch-Godfrey and Ljung-Box p-values are all above 0.4), no departure from normality (Shapiro-Wilk p = 0.42) and no sign of unstable coefficients (OLS-CUSUM p = 0.96, sup-F p = 0.31). Three findings do need attention, as the diagnostics summary table shows. The Breusch-Pagan test rejects constant error variance (p = 0.025), although the White test does not (p = 0.49). The RESET test rejects the linear functional form (p = 0.020). And seven months have a Cook's distance above 4/n, one of them, March 2020, at 1.55, against 0.28 for the next largest.

Each problem calls for a remedy that fits its nature. Unequal error variance leaves the OLS coefficients unbiased and distorts only their standard errors, so we correct the standard errors with the HC3 estimator of MacKinnon & White (1985), a refinement of the heteroskedasticity-consistent covariance matrix of White (1980). Weighted least squares would need a model of how the variance depends on the regressors, which a rejection by one of two tests cannot identify. Autocorrelation is absent, and Newey-West standard errors, which are compared with HC3 below, change no significance decision, so they are not needed as a remedy. Further transformation is not called for, because the regressors already have the stationary forms built in Section 3. The influence and specification findings call for a respecification, a dummy variable for March 2020, which we cross-check with robust regression (Huber, 1973; Yohai, 1987) and compare with a reduced model.

We analyze on one central finding. March 2020, the month of the COVID-19 market crash, has the highest leverage of the sample (0.57, against the 2k/n cut-off of 0.21), and the S&P 500, the VIX and WTI all take their most extreme values in it. We therefore ask of every remedy what survives once the pull of that month is controlled, and judge each by four criteria: valid inference, stable estimates, accuracy on held-out months, and parsimony. The Remedy Summary applies these criteria to the initial model and to all the remedies together.

Robust Standard Errors (HC3)

Table 44: Initial model with HC3 standard errors (95% CIs use HC3)

term	estimate	se_ classical	se_hc3	se_ratio	t_hc3	p_ classical	p_hc3	ci95_low	ci95_high	decision_ changes
(Intercept)	6.58e-03	1.93e-02	3.28e-02	1.700	0.200	0.734	0.842	-0.058700	0.071900	FALSE
USD_PHP	-2.05e+00	3.04e-01	2.95e-01	0.971	-6.940	<0.0001	<0.0001	-2.630000	-1.460000	FALSE
PH_10Y_GS_Yield	-4.05e-02	1.28e-02	1.39e-02	1.090	-2.920	0.00215	0.00453	-0.068200	-0.012900	FALSE
WTI	1.30e-01	3.65e-02	8.86e-02	2.430	1.460	0.000617	0.147	-0.046400	0.306000	TRUE
SP500	1.34e-01	1.14e-01	1.20e-01	1.050	1.120	0.244	0.267	-0.104000	0.373000	FALSE
VIX	-2.43e-04	6.98e-04	1.47e-03	2.100	-0.166	0.729	0.869	-0.003160	0.002670	FALSE
BSP_RRP_Rate	-1.20e-02	2.71e-02	3.12e-02	1.150	-0.385	0.659	0.701	-0.074100	0.050000	FALSE
PH_CPI	6.36e-04	3.31e-03	3.22e-03	0.973	0.197	0.848	0.844	-0.005770	0.007040	FALSE
PH_M3	-1.12e-03	8.60e-04	1.06e-03	1.230	-1.060	0.195	0.292	-0.003230	0.000984	FALSE
PH_Manufacturing_Production	3.31e-05	4.46e-05	9.34e-05	2.090	0.354	0.461	0.724	-0.000153	0.000219	FALSE

HC3 leaves the OLS coefficients as they were and changes only the standard errors, so it asks whether the conclusions of the baseline analysis depend on constant error variance. For eight of the nine regressors they do not. The standard errors of USD/PHP and of the 10-year yield move by less than 10 percent, and both stay significant ($p < 0.0001$ and $p = 0.0045$). The exception is WTI: its standard error rises by a factor of 2.4 and its p-value moves from 0.0006 to 0.147, so the significance it showed in the baseline model does not survive. The standard errors of the VIX and of manufacturing production roughly double as well, by a factor of about 2.1, although neither was significant to begin with. The overall F-test still rejects the null that all slopes are zero ($F = 11.1$, against 12.8 under classical errors), so the model remains jointly significant.

The finding concerns WTI, and the question is why its significance depends on the estimator. HC3 corrects for unequal variance, but the classical standard errors also assume serially uncorrelated errors, and the Newey-West estimator (Newey & West, 1987) relaxes both. Comparing the classical, HC3 and Newey-West standard errors therefore separates two explanations: if a decision changes under HC3 alone, the cause is unequal variance or the leverage of particular months, and if it changes under Newey-West as well, serial dependence is also involved. The comparison below holds the coefficients fixed and varies only the covariance matrix, using a Newey-West bandwidth chosen from the data and, as a check, a fixed lag of six months.

Table 45: Does the significance decision (5%) change with the estimator? (Newey-West automatic bandwidth = 3.22)

term	SE ratio HC3 / classical	SE ratio NW (auto) / classical	sig_ classical	sig_hc3	sig_nw_ auto	sig_nw_ lag6
BSP_RRP_Rate	1.150	0.756	FALSE	FALSE	FALSE	FALSE
PH_10Y_GS_Yield	1.085	0.785	TRUE	TRUE	TRUE	TRUE
PH_CPI	0.973	0.783	FALSE	FALSE	FALSE	FALSE
PH_M3	1.233	1.042	FALSE	FALSE	FALSE	FALSE
PH_Manufacturing_Production	2.094	0.737	FALSE	FALSE	FALSE	FALSE
SP500	1.049	0.983	FALSE	FALSE	FALSE	FALSE
USD_PHP	0.971	0.780	TRUE	TRUE	TRUE	TRUE
VIX	2.101	0.940	FALSE	FALSE	FALSE	FALSE
WTI	2.428	1.016	TRUE	FALSE	TRUE	TRUE

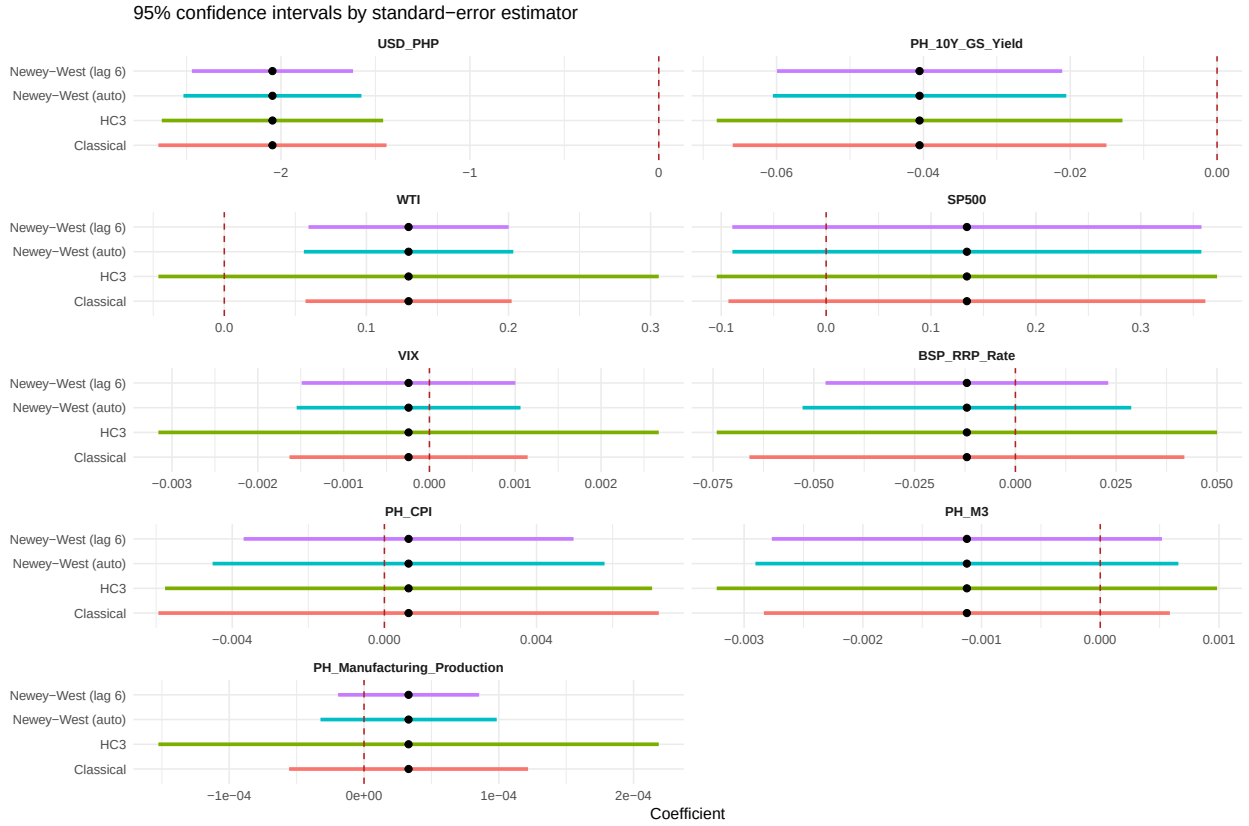


Figure 21: 95% confidence intervals of each coefficient by standard-error estimator (dashed line = 0)

At the 5 percent level, the decision is the same under all four estimators for eight of the nine regressors: the exchange rate and the yield are significant under every estimator, and the other six regressors are insignificant under every estimator. WTI is the only regressor whose decision depends on the estimator. It is significant under the classical and both Newey-West estimators and insignificant under HC3. The standard errors identify the source. The Newey-West standard errors of the exchange rate and the yield are about a fifth smaller than the classical ones and that of WTI is essentially unchanged, so allowing for serial dependence weakens no conclusion, whereas HC3 raises the standard error of WTI by a factor of 2.4. The interval figure

shows the same pattern, with the WTI interval crossing zero under HC3 alone. The fragility of WTI is therefore a matter of unequal variance and influential months and not of autocorrelation, which agrees with the autocorrelation tests in the diagnostics and is why Newey-West is not carried forward as a remedy.

HC3 is the appropriate estimator here for two reasons. It does not require the form of the heteroskedasticity to be known, which the diagnostic tests could not identify, and it is the variant that behaves best in samples below about 250 observations (Long & Ervin, 2000; MacKinnon & White, 1985), where this study has 96. It also treats the month at issue in the right way. HC3 divides each squared residual by the square of one minus the month's leverage, so March 2020, with the highest leverage of the sample, receives about 5.4 times the weight of a month with leverage near zero, which is why the standard errors of WTI and the VIX, whose most extreme values fall in that month, are the ones that inflate. Crisis episodes raise the variance of returns and can distort conventional measures of co-movement (Forbes & Rigobon, 2002). The correction is nonetheless partial, since it repairs the standard errors but leaves the estimates, and the pull of March 2020 on them, untouched. The next remedy addresses the month directly.

March 2020 Dummy Solution

Table 46: Model with a March 2020 dummy (HC3 from the 95-month fit without March 2020; the dummy row uses its classical SE)

term	estimate	se_ classical	se_hc3	t_hc3	p_ classical	p_hc3	ci95_low	ci95_high
(Intercept)	-2.07e-02	1.95e-02	1.87e-02	-1.110	0.292	0.272	-0.057900	0.016500
USD_PHP	-1.98e+00	2.84e-01	2.88e-01	-6.850	<0.0001	<0.0001	-2.550000	-1.400000
PH_10Y_GS_Yield	-3.46e-02	1.21e-02	1.21e-02	-2.870	0.00521	0.00513	-0.058600	-0.010700
WTI	5.04e-02	4.04e-02	3.65e-02	1.380	0.215	0.171	-0.022100	0.123000
SP500	1.48e-01	1.07e-01	1.13e-01	1.310	0.171	0.193	-0.076400	0.372000
VIX	1.04e-03	7.40e-04	7.39e-04	1.410	0.162	0.161	-0.000426	0.002510
BSP_RRP_Rate	-2.32e-02	2.55e-02	2.63e-02	-0.884	0.366	0.379	-0.075500	0.029000
PH_CPI	9.62e-04	3.09e-03	3.02e-03	0.318	0.757	0.751	-0.005050	0.006970
PH_M3	-7.42e-04	8.10e-04	9.74e-04	-0.762	0.362	0.448	-0.002680	0.001190
PH_Manufacturing_Production	3.84e-05	4.17e-05	9.24e-05	0.416	0.36	0.679	-0.000145	0.000222
MARCH_2020	-2.14e-01	5.84e-02	NA	NA	0.000417	NA	NA	NA

Table 47: Effect of the March 2020 dummy on the estimates (significance by HC3 p-values)

term	est_ initial	p_initial	est_march	p_march	pct_change	sig_ initial	sig_march
(Intercept)	6.58e-03	0.842	-2.07e-02	0.272	-414.0	FALSE	FALSE
USD_PHP	-2.05e+00	<0.0001	-1.98e+00	<0.0001	3.4	TRUE	TRUE
PH_10Y_GS_Yield	-4.05e-02	0.00453	-3.46e-02	0.00513	14.6	TRUE	TRUE
WTI	1.30e-01	0.147	5.04e-02	0.171	-61.1	FALSE	FALSE
SP500	1.34e-01	0.267	1.48e-01	0.193	10.1	FALSE	FALSE
VIX	-2.43e-04	0.869	1.04e-03	0.161	530.0	FALSE	FALSE
BSP_RRP_Rate	-1.20e-02	0.701	-2.32e-02	0.379	-93.0	FALSE	FALSE
PH_CPI	6.36e-04	0.844	9.62e-04	0.751	51.3	FALSE	FALSE
PH_M3	-1.12e-03	0.292	-7.42e-04	0.448	33.9	FALSE	FALSE
PH_Manufacturing_Production	3.31e-05	0.724	3.84e-05	0.679	16.2	FALSE	FALSE

Table 48: Is the March 2020 dummy significant?

test	F	p_value
Nested F-test: add MARCH_2020 (H0: coefficient = 0)	13.5	0.000417

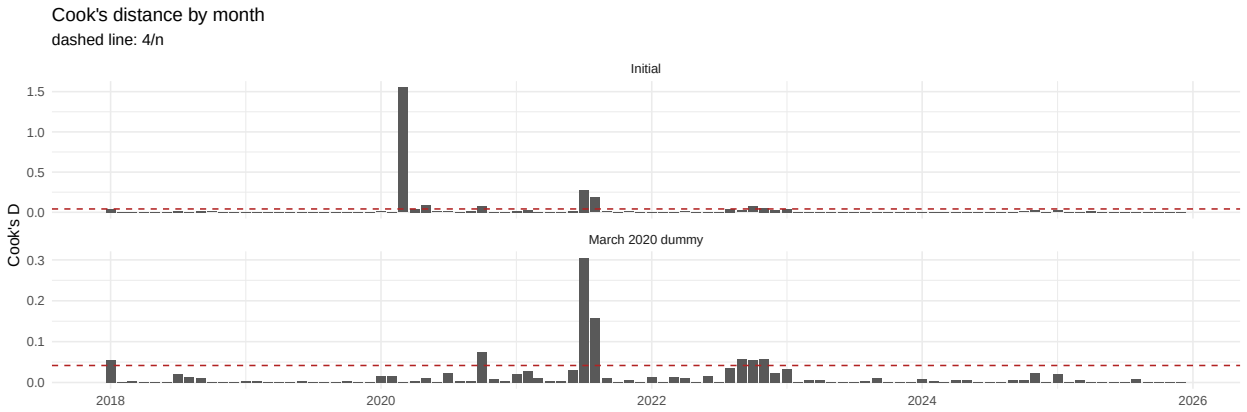


Figure 22: Cook's distance by month, initial model against the model with a March 2020 dummy (dashed line = 4/n)

Table 49: Most influential months after the March 2020 dummy

year_month	cooks_d
2021-07-01	0.3060
2021-08-01	0.1570
2020-10-01	0.0736
2022-11-01	0.0585
2022-09-01	0.0581

Where HC3 corrects the standard errors, we present the dummy to correct the estimates. March 2020 is a real and identifiable event, the COVID-19 crash, so deleting it as though it were a data error would be hard to defend, while leaving it alone lets it dominate the fit. A dummy variable for the month avoids both choices: it gives the month its own intercept, the other coefficients then equal those of a fit without the month (Salkever, 1976), and the month stays in the sample with its effect estimated. The dummy coefficient is -0.214 ($p = 0.0004$), and the nested F-test confirms that it belongs in the model ($F = 13.5$, $p = 0.0004$). Read as a prediction error, it says that the PSEi log return of -24.2 percent in March 2020 fell about 21 percentage points below the -2.7 percent that the other 95 months predict from the same regressors.

The dummy resolves the three problems found in the diagnostics together. The Breusch-Pagan p-value rises from 0.025 to 0.129, the RESET p-value from 0.020 to 0.573, and the largest Cook's distance falls from 1.55 to 0.31, now belonging to July 2021, about a fifth of what March 2020 had; the adjusted R-squared rises from 0.528 to 0.588. The cost is one degree of freedom and our judgment that one month is exceptional. The effect on the coefficients is the more important result. The exchange-rate coefficient changes by 3 percent and the yield coefficient by 15 percent, and both stay significant under HC3, but the WTI coefficient falls by 61 percent, from 0.130 to 0.050, and is no longer significant ($p = 0.17$). Its apparent effect came from March 2020 and not from the other 95 months: a 10 percent rise in oil goes with a 1.3 percentage-point higher monthly return in the baseline model and with 0.5 once the month is controlled.

Robust Regression

Table 50: OLS vs Robust Regression (OLS p-values use HC3)

term	OLS	p_OLS_hc3	Huber	p_Huber	MM	p_MM
(Intercept)	6.58e-03	0.842	2.38e-03	0.895	-0.015500	0.398
USD_PHP	-2.05e+00	<0.0001	-2.00e+00	<0.0001	-1.940000	<0.0001
PH_10Y_GS_Yield	-4.05e-02	0.00453	-3.87e-02	0.00164	-0.035800	0.00399
WTI	1.30e-01	0.147	1.04e-01	0.00279	0.054000	0.121
SP500	1.34e-01	0.267	1.59e-01	0.138	0.155000	0.154
VIX	-2.43e-04	0.869	1.63e-04	0.802	0.001020	0.127
BSP_RRP_Rate	-1.20e-02	0.701	-2.03e-02	0.422	-0.025100	0.332
PH_CPI	6.36e-04	0.844	9.57e-05	0.975	0.000150	0.962
PH_M3	-1.12e-03	0.292	-1.33e-03	0.101	-0.000968	0.237
PH_Manufacturing_Production	3.31e-05	0.724	6.01e-05	0.15	0.000080	0.0612

Table 51: Residual scale: robust fits are less inflated by outliers

estimator	residual_ scale
OLS	0.04102
Huber	0.03863
MM	0.03737

Table 52: Ten most downweighted months (weight 1 = full weight)

year_month	residual_ ols	weight_ huber	weight_mm
2020-03-01	-0.0925	0.406	0.000
2021-07-01	-0.0949	0.475	0.288
2025-01-01	-0.1060	0.485	0.408
2020-10-01	0.1170	0.464	0.518
2024-11-01	-0.0736	0.680	0.672
2018-01-01	0.0631	0.797	0.702
2021-01-01	-0.0649	0.825	0.718
2018-07-01	0.0659	0.792	0.734
2022-08-01	0.0690	0.762	0.747
2021-02-01	0.0570	0.862	0.782

Robust regression gives a second answer to the same question without naming the month in advance. The Huber estimator downweights observations with large residuals (Huber, 1973), and the MM estimator also resists high-leverage points (Yohai, 1987); the weights show which months each treats as outliers. Huber lowers March 2020 to a weight of 0.41 and keeps it in the sample, while MM gives it a weight of zero, which treats the month as the dummy does. The residual scale falls from 0.041 under OLS to 0.039 under Huber and 0.037 under MM, which shows how much the outlying months inflated the error variance.

Two regressors give the same answer as in the dummy model and the third a graded one. USD/PHP (-2.00 under Huber, -1.94 under MM) and the 10-year yield (-0.039 and -0.036) are significant under both estimators, with estimates close to the OLS values. WTI shows how the answer depends on the treatment of the month. Under MM, which discards March 2020, its coefficient is 0.054 and insignificant ($p = 0.12$), as in

the dummy model; under Huber, which keeps the month at a reduced weight, it is 0.104 and significant (p = 0.003), between the OLS value and the dummy value. The significance of WTI therefore fades as the weight of March 2020 goes to zero, which is what one expects if the oil effect is carried by that single observation.

Reduced Model

Table 53: Reduced model with the March 2020 dummy (HC3 from the 95-month fit without March 2020)

term	estimate	se_ classical	se_hc3	t_hc3	p_ classical	p_hc3	ci95_low	ci95_high
(Intercept)	0.000429	0.00396	0.00400	0.107	0.914	0.915	-0.00752	0.00837
USD_PHP	-2.040000	0.25800	0.24200	-8.440	<0.0001	<0.0001	-2.52000	-1.56000
PH_10Y_GS_Yield	-0.040800	0.01120	0.00965	-4.230	0.000458	<0.0001	-0.06000	-0.02170
WTI	0.066700	0.03620	0.03430	1.940	0.0688	0.055	-0.00147	0.13500
MARCH_2020	-0.179000	0.04830	NA	NA	0.000364	NA	NA	NA

Table 54: Can the six regressors be dropped?

test	F	p_value
Nested F-test: drop the six insignificant regressors (H0: all zero)	0.9827	0.442

The last remedy is a statement about parsimony and not a direct repair of a diagnostic problem. The reduced model keeps the three regressors retained in the model-evaluation section, USD/PHP, the 10-year yield and WTI, and adds the March 2020 dummy. The nested F-test does not reject that the six omitted regressors are jointly zero (F = 0.98, p = 0.44), so little is lost by dropping them, and the fit is no worse with six fewer parameters: the adjusted R-squared is 0.588, equal to that of the dummy model, and the AIC (-347.0) and BIC (-331.7) are the lowest of all the OLS models, the BIC by 21 points. The residual diagnostics are clean (Breusch-Pagan p = 0.33, RESET p = 0.60, Shapiro-Wilk p = 0.99), and only two months have a Cook's distance above 4/n, the largest being 0.06.

The coefficients are close to those of the full model with the dummy and are estimated more tightly. USD/PHP is -2.04 and the yield -0.041, both with HC3 p-values below 0.0001, and the dummy is -0.179, against -0.214 in the full model. WTI is 0.067 with an HC3 p-value of 0.055, on the 5 percent line, so the model keeps WTI because the information criteria and the nested tests favor it and not because the evidence for it is firm. We apply two cautions. The three regressors were chosen on this same sample, so the fit statistics and the cross-validated error reported next are somewhat optimistic (Heinze et al., 2018). And a reduced model is a statement about parsimony, so it must keep the dummy and the HC3 standard errors to carry the same protection as the full model.

Remedy Summary

Table 55: Fit and 10-fold cross-validated RMSE (robust fits: cross-validated error only)

model	n_ regressors	adj_r_ squared	AIC	BIC	cv_rmse_ all	cv_rmse_ excl_ mar2020
Initial (OLS)	9	0.5277	-329.3	-301.1	0.04936	0.04442
March 2020 dummy	10	0.5876	-341.5	-310.7	0.04732	0.04212
Robust: Huber	NA	NA	NA	NA	0.04845	0.04386
Robust: MM	NA	NA	NA	NA	0.04729	0.04245
Reduced (core 3 + dummy)	4	0.5881	-347.0	-331.7	0.04280	0.03877

Table 56: Diagnostics re-run on each OLS model (Breusch-Pagan, RESET, Shapiro-Wilk, Durbin-Watson p-values)

model	bp_p	reset_p	shapiro_p	dw_p	max_cooks_ d	n_cooks_ flagged
Initial (OLS)	0.0254	0.0201	0.421	0.896	1.5500	7
March 2020 dummy	0.1290	0.5730	0.903	0.731	0.3060	7
Reduced (core 3 + dummy)	0.3310	0.6030	0.991	0.755	0.0586	2

Table 57: Coefficients by model (p-values: HC3 for OLS rows; the March dummy rows use classical)

term	Initial (OLS)	March 2020 dummy	Reduced (core 3 + dummy)	Robust: Huber	Robust: MM
(Intercept)	0.00658 (p=0.842)	-0.02068 (p=0.272)	0.000429 (p=0.915)	0.002381 (p=0.895)	-0.01549 (p=0.398)
USD_PHP	-2.045 (p=<0.0001)	-1.976 (p=<0.0001)	-2.041 (p=<0.0001)	-2.004 (p=<0.0001)	-1.944 (p=<0.0001)
PH_10Y_GS_Yield	-0.04052 (p=0.00453)	-0.03462 (p=0.00513)	-0.04083 (p=<0.0001)	-0.03866 (p=0.00164)	-0.03584 (p=0.00399)
WTI	0.1297 (p=0.147)	0.05039 (p=0.171)	0.06668 (p=0.055)	0.1043 (p=0.00279)	0.05403 (p=0.121)
SP500	0.1342 (p=0.267)	0.1477 (p=0.193)	NA	0.1591 (p=0.138)	0.1555 (p=0.154)
VIX	-0.000243 (p=0.869)	0.001044 (p=0.161)	NA	0.0001629 (p=0.802)	0.001018 (p=0.127)
BSP_RRP_Rate	-0.01203 (p=0.701)	-0.02323 (p=0.379)	NA	-0.02031 (p=0.422)	-0.02507 (p=0.332)
PH_CPI	0.0006358 (p=0.844)	0.000962 (p=0.751)	NA	9.572e-05 (p=0.975)	0.00015 (p=0.962)
PH_M3	-0.001124 (p=0.292)	-0.0007423 (p=0.448)	NA	-0.001325 (p=0.101)	-0.0009684 (p=0.237)
PH_Manufacturing_Production	3.305e-05 (p=0.724)	3.842e-05 (p=0.679)	NA	6.008e-05 (p=0.15)	8.004e-05 (p=0.0612)
MARCH_2020	NA	-0.2144 (p=0.000417)	-0.1791 (p=0.000364)	NA	NA

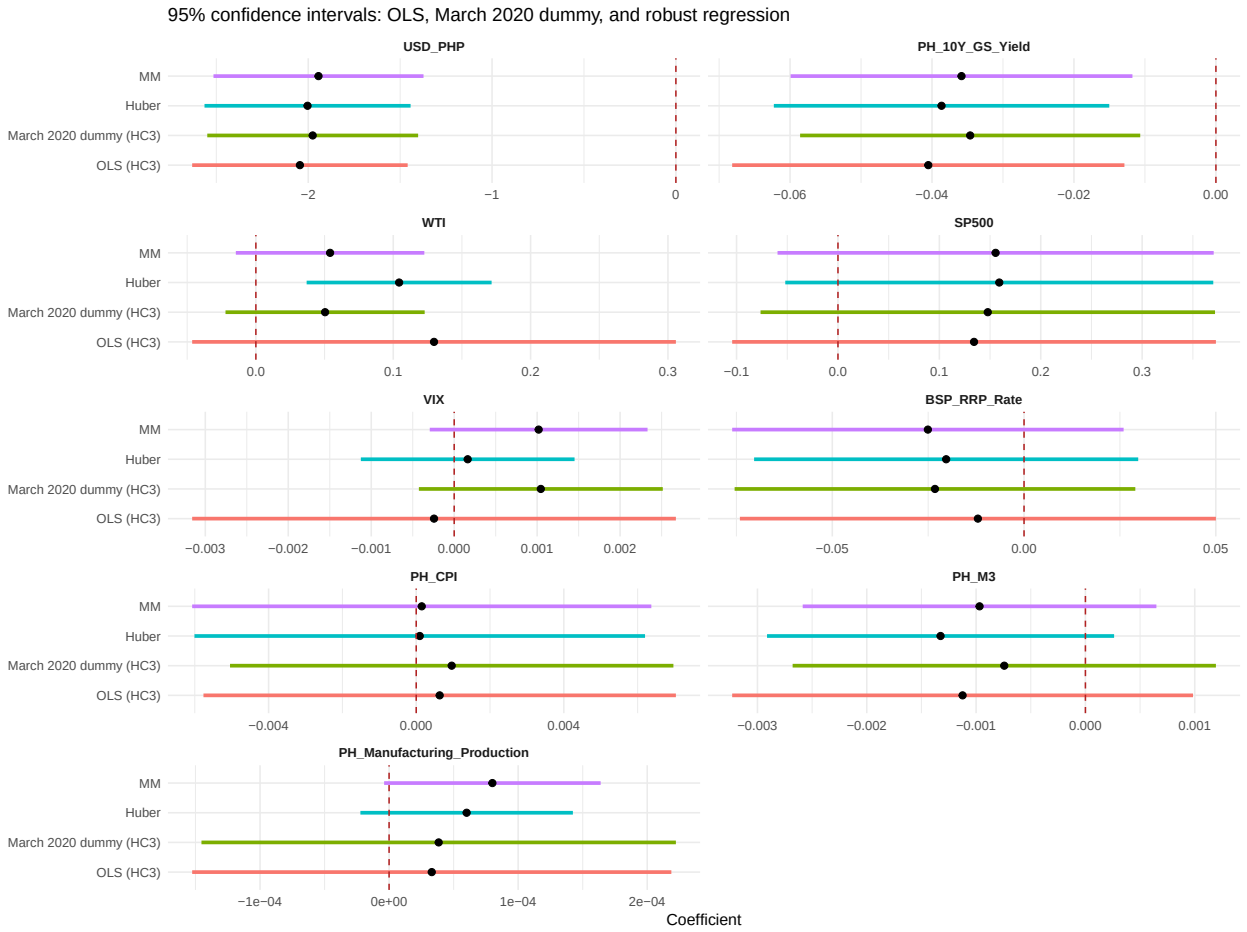


Figure 23: 95% confidence intervals of each coefficient: OLS (HC3), March 2020 dummy (HC3), Huber and MM robust regression (dashed line = 0)

Judged by the four criteria, the remedies line up in a clear order. On valid inference, HC3 removes the unreliable significance of WTI, and the dummy and the reduced model carry the correction forward. On stable estimates, the dummy and the reduced model remove the dependence on March 2020: between the initial and the reduced model the Breusch-Pagan p-value rises from 0.025 to 0.33, the RESET p-value from 0.020 to 0.60, and the largest Cook's distance falls from 1.55 to 0.06. On accuracy for held-out months, the 10-fold cross-validated RMSE falls from 0.0494 for the initial model to 0.0473 for the dummy model and 0.0428 for the reduced model, a reduction of 13 percent, and the ranking is the same when March 2020 is excluded from the error (0.0444, 0.0421 and 0.0388). Of the robust fits, MM matches the dummy model (0.0473) and Huber is slightly worse (0.048). On parsimony, the reduced model has four regressors against ten and the lowest BIC. The reduced model with the March 2020 dummy and HC3 standard errors is therefore the model carried forward, with the caution that its regressors were chosen on the same data, so cross-validation is a fairer test than any in-sample statistic but not an independent one.

What the comparison means for an investor is clearest coefficient by coefficient. The exposure to the exchange rate is reliable: a one percent depreciation of the peso goes with a monthly return about 2.0 percentage points lower under every remedy, with estimates between -1.94 and -2.05 and significance throughout. The exposure to the 10-year yield is also reliable, a 10 basis point rise going with a return 0.35 to 0.41 percentage points lower. The exposure to oil is conditional. The baseline relationship, a 10 percent rise in oil with a 1.3 percentage-point higher return, is carried by March 2020, and once the month is controlled the effect is about 0.5 points and no longer distinguishable from zero, so an oil hedge should not be sized from the baseline coefficient. The other six regressors remain unestablished, since the evidence neither supports nor

rules out effects. One limit remains. The dummy rests on the judgment that one month is exceptional, and that judgment has a cost, which falls on tail risk that a model of the average return cannot describe. The next section therefore turns from the size of the average return to the probability of a rising or sharply falling month.

Section 9: Logistic Regression

Sections 5 to 8 model how large the monthly PSEi return is. The linear model explains the conditional mean and, once March 2020 is handled, finds that the peso and the 10-year yield carry most of the explanatory power. We further show that an investor also needs to know how likely a month is to be an up month and how likely it is to be a crash, and the mean of a return distribution says little about its tail. The PSEi return is left-skewed (skewness -0.94, Section 4), so the factors that move a typical month need not be the ones that separate a crash from the rest. This section answers that question with logistic regression, which models the log-odds of a yes/no outcome and is the advanced regression method of the report. Model A takes the value 1 when the PSEi return is positive (48 of the 96 months) and Model B when it falls below -5% (21 months, 22%), a threshold whose sensitivity is tested below. Both use the same 96 months and the same nine regressors as the linear model, so any difference in results comes from the question asked and not from the data.

The logit is the right tool for three reasons. First, the outcomes are binary, and a linear model of a 0/1 variable can predict probabilities outside the range of 0 to 1 and has unequal error variance by construction, whereas the logit keeps probabilities inside the range and states each effect as a multiplier of the odds, which is how risk is usually discussed. Second, the question is a financial one: binary crash models with macroeconomic and financial regressors are standard in the literature on currency crashes and early-warning systems for emerging markets (Bussiere & Fratzscher, 2006; Frankel & Rose, 1996). Third, the data support it, because both outcomes are defined from series already in the dataset and 21 crash months are enough for a core model of three regressors (7 events per regressor) if the estimator is chosen with care. That care is the Firth penalized likelihood (Firth, 1993; Heinze & Schemper, 2002) for Model B. With 2.3 events per regressor in the full model, far below the rule of 10 (Peduzzi et al., 1996), ordinary maximum likelihood is biased, and the penalty removes the bias and keeps the estimates finite even if the outcome were perfectly separated. Model A, with 48 events, needs only the standard logit. Turning a return into a yes/no outcome discards the size of the move, which is the price of asking the question, and the linear model of the earlier sections, which keeps that information, should be read alongside it.

The other advanced methods were considered and are not needed, because the data do not show the problem that each one solves. Regularization, whether ridge (Hoerl & Kennard, 1970) or the lasso (Tibshirani, 1996), helps when regressors are collinear or too many for the sample. Here the largest variance inflation factor is 2.16 and the condition number 2.9 (Section 7), nine regressors leave 86 residual degrees of freedom, and the nested tests of Sections 6 and 8 already show that the six weakest regressors can be dropped together (F-test $p = 0.44$), so selection is handled without a penalty parameter that 96 months could not estimate reliably. Nonlinear functional forms were the concern raised by the RESET test ($p = 0.020$), but the rejection disappeared once a March 2020 dummy was added ($p = 0.573$), which points to one month and not to curvature, and the regressors are already log returns and changes. Interaction effects have no anchor in the channels of Section 2, and with 96 months any interaction would rest on a handful of observations. The only categorical variable the data call for is the March 2020 event dummy, used in Section 8, where robust estimation was also applied (HC3 standard errors, Huber and MM regression). Panel methods need several cross-sectional units, while this study follows one market. We stress that forcing any of these into the analysis would add parameters without answering a question that the data raise.

As in the linear model, the analysis is explanatory: the regressors are contemporaneous and the models describe what moves with the market in the same month, not what will move it next month. The subsections below estimate and interpret Model A and Model B, check their diagnostics, the choice of threshold and the influence of March 2020, test the fit on months the models did not see, and compare the two models with each other and with the linear model.

Table 58: Binary outcomes (EPV = events per regressor; below 10 is thin, below 5 is very thin)

outcome	n	events	non_events	pct_events	epv_full	epv_core
A. Up month	96	48	48	50.0	5.3	16
B. Crash month (return < -5%)	96	21	75	21.9	2.3	7

Logistic Regression A: Up-month logit

Table 59: A. Up-month logit, full model (log-odds; odds ratios are per 1 unit of each regressor)

term	estimate	std_error	p_value	p_hc3	odds_ratio	or_low	or_high	signif
(Intercept)	-0.457300	1.317000	0.728	0.74	0.63300	0.046020	8.2770	
USD_PHP	-1.042000	0.272800	0.000134	0.00381	0.35290	0.193700	0.5716	***
PH_10Y_GS_Yield	-2.663000	0.995100	0.00744	0.0241	0.06972	0.008032	0.4235	**
WTI	0.030710	0.027080	0.257	0.243	1.03100	0.984200	1.0950	
SP500	0.040050	0.073000	0.583	0.63	1.04100	0.903800	1.2080	
VIX	0.045010	0.050960	0.377	0.381	1.04600	0.948100	1.1590	
BSP_RRP_Rate	2.057000	1.992000	0.302	0.379	7.82200	0.169700	454.0000	
PH_CPI	-0.042870	0.213900	0.841	0.845	0.95800	0.622400	1.4550	
PH_M3	-0.030820	0.051890	0.553	0.621	0.96960	0.873500	1.0730	
PH_Manufacturing_Production	0.001959	0.002621	0.455	0.53	1.00200	0.996700	1.0070	

Table 60: A. Up-month logit, core model

term	estimate	std_error	p_value	p_hc3	odds_ratio	or_low	or_high	signif
(Intercept)	0.11940	0.25640	0.641	0.652	1.12700	0.68210	1.8800	
USD_PHP	-0.93890	0.23160	<0.0001	<0.0001	0.39110	0.23620	0.5913	***
PH_10Y_GS_Yield	-2.45000	0.89530	0.0062	0.00301	0.08626	0.01246	0.4372	**
WTI	0.03506	0.02456	0.153	0.113	1.03600	0.99460	1.0940	

Table 61: A. Likelihood-ratio test against the intercept-only model, pseudo-R-squared and information criteria

spec	LR_chisq	df	p_value	mcfadden_r2	nagelkerke_r2	AIC	BIC
full (9 regressors)	44.48	9	<0.0001	0.3342	0.4945	108.6	134.2
core (USD_PHP, 10Y, WTI)	40.55	3	<0.0001	0.3047	0.4594	100.5	110.8

Table 62: A. Can the six non-core regressors be dropped?

test	chisq	df	p_value
Nested LR test: drop the six non-core regressors (H0: all zero)	3.934	6	0.686

Table 63: A. Odds of an UP month for a realistic move (core model; OR below 1 lowers the odds)

model	term	move	odds_ratio	or_low	or_high
A. Up month	USD_PHP	USD_PHP +1%	0.391	0.236	0.591
A. Up month	PH_10Y_GS_Yield	10Y yield +10 bp	0.783	0.645	0.921
A. Up month	WTI	WTI +5%	1.190	0.973	1.570

Model A asks what separates a month in which the PSEi rises from one in which it falls. The full model is jointly significant (likelihood-ratio chi-square(9) = 44.5, p < 0.0001; McFadden R-squared 0.33) and ranks months well: its in-sample AUC of 0.87 means that a randomly chosen up month receives a higher predicted probability than a randomly chosen down month in 87% of pairs. Only two regressors matter. A 1% depreciation of the peso multiplies the odds of an up month by 0.39 (95% CI 0.24 to 0.59) and a 10 basis point rise in the 10-year yield multiplies them by 0.78 (95% CI 0.64 to 0.92), and both remain significant with HC3 standard errors (p = 0.004 and p = 0.024). WTI, the S&P 500, the VIX, inflation, M3, manufacturing production and the policy rate do not change the odds, and dropping all six non-core regressors is not rejected (chi-square = 3.93, p = 0.69). The core model that keeps the peso, the yield and WTI therefore fits almost as well (McFadden R-squared 0.30) and is preferred by the information criteria (AIC 100.5 against 108.6, BIC 110.8 against 134.2). In financial terms, an ordinary down month is a month in which the peso weakens and long-term yields rise, the same two channels that the linear model found, now expressed as odds.

Table 64: A. Variance inflation factors (regressors as in the MLR)

statistic	value
Largest VIF	1.962
Mean VIF	1.388

Table 65: A. Calibration by quintile of predicted probability (full model)

bin	n	mean_ predicted	observed_ rate	events	expected
1	20	0.0789	0.100	2	1.58
2	19	0.2730	0.158	3	5.18
3	19	0.5080	0.526	10	9.66
4	19	0.7370	0.842	16	14.00
5	19	0.9250	0.895	17	17.60

Table 66: A. Goodness-of-fit, functional-form and autocorrelation tests

test	statistic	df	p_value	decision
Hosmer-Lemeshow, 5 groups (H0: well calibrated)	2.7560000	3	0.431	Fail to reject H0
Link test: squared linear predictor (H0: logit link adequate)	0.0001668	1	0.99	Fail to reject H0
Ljung-Box, 12 lags (deviance residuals)	5.1280000	12	0.954	Fail to reject H0

Table 67: A. Five most influential months (Cook's D cut-off 4/n = 0.042)

year_month	cooks_d	psei_ return	up
2022-12-01	0.1650	-0.0186	0
2021-02-01	0.1150	0.0133	1
2021-06-01	0.1130	0.0155	1
2022-08-01	0.0634	0.0254	1
2020-07-01	0.0510	-0.0339	0

The diagnostics give no reason to doubt Model A. The regressors are not collinear (largest VIF 1.96, mean 1.39), the Hosmer-Lemeshow test does not reject calibration (p = 0.43), the squared linear predictor adds nothing, so the logit link is adequate (p = 0.99), and the deviance residuals show no autocorrelation (Ljung-Box p = 0.95). In the calibration table the predicted and observed up-month rates rise together, from roughly 8% and 10% in the lowest quintile to roughly 93% and 89% in the highest, although the second quintile is over-predicted (27% against 16%), a gap of about two months in a group of 19. No month is influential: the largest Cook's distance, 0.165 for December 2022, is far below 1, and March 2020 is not among the five largest, because a binary outcome records only that the market fell and not by how much.

Logistic Regression B: Crash-month logit (Firth penalized likelihood)

Table 68: B. Crash-month logit (Firth), full model: penalized-likelihood p-values and 95% CIs

term	estimate	std_error	p_value	ci_low	ci_high	odds_ratio	signif
(Intercept)	0.825700	1.630000	0.64	-2.625000	4.530000	2.2830	
USD_PHP	0.883500	0.246400	<0.0001	0.418200	1.484000	2.4190	***
PH_10Y_GS_Yield	2.300000	1.119000	0.0398	0.099240	5.019000	9.9750	*
WTI	-0.068850	0.030800	0.0267	-0.140700	-0.007210	0.9335	*
SP500	-0.106400	0.091880	0.289	-0.316400	0.090870	0.8991	
VIX	-0.094310	0.062260	0.155	-0.243300	0.034470	0.9100	
BSP_RRP_Rate	3.156000	2.191000	0.173	-1.369000	8.072000	23.4900	
PH_CPI	-0.367800	0.269200	0.189	-0.995300	0.173200	0.6923	
PH_M3	0.036280	0.063420	0.591	-0.099700	0.171300	1.0370	
PH_Manufacturing_Production	0.001669	0.002559	0.549	-0.004152	0.007102	1.0020	

Table 69: B. Crash-month logit (Firth), core model

term	estimate	std_error	p_value	ci_low	ci_high	odds_ratio	signif
(Intercept)	-2.07400	0.40620	<0.0001	-3.0520	-1.346000	0.1257	***
USD_PHP	0.97340	0.25260	<0.0001	0.5129	1.574000	2.6470	***
PH_10Y_GS_Yield	2.00900	0.97440	0.0372	0.1112	4.260000	7.4550	*
WTI	-0.05736	0.02983	0.0155	-0.1422	-0.008719	0.9443	*

Table 70: B. Penalized likelihood-ratio test against the intercept-only model and pseudo-R-squared (AIC/BIC are not defined for a penalized fit)

spec	LR_chisq	df	p_value	mcfadden_r2	nagelkerke_r2
full (9 regressors)	35.76	9	<0.0001	0.4213	0.5500
core (USD_PHP, 10Y, WTI)	34.72	3	<0.0001	0.3762	0.5021

Table 71: B. Firth vs maximum likelihood. ML converged: TRUE; warnings: 0; max fitted probability: 0.997

term	ml_estimate	firth_estimate	shrinkage_pct
(Intercept)	1.19000	0.82600	30.4
USD_PHP	1.10000	0.88300	19.7
PH_10Y_GS_Yield	2.96000	2.30000	22.4
WTI	-0.09100	-0.06890	24.3
SP500	-0.12900	-0.10600	17.7
VIX	-0.11800	-0.09430	19.8
BSP_RRP_Rate	3.77000	3.16000	16.2
PH_CPI	-0.48400	-0.36800	24.1
PH_M3	0.04090	0.03630	11.3
PH_Manufacturing_Production	0.00185	0.00167	10.0

Table 72: B. Odds of a CRASH month for a realistic move (core model; OR above 1 raises the odds)

model	term	move	odds_ratio	or_low	or_high
B. Crash month	USD_PHP	USD_PHP +1%	2.650	1.670	4.830
B. Crash month	PH_10Y_GS_Yield	10Y yield +10 bp	1.220	1.010	1.530
B. Crash month	WTI	WTI +5%	0.751	0.491	0.957

Model B asks what separates a crash month, a return below -5%, from the other 75. The Firth fit is jointly significant (penalized likelihood-ratio chi-square(9) = 35.8, $p < 0.0001$; McFadden R-squared 0.42; AUC 0.91), and its pattern mirrors Model A with the signs reversed, as it should: the drivers that lower the odds of an up month raise the odds of a crash. A 1% depreciation of the peso multiplies the odds of a crash month by 2.65 (95% CI 1.67 to 4.83) and a 10 basis point rise in the 10-year yield by 1.22 (95% CI 1.01 to 1.53), the peso effect being the stronger and the more precisely estimated. WTI enters with a negative sign: a 5% rise in oil multiplies the odds of a crash by 0.75 (95% CI 0.49 to 0.96), so crashes tend to coincide with falling oil, the pattern of a demand shock and not of the cost channel that makes oil a burden for an importer. The up-month model found no oil effect, so oil is a tail variable, a point tested below because it rests on one month.

Our choice of the Firth estimator is supported by the data. The comparison table shows that the ordinary maximum-likelihood fit converges without warning and that its highest fitted probability is 0.997, so there is no complete separation, yet every coefficient is 10% to 30% larger in absolute value than its penalized counterpart (the peso coefficient by 20%, the yield by 22% and oil by 24%). This is the small-sample bias that Firth (1993) corrects and that Peduzzi et al. (1996) warn about when events per variable are few, so maximum likelihood would have overstated every effect. Puhr et al. (2017) note that the penalty can inflate predicted probabilities when events are very rare; with crashes in 22% of months this concern is small, and the calibration check below shows no systematic over-prediction in the quintiles that hold the crashes.

Table 73: B. Crash-threshold sensitivity (core model, Firth)

threshold	cutoff	events	LR_p	auc	b_USD_PHP	p_USD_PHP	b_10Y	p_10Y	b_WTI	p_WTI
-4%	-0.0400	24	<0.0001	0.911	1.060	<0.0001	3.22	0.00158	-0.0877	0.00575
-5%	-0.0500	21	<0.0001	0.891	0.973	<0.0001	2.01	0.0372	-0.0574	0.0155
-6%	-0.0600	12	0.000629	0.844	0.500	0.0146	1.94	0.0588	-0.0391	0.0442
bottom 20%	-0.0536	19	<0.0001	0.868	0.773	0.000128	2.29	0.0178	-0.0614	0.0152

Table 74: B. Calibration by quintile of predicted probability (full model)

bin	n	mean_ predicted	observed_ rate	events	expected
1	20	0.00964	0.0000	0	0.193
2	19	0.05340	0.0526	1	1.010
3	19	0.12400	0.0000	0	2.360
4	19	0.29600	0.3160	6	5.620
5	19	0.69200	0.7370	14	13.100

Table 75: B. Five highest-leverage months (Firth hat values; 2k/n = 0.208)

year_month	hat	psei_ return	crash
2020-03-01	0.776	-0.2420	1
2021-07-01	0.406	-0.1190	1
2021-06-01	0.326	0.0155	0
2022-08-01	0.310	0.0254	0
2018-10-01	0.299	-0.0078	0

Three checks test whether Model B depends on how it is set up. First, the -5% threshold is arbitrary, and the sensitivity table shows that it matters little. At -4% (24 crash months), at -5% and at the bottom 20% of months (a cut-off of -5.4%, 19 months) the peso, yield and oil effects are all significant at the 5% level. At -6% (12 months) the peso effect stays significant (p = 0.015), oil does too (p = 0.044) and the yield is borderline (p = 0.059), which is what a dozen events can support. The in-sample AUC falls from 0.91 to 0.84 as the events thin out. Second, calibration is reasonable where it matters: in the two quintiles that hold the crashes the model expects 5.6 and 13.1 crashes and observes 6 and 14, and it over-predicts only in two low-risk quintiles in which no crash occurred (0.2 and 2.4 expected), the scatter to be expected with 19 months per group. Third, March 2020 has a Firth leverage of 0.78 against a cut-off of 0.21, more than three times the cut-off and almost twice the next highest month (July 2021, 0.41), so the fit leans on that month more than on any other, and the next step tests how much.

Table 76: p-values of the core drivers with and without March 2020

model	sample	USD_PHP	PH_10Y_ GS_Yield	WTI
A. Up month	all months	<0.0001	0.0062	0.153
A. Up month	without March 2020	<0.0001	0.00684	0.203
B. Crash month	all months	<0.0001	0.0372	0.0155
B. Crash month	without March 2020	<0.0001	0.0338	0.084

Re-estimating the core models without March 2020 shows which findings are specific to that month. The

peso is not: its p-value is below 0.0001 in both models with and without the month. The yield is almost unchanged (p = 0.006 against 0.007 in Model A and 0.037 against 0.034 in Model B). Oil is different. In Model A, where it was never significant, it stays insignificant (p = 0.20), but in Model B its p-value moves from 0.016 to 0.084, so the effect of oil on the odds of a crash disappears once March 2020 is set aside. The linear model reached the same conclusion by a different route, since WTI was significant there only while March 2020 was given weight (Section 8), and it is the one result of this section that rests on a single episode. A binary outcome does limit the damage that one month can do, because March 2020 counts as one crash like any other, but it does not remove it in Model B, where the month is both a crash and the most leveraged observation. Our reading that follows is that the peso and the yield describe crashes in general, whereas oil describes the pandemic crash.

Cross-Validation and Out-Of-Time Check

Table 77: Repeated stratified 10-fold cross-validation (30 repeats)

outcome	spec	cv_skill	cv_auc	cv_accuracy
A. Up month	full (9 regressors)	0.145	0.797	0.739
A. Up month	core (USD_PHP, 10Y, WTI)	0.254	0.821	0.755
B. Crash month	full (9 regressors)	0.218	0.831	0.823
B. Crash month	core (USD_PHP, 10Y, WTI)	0.318	0.865	0.817

Table 78: Out-of-time check: fit on 2018-2022, test on 2023-2025

outcome	spec	oot_train_n	oot_test_n	oot_events	oot_auc	oot_accuracy	oot_majority
A. Up month	full (9 regressors)	60	36	18	0.827	0.778	0.50
A. Up month	core (USD_PHP, 10Y, WTI)	60	36	18	0.892	0.833	0.50
B. Crash month	full (9 regressors)	60	36	9	0.827	0.833	0.75
B. Crash month	core (USD_PHP, 10Y, WTI)	60	36	9	0.947	0.861	0.75

In-sample fit flatters a model, so the logits are also tested on months they did not see. Repeated stratified ten-fold cross-validation measures skill as the improvement in log-loss over always predicting the base rate. The full models beat the base rate (skill 0.14 for Model A and 0.22 for Model B) and the core models do so by more (0.25 and 0.32), with cross-validated AUCs from 0.80 (Model A, full) to 0.87 (Model B, core). The core figures are the more optimistic because the three core regressors were chosen on the same data, and the full-model figures, which involve no selection, are the fair comparison. The out-of-time check fits the models on 2018-2022 (60 months) and tests them on 2023-2025 (36 months), a later period that they never saw. The AUC holds up (0.83 for both full models, 0.89 and 0.95 for the core models) and accuracy exceeds the base rate in every case (for the full models 0.78 against 0.50 for Model A and 0.83 against 0.75 for Model B), but the test window holds only nine crash months, so the Model B result is indicative only.

Table 79: Same-month versus one-month-lagged regressors (full models)

outcome	regressors	n	LR_p	mcfadden_ r2	auc
A. Up month	same month	96	<0.0001	0.334	0.868
A. Up month	lagged one month	95	0.938	0.027	0.594
B. Crash month	same month	96	<0.0001	0.421	0.909
B. Crash month	lagged one month	95	0.834	0.062	0.662

The lag check separates explanation from prediction. If the regressors anticipated the market, last month's values would explain this month's outcome, and they do not. Lagged by one month, the full models lose almost all of their fit: McFadden R-squared falls from 0.33 to 0.03 for Model A and from 0.42 to 0.06 for Model B, the likelihood-ratio tests no longer reject ($p = 0.94$ and 0.83), and the AUCs fall to 0.59 and 0.66 from 0.87 and 0.91 in the same month. The out-of-time results above therefore show that the same-month relationship is stable between 2018-2022 and 2023-2025, not that the models can forecast. This agrees with the conclusion of Section 8 that models of this kind should not be relied on to anticipate crashes: the peso, the yield and oil move with the PSEi within the month, and the logit describes that co-movement, which is the purpose of our explanatory analysis.

Model Comparison

Table 80: Models A and B side by side (LR_p: likelihood-ratio test against the intercept-only model)

outcome	spec	events	LR_p	mcfadden_ r2	auc_in_ sample	cv_skill	cv_auc	oot_auc
A. Up month	full (9 regressors)	48	<0.0001	0.334	0.868	0.145	0.797	0.827
A. Up month	core (USD_ PHP, 10Y, WTI)	48	<0.0001	0.305	0.843	0.254	0.821	0.892
B. Crash month	full (9 regressors)	21	<0.0001	0.421	0.909	0.218	0.831	0.827
B. Crash month	core (USD_ PHP, 10Y, WTI)	21	<0.0001	0.376	0.891	0.318	0.865	0.947

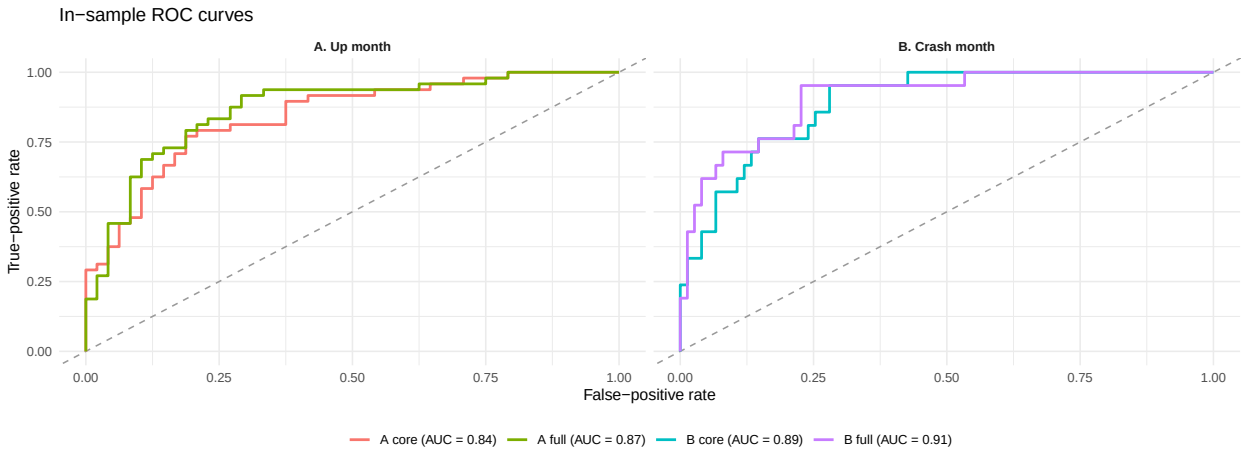


Figure 24: In-sample ROC curves of the logistic models, full and core specifications)

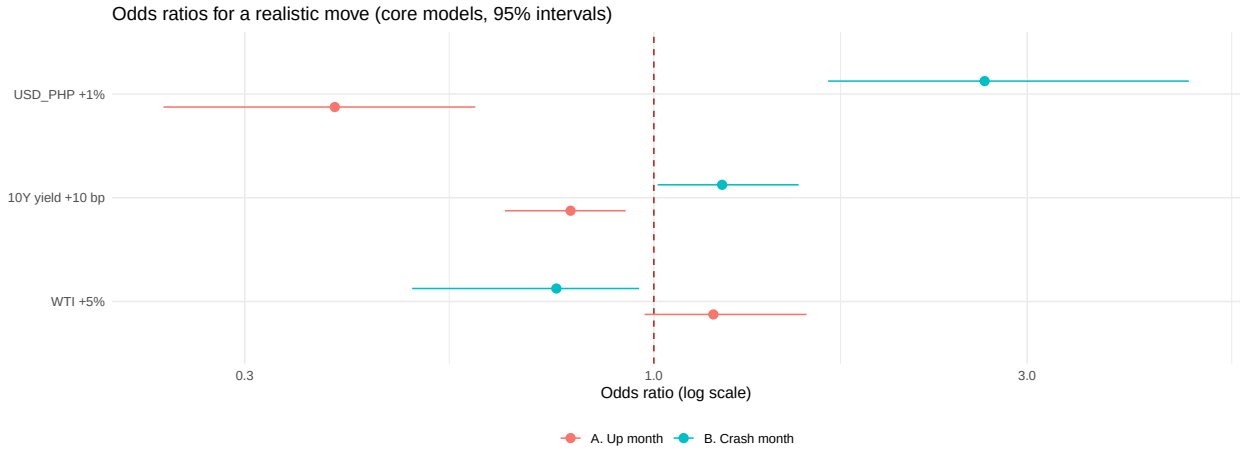


Figure 25: Logistic models: odds ratios for a realistic move in each driver, 95% intervals on a log scale

The ROC curves summarise discrimination. Each curve plots the share of true events that the model flags against the share of false alarms as the probability cut-off moves, and a curve well above the diagonal means good separation. Both outcomes are well separated, Model B (AUC 0.89 core, 0.91 full) more than Model A (0.84 and 0.87), and the core curves lie close to the full ones, which supports the core specification. The odds-ratio plot carries the financial message. The peso and the yield sit on opposite sides of 1 for the two outcomes, below 1 for an up month and above 1 for a crash, with intervals that exclude 1 in both models, so the conditions that make an ordinary good month less likely also make a crash more likely. Oil is the exception: its interval for an up month spans 1 (0.97 to 1.57) and for a crash lies just below 1 (0.49 to 0.96), an effect that depends on March 2020 as shown above.

- **A. Up month** (48 of 96 months). Full model: LR chi-square(9) = 44.5, p <0.0001; McFadden R-squared = 0.33; in-sample AUC = 0.87. Significant at 5%: USD_PHP, PH_10Y_GS_Yield. Core model odds of an up month: x0.39 for USD_PHP +1%, x0.78 for the 10Y yield +10 bp.
- **B. Crash month** (21 of 96 months, return below -5%). Full model (Firth): penalized LR chi-square(9) = 35.8, p <0.0001; McFadden R-squared = 0.42; in-sample AUC = 0.91. Significant at 5%: USD_PHP, PH_10Y_GS_Yield, WTI. Core model odds of a crash: x2.65 for USD_PHP +1%, x1.22 for the 10Y yield +10 bp, x0.75 for WTI +5%.

- **Out of sample.** Cross-validated skill over the base rate, full / core: A 0.14 / 0.25; B 0.22 / 0.32. Out-of-time AUC (2023-2025), full / core: A 0.83 / 0.89; B 0.83 / 0.95 (B has only 9 crash months in the test window). Core figures are optimistic since the core set was chosen on the same data.

Taken together, these figures show that the logistic models separate months well and that the same two drivers do the work in both. Discrimination is strong in sample (AUC 0.87 for Model A and 0.91 for Model B) and falls by about 7 to 8 points in cross-validation (0.80 and 0.83), the ordinary loss when a model meets data it has not seen, and the models beat the base rate in every validation. Model A has more events (48) and steadier inference, whereas Model B discriminates better but rests on 21 events, a penalized estimator and one dominant month, which is why its intervals are wider and its oil result is the least secure.

Thus, in review, The logit was the appropriate advanced method for this study because it answered a question that the linear model could not, how likely a crash is and not only how large the average return is; because the diagnostics support its specification and the Firth estimator suits its thin events; and because its findings are consistent with the linear model while adding information about the tail. The Results below set the two sets of findings side by side, and the Analysis that follows states the limits of the evidence.

Results

The MLR of Section 5 explains the size of the PSEi return (R-squared = 0.57). Only three of its nine regressors are individually significant, and AIC, AICc and BIC all prefer the three-regressor model. Its slopes, in percentage points of PSEi return, are -2.05 per 1% peso depreciation, -0.41 per 10 bp rise in the 10Y yield and 0.65 per 5% rise in WTI. The logistic models above restate the same relationships as odds, and they agree with the MLR on which variables matter and on the signs.

We observe that the Peso depreciation is the dominant channel. In the core models a 1% depreciation multiplies the odds of an up month by 0.39 (95% CI 0.24 to 0.59) and the odds of a crash month by 2.65 (95% CI 1.67 to 4.83). A weaker peso both signals and accompanies foreign selling, because foreign flows are an important driver of PSE prices and the peso moves with global risk appetite. This is why `SP500` and `VIX` add little once `USD_PHP` is in the model. Higher yields work against equities through the discount-rate channel: a 10 bp rise in the 10Y yield multiplies the odds of an up month by 0.78 ($p = 0.0062$) and the odds of a crash month by 1.22 ($p = 0.0372$). The yield effect is smaller and less precisely estimated than the peso's, and for crashes the interval (1.01 to 1.53) comes close to 1.

Oil matters for the tail, not for direction, and mostly because of March 2020. A 5% rise in WTI does not significantly change the odds of an up month (odds ratio 1.19, 95% CI 0.97 to 1.57), but it multiplies the odds of a crash month by 0.75 (95% CI 0.49 to 0.96). March 2020 is the one month in which a large oil collapse (a -78% log change) coincided with a PSEi crash, a global-demand and risk-off shock, whereas the other large oil drops (November 2018, November 2021 and April 2025) came with PSEi gains. The oil result therefore registers as a single-episode demand shock, not as a stable cost channel. The domestic macro variables add nothing, since `PH_CPI`, `PH_M3`, `PH_Manufacturing_Production` and the BSP policy-rate change are not significant in either logit. Dropping all six non-core regressors from model A is not rejected (likelihood-ratio test, $p = 0.686$), and a plausible reason is that these are slow-moving year-on-year series that markets price before the monthly figure is published.

The logit adds to the MLR because the same variables that explain ordinary up and down months also explain crashes, while oil only enters the crash model, a distinction the MLR cannot show. A binary outcome also caps the weight of one extreme month, so model A is unaffected by dropping March 2020. Model B is not immune, because March 2020 is its highest-leverage month.

Analysis

Reiterating, we stress that the models are explanatory, not forecasting, because the regressors are contemporaneous, as in the MLR. A check that lags every regressor by one month (see the lag-check table above)

leaves almost no fit for the up-month model (McFadden R-squared 0.03, likelihood-ratio p = 0.938, AUC 0.59). The models therefore describe what moves with the market in the same month, and they do not predict next month.

The sample has 21 crash months in 96, about 2.33 events per regressor in the full model, so the crash results lean on the core model and on Firth's penalized likelihood, and the intervals are wide. Cross-validated skill over the base rate is 0.25 (A) and 0.32 (B) for the core models, and 0.14 and 0.22 for the full models. The core set was chosen from the OLS results on the same data, so the full-model figures are the fair comparison. On the 2023-2025 test window the core models reach an AUC of 0.89 (A) and 0.95 (B), but the window holds only 9 crash months, so this is indicative only.

Dropping March 2020 leaves `USD_PHP` and `PH_10Y_GS_Yield` essentially unchanged in both models, but the oil effect in model B loses significance (p = 0.084), so the oil finding is the least robust result. The results hold at -4%, -5% and the bottom 20% of months, while at -6% (12 events) the peso effect remains but the yield effect becomes borderline (p = 0.0588). Finally, the peso, yields and the equity index all respond to global risk and to BSP policy at the same time, so the estimates show co-movement, not that a peso move causes a market move.

Section 10: Financial Interpretation

Over 96 months, from January 2018 to December 2025, the PSEi is tied more consistently to the USD/PHP exchange rate (the peso) and the Philippine 10-year government bond yield (the 10-year yield) than to any other variable studied. The nine regressors are jointly significant ($F(9, 86) = 12.79$, $p < 0.0001$) and explain 57.2 percent of the variation in monthly PSEi returns (adjusted R-squared 0.528), yet only the peso, the 10-year yield and oil are individually significant. The diagnostics then show that March 2020 dominates the fit and that the error variance is not constant, and after the remedies only the peso and the yield stay significant in every linear fit. The reduced model, with the peso, the 10-year yield, oil and a March 2020 dummy, accounts for 58.8 percent of the variation (adjusted R-squared). The regressors were chosen on the same sample, so the figure is optimistic, though cross-validated error is 13 percent below that of the nine-regressor model.

The peso has a negative relationship with PSEi returns ($p < 0.0001$). When the peso falls against the US dollar, the PSEi usually records negative returns: a 1 percent depreciation goes with a monthly return about 2.0 percentage points lower in every fit, and a one-standard-deviation depreciation with 3.25 points lower (95 percent interval -4.21 to -2.29), just over half a standard deviation of the return itself and the largest economic effect in the study. A weaker peso may signal slipping investor confidence or foreign investors pulling money from the market, and exchange rates and equity prices move together through international portfolio flows (Hau & Rey, 2006). It also raises the cost of imported goods and foreign-currency payments, so companies may bear expenses that hurt their earnings and share prices (Bartov & Bodnar, 1994). Because capital flows, import costs and risk appetite all move the peso, it is best read as a price in which several channels meet, and the data cannot separate them.

The 10-year yield also has a negative, smaller relationship: a 10 basis point rise goes with a return 0.35 to 0.41 percentage points lower, and a one-standard-deviation rise with 1.49 points lower, where even the conservative interval end of 0.56 points is about a tenth of the index's monthly volatility (5.97 points), a size that matters to an investor. Investors may prefer government bonds when yields increase, since they pay a return with less risk than stocks. A higher yield raises the rate at which expected earnings are discounted, which lowers present values (Campbell & Shiller, 1988), and raises borrowing costs, which weigh on business operations and profits. The policy rate carries no separate signal: it is unchanged in 68 percent of months, and stocks respond mainly to the unanticipated part of policy (Bernanke & Kuttner, 2005), so its effect may already sit in the yield and the peso, which are jointly significant with it ($F = 21.44$, $p < 0.0001$) under classical, HC3 and Newey-West errors.

Oil is the conditional link. In the baseline model a rise in WTI goes with a higher return, but March 2020, when oil fell by more than half in a global demand collapse, carries that result. Its p-value rises from 0.0006

under classical errors to 0.147 under HC3, and once the month is controlled the effect is no longer significant, which fits evidence that oil affects equities differently when demand and when supply drives it (Kilian & Park, 2009). Oil therefore reads as a signal of a global demand shock in an extreme month.

The other six regressors add little. Inflation, M3 growth and manufacturing production are slow year-on-year rates that markets price before the monthly release (N.-F. Chen et al., 1986), and as a group they are not jointly significant ($F = 1.07$, $p = 0.366$). The S&P 500, although correlated 0.49 with the PSEi, adds little once the peso is included ($p = 0.244$), so its information appears to reach the PSEi through the peso, and the global factor group (the S&P 500, oil and the VIX) is significant under classical ($p < 0.0001$) and Newey-West ($p = 0.003$) errors but only at the 10 percent level under HC3 ($p = 0.090$), which suggests that it rests largely on oil. The intervals of the six allow effects of one to two percentage points per standard deviation, so they are inconclusive, not irrelevant.

Furthermore, the Core logistic models carry the same two drivers into the tail. A 1 percent depreciation multiplies the odds of an up month by 0.39 and of a crash month, a fall of more than 5 percent, by 2.65 (95 percent interval 1.67 to 4.83), and a 10 basis point rise in the yield multiplies them by 0.78 and 1.22. From the sample frequencies of 50 percent for an up month and 22 percent for a crash, a 1 percent depreciation would roughly cut the chance of an up month to 28 percent and raise the chance of a crash to 43 percent. The conditions that make a good month less likely therefore also make a crash more likely, so the exposure is to the tail as well as to the average return. The full models separate months well (cross-validated AUC 0.80 and 0.83), and neither driver depends on March 2020 in the logistic fits.

We note that these findings need caution. Only 21 of 96 months are crashes, about 2.3 per regressor in the full model against the rule of 10 (Peduzzi et al., 1996), so the crash model relies on a penalized likelihood (Firth, 1993) and its intervals are wide: without March 2020 the oil effect loses significance ($p = 0.084$), and at a -6 percent threshold the yield effect turns borderline ($p = 0.059$). The models are explanatory, not predictive: lagging the regressors one month leaves the up-month model almost no fit (McFadden R-squared 0.03), and in-sample fit seldom survives out of sample (Welch & Goyal, 2008). No break is found in the coefficients (sup-F $p = 0.306$), but with 96 months the test has little power, and the exposure of an emerging market to global conditions changes over time (Bekaert & Harvey, 1995), so 2018 to 2025 need not describe other periods. The peso, the yield and the equity index also respond together to global risk and Bangko Sentral policy, so the results show co-movement, not that a peso or yield move causes a market move.

For an investor, the evidence supports monitoring the peso and the 10-year yield: months in which the peso weakens and the yield climbs go with lower returns and a higher chance of a sharp fall. We suggest watching both closely when evaluating the stock market, because a Philippine equity position carries exposures to both that the index return alone hides. The reduced model still leaves about 41 percent of the variation unexplained, so these two prices are not signals to buy or sell on their own, and company performance, financial statements and the broader market climate lie outside the model.

Section 11: Financial Recommendation

The evidence supports one recommendation with three parts: investment analysis under Philippine equity exposure should monitor the peso and the 10-year yield, should treat a month in which the peso weakens and the yield rises together as a reason to review risk, and should not rest decisions on the other regressors studied. The two prices are the only ones that stayed significant in every linear fit and in both logistic models, whereas the oil association depended on March 2020, the joint significance of the global factors depended on the covariance estimator, and the domestic macroeconomic indicators were not jointly significant. Our recommendation is therefore a way of managing exposure and not a forecast of returns, because the estimates are contemporaneous associations from a single sample and do not establish causation.

The USD/PHP exchange rate should be monitored first, since it has the strongest association with the market: a 1 percent depreciation goes with a monthly return about 2.0 percentage points lower and multiplies the odds of a crash month by 2.65. A sustained weakening may signal declining investor confidence or foreign capital outflows, and it raises the cost of imported inputs and foreign-currency obligations, which cuts into

earnings (Bartov & Bodnar, 1994). The index-level result cannot show which holdings carry that exposure, so a depreciation should lead the investor to examine the companies held, in particular their foreign-currency liabilities, reliance on imported materials and dependence on foreign capital, and to consider reducing the weight of the most exposed.

The Philippine 10-year government bond yield should be monitored alongside it. A 10 basis point rise goes with a return 0.35 to 0.41 percentage points lower, a smaller but consistent effect that works through two channels. Higher yields lower the present value of expected earnings (Campbell & Shiller, 1988) and make government securities more attractive relative to shares, because they pay a return with less risk. They also raise borrowing costs, which fall hardest on firms with substantial debt or dependence on external financing (Bernanke & Gertler, 1995), so the debt and refinancing profile of each holding deserves attention when yields climb. The policy rate need not be tracked separately, since it carried no signal beyond the yield and the peso.

Crude oil prices should not serve as a standalone indicator of PSEi performance. The association between WTI and returns lost significance under HC3 errors and once March 2020 was controlled, so the sample does not support using oil movements as a primary basis for anticipating returns or for investment decisions. The S&P 500, the VIX and the domestic indicators of inflation, M3 growth and manufacturing production are in the same position: the data leave them inconclusive, not irrelevant, and so give no basis for a rule.

The strongest case for action arises when the two signals coincide. Applying the estimated odds ratios to the sample frequencies, a month with a 1 percent depreciation and a 10 basis point rise in the yield would cut the chance of an up month from 50 percent to about 23 percent and raise the chance of a crash month from 22 percent to about 48 percent, illustrative figures and not forecasts. In such a month the investor should reassess portfolio risk, favor companies with strong balance sheets, manageable debt, stable cash flows and resilient business models, and diversify beyond equities, since equity markets tend to move together more closely in downturns (Longin & Solnik, 2001). Because bond prices fall when yields rise, the added fixed-income exposure is better held in short-dated government securities than in long-dated ones.

In conclusion, we stress that these indicators supplement, and do not replace, the analysis of company fundamentals and the broader economic outlook. The models leave a large share of return variation unexplained, describe only the same month and rest on 21 crash months in one sample from 2018 to 2025, so the peso and the yield indicate when to review exposure and not when to buy or sell. We recommend therefore to use them as triggers for a deliberate review and should re-estimate the relationships as new data accumulate.

References

Abd-Allah, M. H. (2026). The impact of exchange rates on stock market performance: Evidence from the Egyptian stock exchange. *Copernican Journal of Finance & Accounting*, *14*(4), 9–24. https://doi.org/10.12775/cjfa.2025.015

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, *19*(6), 716–723. https://doi.org/10.1109/TAC.1974.1100705

Akter, N., & Nobi, A. (2018). Investigation of the financial stability of S&P 500 using realized volatility and stock returns distribution. *Journal of Risk and Financial Management*, *11*(2), Article 22. https://doi.org/10.3390/jrfm11020022

Allen, D. M. (1974). The relationship between variable selection and data augmentation and a method for prediction. *Technometrics*, *16*(1), 125–127. https://doi.org/10.1080/00401706.1974.10489157

Alloul, F., & Ferrouhi, E. M. (2024). Macroeconomic indicators and stock market returns: A comparative analysis. *International Journal of Revenue Management*, *14*(4), 333–360. https://doi.org/10.1504/IJRM.2024.143451

Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, *61*(4), 821–856. https://doi.org/10.2307/2951764

Babar, H., Hamid, K., Rasheed, M., Rasool, F., & Saeed, M. Y. (2024). Do macroeconomic factors are interlinked with stock markets in Asian emerging economies: A panel data approach. *International Journal of Business and Economic Affairs*, *9*(2), 42–51. https://doi.org/10.24088/IJBEA-2024-92004

Baker, S. R., Bloom, N., Davis, S. J., Kost, K., Sammon, M., & Viratyosin, T. (2020). The unprecedented stock market reaction to COVID-19. *The Review of Asset Pricing Studies*, *10*(4), 742–758. https://doi.org/10.1093/rapstu/raaa008

Bartov, E., & Bodnar, G. M. (1994). Firm valuation, earnings expectations, and the exchange-rate exposure effect. *The Journal of Finance*, *49*(5), 1755–1785. https://doi.org/10.1111/j.1540-6261.1994.tb04780.x

Bekaert, G., & Harvey, C. R. (1995). Time-varying world market integration. *The Journal of Finance*, *50*(2), 403–444. https://doi.org/10.2307/2329414

Bekaert, G., & Hoerova, M. (2014). The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, *183*(2), 181–192. https://doi.org/10.1016/j.jeconom.2014.05.008

Bernanke, B. S., & Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, *9*(4), 27–48. https://doi.org/10.1257/jep.9.4.27

Bernanke, B. S., & Kuttner, K. N. (2005). What explains the stock market's reaction to Federal Reserve policy? *The Journal of Finance*, *60*(3), 1221–1257. https://doi.org/10.1111/j.1540-6261.2005.00760.x

Bhattacharjee, A., & Das, J. (2021). Investigating the effect of broad money supply on stock market index and market capitalization: Evidence from liberalized India. *Jindal Journal of Business Research*, *10*(2), 185–198. https://doi.org/10.1177/22786821211047615

Bonga-Bonga, L., & Kabwe wa Kabwe, M. (2026). The role of key factors in the relationship between exchange rate and stock market. *Risk and Decision Analysis*, *12*(3), 123–140. https://doi.org/10.1177/15697371251415112

Breusch, T. S. (1978). Testing for autocorrelation in dynamic linear models. *Australian Economic Papers*, *17*(31), 334–355. https://doi.org/10.1111/j.1467-8454.1978.tb00635.x

Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica*, *47*(5), 1287–1294. https://doi.org/10.2307/1911963

Burnham, K. P., & Anderson, D. R. (2004). Multimodel inference: Understanding AIC and BIC in model selection. *Sociological Methods & Research*, *33*(2), 261–304. https://doi.org/10.1177/0049124104268644

Bussiere, M., & Fratzscher, M. (2006). Towards a new early warning system of financial crises. *Journal of International Money and Finance*, *25*(6), 953–973. https://doi.org/10.1016/j.jimonfin.2006.07.007

Cakici, N., & Zaremba, A. (2024). What drives stock returns across countries? Insights from machine learning models. *International Review of Financial Analysis*, *96*, Article 103569. https://doi.org/10.1016/j.irfa.2024.103569

Camino, C. C., Del Pilar, M. D., Fuentebella, P. A., & Hong, B. A. (2023). *The relationship between economic indicators and stock exchange index of ASEAN-4 countries: Indonesia, Malaysia, Philippines, and Thailand* [Thesis]. De La Salle University. https://animorepository.dlsu.edu.ph/etdb_finman/64/

Campbell, J. Y., & Shiller, R. J. (1988). The dividend-price ratio and expectations of future dividends and discount factors. *The Review of Financial Studies*, *1*(3), 195–228. https://doi.org/10.1093/rfs/1.3.195

Castro, C., & Jiménez-Rodríguez, R. (2024). The impact of oil shocks on the stock market. *Global Finance Journal*, *60*, Article 100967. https://doi.org/10.1016/j.gfj.2024.100967

Checo, A., Grigoli, F., & Sandri, D. (2024). *Monetary policy transmission in emerging markets: Proverbial concerns, novel evidence* (IMF Working Paper 2024/093). International Monetary Fund.

Chen, K. (2026). The cointegration between gold price and stock market volatilities: Evidence from S&P500. *Advances in Economics Management and Political Sciences*, *283*(1), 21–29. https://doi.org/10.54254/2754-1169/2026.34908

Chen, N.-F., Roll, R., & Ross, S. A. (1986). Economic forces and the stock market. *The Journal of Business*, *59*(3), 383–403. https://doi.org/10.1086/296344

Cieslak, A., & Pflueger, C. (2023). Inflation and asset returns. *Annual Review of Financial Economics*, *15*, 433–448. https://doi.org/10.1146/annurev-financial-110921-104726

Cook, R. D. (1977). Detection of influential observation in linear regression. *Technometrics*, *19*(1), 15–18. https://doi.org/10.1080/00401706.1977.10489493

Dilpriya, T. A. H., Lanel, G. H. J., Perera, M. T. M., & Vidanage, B. V. N. C. (2025). Analysing the S&P 500 index in relation to the Google trends of stock market related words in the United States. In D. Dahanayake & M. Rabindrakumar (Eds.), *Transformative applied research in computing, engineering, science and technology* (pp. 133–140). CRC Press. https://doi.org/10.1201/9781003616368-17

Drechsler, I., Savov, A., & Schnabl, P. (2018). Liquidity, risk premia, and the financial transmission of monetary policy. *Annual Review of Financial Economics*, *10*, 309–328. https://doi.org/10.1146/annurev-

financial-110217-022833
Durbin, J., & Watson, G. S. (1950). Testing for serial correlation in least squares regression. I. *Biometrika*, *37*(3-4), 409–428. https://doi.org/10.1093/biomet/37.3-4.409
El-Diftar, D. (2023). The impact of exchange rates on stock market performance of the emerging 7. *Journal of Capital Markets Studies*, *7*(2), 125–139. https://doi.org/10.1108/JCMS-03-2023-0005
Endres, E. J. E. (2020). Impact of macroeconomic factors on stock market: Empirical evidence from the Philippines. *International Journal of Research Publication*, *50*(1), 114–125. https://doi.org/10.47119/IJRP100501420201059
Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, *50*(4), 987–1007. https://doi.org/10.2307/1912773
Escribano, A., Koczar, M. W., Jareño, F., & Esparcia, C. (2023). Shock transmission between crude oil prices and stock markets. *Resources Policy*, *83*, Article 103754. https://doi.org/10.1016/j.resourpol.2023.103754
Fama, E. F. (1981). Stock returns, real activity, inflation, and money. *The American Economic Review*, *71*(4), 545–565. https://www.jstor.org/stable/1806180
Fernandez, N. B., & Li, R. C. (2020). The influence of macroeconomic variables on Philippine stock market indices: A structural equation model approach. *2020 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 1027–1031. https://doi.org/10.1109/IEEM45057.2020.9309816
Firth, D. (1993). Bias reduction of maximum likelihood estimates. *Biometrika*, *80*(1), 27–38. https://doi.org/10.1093/biomet/80.1.27
Fitch Solutions Group Limited. (2019). *Philippines banking & financial services report - Q4 2019* [Fitch Solutions Country Industry Reports]. Fitch Solutions Group Limited.
Fitch Solutions Group Limited. (2022). *Philippines banking & financial services report - Q3 2022* [Fitch Solutions Country Industry Reports]. Fitch Solutions Group Limited.
Folorunso, A. T. (2023). Investigating the impact of inflation and other macroeconomic variables on stock returns in Nigeria. *Asian Journal of Economics, Business and Accounting*, *23*(14), 8–26. https://doi.org/10.9734/ajeba/2023/v23i141001
Forbes, K. J., & Rigobon, R. (2002). No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, *57*(5), 2223–2261. https://doi.org/10.1111/0022-1082.00494
Frankel, J. A., & Rose, A. K. (1996). Currency crashes in emerging markets: An empirical treatment. *Journal of International Economics*, *41*(3-4), 351–366. https://doi.org/10.1016/S0022-1996(96)01441-9
Gallardo, M., & Viray, J. (2025). Philippine stock exchange index (PSEi) and its selected macroeconomic indicators: Catalyst for sustainable economic growth. *Bedan Research Journal*, *10*(1), 82–116. https://doi.org/10.58870/berj.v10i1.81
Godfrey, L. G. (1978). Testing against general autoregressive and moving average error models when the regressors include lagged dependent variables. *Econometrica*, *46*(6), 1293–1301. https://doi.org/10.2307/1913829
Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, *2*(2), 111–120. https://doi.org/10.1016/0304-4076(74)90034-7
Greenland, S., Senn, S. J., Rothman, K. J., Carlin, J. B., Poole, C., Goodman, S. N., & Altman, D. G. (2016). Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. *European Journal of Epidemiology*, *31*(4), 337–350. https://doi.org/10.1007/s10654-016-0149-3
Hanif, W., Hadhri, S., & El Khoury, R. (2024). Quantile spillovers and connectedness between oil shocks and stock markets of the largest oil producers and consumers. *Journal of Commodity Markets*, *34*, Article 100404. https://doi.org/10.1016/j.jcomm.2024.100404
Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, *29*(1), 5–68. https://doi.org/10.1093/rfs/hhv059
Hau, H., & Rey, H. (2006). Exchange rates, equity prices, and capital flows. *The Review of Financial Studies*, *19*(1), 273–317. https://doi.org/10.1093/rfs/hhj008
Heinze, G., & Schemper, M. (2002). A solution to the problem of separation in logistic regression. *Statistics in Medicine*, *21*(16), 2409–2419. https://doi.org/10.1002/sim.1047
Heinze, G., Wallisch, C., & Dunkler, D. (2018). Variable selection – A review and recommendations for the practicing statistician. *Biometrical Journal*, *60*(3), 431–449. https://doi.org/10.1002/bimj.201700067

Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, *12*(1), 55–67. https://doi.org/10.1080/00401706.1970.10488634

Huber, P. J. (1973). Robust regression: Asymptotics, conjectures and Monte Carlo. *The Annals of Statistics*, *1*(5), 799–821. https://doi.org/10.1214/aos/1176342503

Hudson, R. S., & Gregoriou, A. (2015). Calculating and comparing security returns is harder than you think: A comparison between logarithmic and simple returns. *International Review of Financial Analysis*, *38*, 151–162. https://doi.org/10.1016/j.irfa.2014.10.008

Hurvich, C. M., & Tsai, C.-L. (1989). Regression and time series model selection in small samples. *Biometrika*, *76*(2), 297–307. https://doi.org/10.1093/biomet/76.2.297

Isnandari, M. L., & Chalid, D. A. (2017). Stock returns and inflation: Evidence from emerging markets in Asia. *The Indonesian Capital Market Review*, *9*(1). https://doi.org/10.21002/icmr.v9i1.8554

Jarque, C. M., & Bera, A. K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, *6*(3), 255–259. https://doi.org/10.1016/0165-1765(80)90024-5

Jurkowska, A., Ozcelebi, O., & Fijorek, K. (2024). Stock market TVP-VAR dynamic connectedness and VIX shocks spillovers: Evidence from a sectoral analysis of the Fragile Five. *International Entrepreneurship Review*, *10*(1), 97–126. https://doi.org/10.15678/IER.2024.1001.07

Karunanayake, I. (2014). Exchange rate influences on stock market returns and volatility dynamics: Empirical evidence from the Australian stock market. *Review of Applied Economics*, *10*(1–2), 83–97. https://doi.org/10.22004/ag.econ.264592

Kilian, L., & Park, C. (2009). The impact of oil price shocks on the U.S. stock market. *International Economic Review*, *50*(4), 1267–1287. https://doi.org/10.1111/j.1468-2354.2009.00568.x

Kirci Altinkeski, B., Dibooglu, S., Cevik, E. I., Kılıç, Y., & Bugan, M. F. (2024). Quantile connectedness between VIX and global stock markets. *Borsa Istanbul Review*, *24*(Suppl. 1), 71–79. https://doi.org/10.1016/j.bir.2024.07.006

Koenker, R. (1981). A note on studentizing a test for heteroscedasticity. *Journal of Econometrics*, *17*(1), 107–112. https://doi.org/10.1016/0304-4076(81)90062-2

Krispin, R. (2019). *Hands-on time series analysis with R: Perform time series analysis and forecasting using R* (1st ed.). Packt Publishing.

Kruel, M., & Ceretta, P. S. (2025). Extreme return spillover between the WTI, the VIX, and six Latin American stock markets: A quantile connectedness approach. *Studies in Nonlinear Dynamics and Econometrics*, *29*(3), 301–316. https://doi.org/10.1515/snde-2023-0076

Le, T.-H., Sabri, B., Bui, M. T., & Park, D. (2022). On the volatility of WTI crude oil prices: A time-varying approach with stochastic volatility. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4066338

Li, X., Li, B., Singh, T., & Shi, K. (2020). Predicting stock market returns in the US: Evidence from an average correlation approach. *Accounting Research Journal*, *33*(2), 411–433. https://doi.org/10.1108/ARJ-10-2018-0168

Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, *65*(2), 297–303. https://doi.org/10.1093/biomet/65.2.297

Lo, A. W., & MacKinlay, A. C. (1988). Stock market prices do not follow random walks: Evidence from a simple specification test. *Review of Financial Studies*, *1*(1), 41–66. https://doi.org/10.1093/rfs/1.1.41

Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, *54*(3), 217–224. https://doi.org/10.1080/00031305.2000.10474549

Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *The Journal of Finance*, *56*(2), 649–676. https://doi.org/10.1111/0022-1082.00340

MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, *29*(3), 305–325. https://doi.org/10.1016/0304-4076(85)90158-7

Mallik, A., Khan, A. T. M. J., & Chowdhury, A. A. K. (2024). Impact of macroeconomic variables on stock market performance: Evidence from Dhaka Stock Exchange. *Journal of Business Administration*, *45*(1), 47–63. https://doi.org/10.58964/JBA45N102

McMillan, D. G. (2022). The time-varying relation between stock returns and monetary variables. *Journal of Risk and Financial Management*, *15*(1), Article 9. https://doi.org/10.3390/jrfm15010009

Molenaar, R., Sénéchal, E., Swinkels, L., & Wang, Z. (2024). Empirical evidence on the stock–bond correlation. *Financial Analysts Journal*, *80*(3), 17–36. https://doi.org/10.1080/0015198X.2024.2317333

Narang, S., Savita, Pradhan, R. P., & Singh, B. (2023). COVID-19 pandemic and stock market response: The role of COVID-induced fear, investor attention, and firm-specific characteristics. *Asian Academy of Management Journal*, *28*(1), 311–344. https://doi.org/10.21315/aamj2023.28.1.13

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610

Nguyen, C. P., Schinckus, C., Su, T. D., & Chong, F. H. L. (2022). Determinants of stock market returns in emerging markets: The linkage between institutional quality and macro liquidity. *International Journal of Finance & Economics*, *27*(4), 4472–4486. https://doi.org/10.1002/ijfe.2382

O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity*, *41*(5), 673–690. https://doi.org/10.1007/s11135-006-9018-6

Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, *49*(12), 1373–1379. https://doi.org/10.1016/S0895-4356(96)00236-3

Prabheesh, K. P., Padhan, R., & Bhat, J. A. (2024). Do financial markets react to emerging economies' asset purchase program? Evidence from the COVID-19 pandemic period. *Journal of Asian Economics*, *90*, Article 101678. https://doi.org/10.1016/j.asieco.2023.101678

Puhr, R., Heinze, G., Nold, M., Lusa, L., & Geroldinger, A. (2017). Firth's logistic regression with rare events: Accurate effect estimates and predictions? *Statistics in Medicine*, *36*(14), 2302–2317. https://doi.org/10.1002/sim.7273

Ramsey, J. B. (1969). Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *31*(2), 350–371. https://doi.org/10.1111/j.2517-6161.1969.tb00796.x

Sadeghi, A., & Roudari, S. (2022). Heterogeneous effects of oil structure and oil shocks on stock prices in different regimes: Evidence from oil-exporting and oil-importing countries. *Resources Policy*, *76*, Article 102596. https://doi.org/10.1016/j.resourpol.2022.102596

Salkever, D. S. (1976). The use of dummy variables to compute predictions, prediction errors, and confidence intervals. *Journal of Econometrics*, *4*(4), 393–397. https://doi.org/10.1016/0304-4076(76)90027-0

Schrank, J. (2024). The impact of a crisis on monetary policy's influence on financial markets: Evidence from the COVID-19 pandemic. *Cogent Economics & Finance*, *12*(1), Article 2322874. https://doi.org/10.1080/23322039.2024.2322874

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, *6*(2), 461–464. https://doi.org/10.1214/aos/1176344136

Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, *52*(3-4), 591–611. https://doi.org/10.1093/biomet/52.3-4.591

Sharma, S., & Malik, K. (2022). Comovement of fear index, stock returns, Brent oil prices in BRIC countries: The case of COVID-19. *The Indian Economic Journal*, *70*(4), 559–576. https://doi.org/10.1177/00194662221082188

Sigfrid, I., & Rozkov, A. (2024). *Impact of interest rates on S&P 500 stock returns: A comparative analysis of value and growth stocks* [Student thesis]. Blekinge Tekniska Högskola, Institutionen för industriell ekonomi.

Smales, L. A. (2021). Geopolitical risk and volatility spillovers in oil and stock markets. *The Quarterly Review of Economics and Finance*, *80*, 358–366. https://doi.org/10.1016/j.qref.2021.03.008

Sreenu, N. (2023). Effect of exchange rate volatility and inflation on stock market returns dynamics - evidence from India. *International Journal of System Assurance Engineering and Management*, *14*(3), 836–843. https://doi.org/10.1007/s13198-023-01914-3

Statista Ltd. (2026). *Philippines: Central bank policy rate 1995-2026: Monthly overnight reverse repurchase rate of the Bangko Sentral ng Pilipinas from January 1995 to August 2026* [Data set]. Statista Ltd.

Su, J.-B. (2022). Structural change in the correlation, return and volatility spillovers: Evidence from the oil, stock and exchange rate markets in the United States. *Economic Research - Ekonomska Istraživanja*, *35*(1), 6918–6944. https://doi.org/10.1080/1331677X.2022.2054453

Sy, D. K. S. (2020). Fiscal policy and stock market movement in the Philippines. *DLSU Business & Economics Review*, *29*(2), Article 12. https://doi.org/10.59588/2243-786X.1138

Tala, M. V. (2025). *Assessing the sensitivity of the Philippine financial index to macroeconomic fluctuations: An empirical analysis (2009–2024)* [Thesis]. De La Salle University. https://animorepository.dlsu.edu.

ph/etdm_finman/20

The Philippine Stock Exchange, Inc. (n.d.). *Indices*. Retrieved https://www.pse.com.ph/indices/

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *58*(1), 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

Umoru, D., Igbinovia, B., & Igbafe, T. A. (2024). Returns, interest rate variations and changes in foreign exchange rates in Asia. *Development Management*, *23*(4), 8–22. https://doi.org/10.57111/devt/4.2024.08

Valadkhani, A., & O'Mahony, B. (2024). Market fear and emerging market returns: Analyzing the impacts of VIX and dollar index. *Annals of Financial Economics*, *19*(3), Article 2450015. https://doi.org/10.1142/S2010495224500155

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, *70*(2), 129–133. https://doi.org/10.1080/00031305.2016.1154108

Welch, I., & Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, *21*(4), 1455–1508. https://doi.org/10.1093/rfs/hhm014

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, *48*(4), 817–838. https://doi.org/10.2307/1912934

Wongbangpo, P., & Sharma, S. C. (2002). Stock market and macroeconomic fundamental dynamic interactions: ASEAN-5 countries. *Journal of Asian Economics*, *13*(1), 27–51. https://doi.org/10.1016/S1049-0078(01)00111-7

Yan, X., & Su, X. G. (2009). *Linear regression analysis: Theory and computing*. World Scientific Publishing Co. Pte. Ltd.

Yohai, V. J. (1987). High breakdown-point and high efficiency robust estimates for regression. *The Annals of Statistics*, *15*(2), 642–656. https://doi.org/10.1214/aos/1176350366

Youssef, R. B., & Jouini, F. (2023). The relationship between investor sentiment and stock returns: The case of the S&P 500 companies. *International Journal of Monetary Economics and Finance*, *16*(5), 309–333. https://doi.org/10.1504/IJMEF.2023.135666

Zaremba, A., Cakici, N., Bianchi, R. J., & Long, H. (2023). Interest rate changes and the cross-section of global equity returns. *Journal of Economic Dynamics and Control*, *147*, Article 104596. https://doi.org/10.1016/j.jedc.2023.104596

Zhang, Z. (2021). *Stock returns and inflation redux: An explanation from monetary policy in advanced and emerging markets* (IMF Working Paper 2021/219). International Monetary Fund. https://doi.org/10.5089/9781513586755.001

Ziadat, S. A., McMillan, D., & Herbst, P. (2022). Oil shocks and equity returns during bull and bear markets: The case of oil importing and exporting nations. *Resources Policy*, *75*, Article 102461. https://doi.org/10.1016/j.resourpol.2021.102461

Ziliak, S. T., & McCloskey, D. N. (2004). Size matters: The standard error of regressions in the American Economic Review. *The Journal of Socio-Economics*, *33*(5), 527–546. https://doi.org/10.1016/j.socec.2004.09.024